Day 001 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Add Decimals with Place Value

Standards: 6.NS.B.3

Learning target: I can add decimals by aligning equal place values.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down.

Worked example 1

Use the standard algorithm: 85.9 + 2.9. Show your regrouping or division steps.

1. Line up the decimal points. Add placeholder zeros so the numbers are 85.9 and 2.9.

2. Column 1 from the right: 9 + 9 = 18. Write 8 and carry 1 to the next column.

3. Column 2 from the right: 5 + 2 + 1 carried = 8. Write 8.

4. Column 3 from the right: 8 + 0 = 8. Write 8.

5. Bring the decimal point straight down, keeping 1 decimal place: 88.8. The simplified decimal answer is 88.8.

Answer: 88.8

Worked example 2

Use the standard algorithm: 4.06 + 0.15. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 521/100
B. 421/100
C. 321/100
D. 421/10

1. Line up decimal points so tenths are under tenths and hundredths under hundredths.

2. Align decimal points

3. 4.06 + 0.15 = 4.21.

Answer: B - 421/100

Day 001 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 40.5 + 8.4. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 84.7 + 2. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 59.4 + 1.9. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 1.67 + 2.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 6.34 + 7.9. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 4.93 + 6.7. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 5.68 + 0.8. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 7.99 + 0.47. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 423/5
B. 423/50
C. 473/50
D. 373/50

09 / Multiple choice

Use the standard algorithm: 2.67 + 0.21. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 72/25
B. 144/5
C. 97/25
D. 47/25

10 / Constructed response

A runner walks 4.4 km in the morning and 0.71 km later. What is the total distance?

Day 002 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Subtract Decimals and Check

Standards: 6.NS.B.3

Learning target: I can subtract decimals and explain regrouping.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Align decimal points and add placeholder zeros. Regroup one whole as ten tenths, or one tenth as ten hundredths. Subtract by place and check by adding the difference to the subtrahend.

Worked example 1

Use the standard algorithm: 94.7 - 4.1. Show your regrouping or division steps.

1. Line up the decimal points. Add placeholder zeros so the numbers are 94.7 and 4.1.

2. Column 1 from the right: 7 − 1 = 6. Write 6.

3. Column 2 from the right: 4 − 4 = 0. Write 0.

4. Column 3 from the right: 9 − 0 = 9. Write 9.

5. Bring the decimal point straight down, keeping 1 decimal place: 90.6. The simplified decimal answer is 90.6.

6. Check: 90.6 + 4.1 = 94.7.

Answer: 90.6

Worked example 2

Use the standard algorithm: 5.37 - 0.49. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 122/25
B. 147/25
C. 97/25
D. 244/5

1. Align decimal points and add placeholder zeros.

2. Align decimal points

3. 5.37 - 0.49 = 4.88.

Answer: A - 122/25

Day 002 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 46.2 - 7.3. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 82.5 - 1.5. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 36.6 - 7.2. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 8.2 - 4.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 8.04 - 6. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 6.47 - 1.4. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 5.45 - 0.35. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 8.54 - 0.34. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 46/5
B. 36/5
C. 82
D. 41/5

09 / Multiple choice

Use the standard algorithm: 6.71 - 0.14. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 757/100
B. 657/100
C. 657/10
D. 557/100

10 / Constructed response

A tank holds 5.59 L. A class uses 0.69 L. How much remains?

Day 003 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Multiply Decimals and Estimate

Standards: 6.NS.B.3

Learning target: I can multiply decimals and use an estimate to place the decimal point.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate.

Worked example 1

Use the standard algorithm: 36.8 × 7.3. Show your regrouping or division steps.

1. Estimate the size first. Treat the factors as whole numbers: 36.8 becomes 368, and 7.3 becomes 73.

2. Multiply the whole numbers: 368 × 73 = 26864.

3. Count decimal places: 1 in 36.8 plus 1 in 7.3 gives 2.

4. Place the decimal point 2 places from the right in 26864, adding leading zeros if necessary: 268.64. Remove only unnecessary trailing zeros.

Answer: 268.64

Worked example 2

Use the standard algorithm: 4.06 × 0.27. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 481/5000
B. 5481/5000
C. 5481/500
D. 10481/5000

1. Multiply as whole numbers using the standard algorithm.

2. Compute the whole-number product and restore decimal places: 4.06 × 0.27 = 1.0962.

Answer: B - 5481/5000

Day 003 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 19.2 × 1.8. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 46.8 × 2.3. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 22.3 × 3.7. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 7.17 × 7.4. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 9.39 × 5.1. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 3.82 × 3.1. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 3 × 0.62. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 6.92 × 0.8. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 817/125
B. 692/125
C. 567/125
D. 1384/25

09 / Multiple choice

Use the standard algorithm: 7.12 × 0.52. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 2939/625
B. 1689/625
C. 2314/625
D. 4628/125

10 / Constructed response

Ribbon costs $7.07 per meter. What is the exact cost of 0.44 meters? Do not round.

Day 004 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Divide Whole Numbers

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 304 ÷ 4. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 30 ÷ 4: write 7, subtract 28, remainder 2

3. bring down the next digit: 24 ÷ 4: write 6, subtract 24, remainder 0.

4. Check: 4 × 76 = 304.

Answer: 76

Worked example 2

Use long division: 2,948 ÷ 23. Show the algorithm and check.

1. Estimate the quotient first.

2. 29 ÷ 23: write 1, subtract 23, remainder 6

3. bring down the next digit: 64 ÷ 23: write 2, subtract 46, remainder 18

4. bring down the next digit: 188 ÷ 23: write 8, subtract 184, remainder 4.

5. Check: 23 × 128 + 4 = 2948.

Answer: 128 R 4

Day 004 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 536 ÷ 8. Show the algorithm and check.

02 / Constructed response

Use long division: 256 ÷ 4. Show the algorithm and check.

03 / Constructed response

Use long division: 592 ÷ 8. Show the algorithm and check.

04 / Constructed response

Use long division: 6,194 ÷ 19. Show the algorithm and check.

05 / Constructed response

Use long division: 7,854 ÷ 22. Show the algorithm and check.

06 / Constructed response

Use long division: 4,008 ÷ 12. Show the algorithm and check.

07 / Constructed response

Use long division: 4,255 ÷ 17. Show the algorithm and check.

08 / Model and constructed response

Use long division: 15,080 ÷ 37. Show the algorithm and check.

09 / Model and constructed response

Use long division: 2,900 ÷ 19. Show the algorithm and check.

10 / Constructed response

A club has 5132 cards. Each full packet holds 13. How many full packets can it make, and how many cards remain?

Day 005 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Estimate and Check Whole-Number Division

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 175 ÷ 5. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 17 ÷ 5: write 3, subtract 15, remainder 2

3. bring down the next digit: 25 ÷ 5: write 5, subtract 25, remainder 0.

4. Check: 5 × 35 = 175.

Answer: 35

Worked example 2

Use long division: 5,554 ÷ 35. Show the algorithm and check.

1. Estimate the quotient first.

2. 55 ÷ 35: write 1, subtract 35, remainder 20

3. bring down the next digit: 205 ÷ 35: write 5, subtract 175, remainder 30

4. bring down the next digit: 304 ÷ 35: write 8, subtract 280, remainder 24.

5. Check: 35 × 158 + 24 = 5554.

Answer: 158 R 24

Day 005 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 729 ÷ 9. Show the algorithm and check.

02 / Constructed response

Use long division: 252 ÷ 4. Show the algorithm and check.

03 / Constructed response

Use long division: 430 ÷ 5. Show the algorithm and check.

04 / Constructed response

Use long division: 2,299 ÷ 19. Show the algorithm and check.

05 / Constructed response

Use long division: 2,716 ÷ 14. Show the algorithm and check.

06 / Constructed response

Use long division: 5,928 ÷ 38. Show the algorithm and check.

07 / Constructed response

Use long division: 6,701 ÷ 21. Show the algorithm and check.

08 / Model and constructed response

Use long division: 2,368 ÷ 15. Show the algorithm and check.

09 / Model and constructed response

Use long division: 11,312 ÷ 37. Show the algorithm and check.

10 / Constructed response

A club has 4822 cards. Each full packet holds 13. How many full packets can it make, and how many cards remain?

Day 006 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Write Zero Digits in Quotients

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 525 ÷ 5. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 5 ÷ 5: write 1, subtract 5, remainder 0

3. bring down the next digit: 2 ÷ 5: write 0, subtract 0, remainder 2

4. bring down the next digit: 25 ÷ 5: write 5, subtract 25, remainder 0.

5. Check: 5 × 105 = 525.

Answer: 105

Worked example 2

Use long division: 3,175 ÷ 31. Show the algorithm and check.

1. Estimate the quotient first.

2. 31 ÷ 31: write 1, subtract 31, remainder 0

3. bring down the next digit: 7 ÷ 31: write 0, subtract 0, remainder 7

4. bring down the next digit: 75 ÷ 31: write 2, subtract 62, remainder 13.

5. Check: 31 × 102 + 13 = 3175.

Answer: 102 R 13

Day 006 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 981 ÷ 9. Show the algorithm and check.

02 / Constructed response

Use long division: 1,535 ÷ 5. Show the algorithm and check.

03 / Constructed response

Use long division: 609 ÷ 3. Show the algorithm and check.

04 / Constructed response

Use long division: 8,442 ÷ 21. Show the algorithm and check.

05 / Constructed response

Use long division: 3,360 ÷ 32. Show the algorithm and check.

06 / Constructed response

Use long division: 8,903 ÷ 29. Show the algorithm and check.

07 / Constructed response

Use long division: 3,675 ÷ 36. Show the algorithm and check.

08 / Model and constructed response

Use long division: 6,811 ÷ 33. Show the algorithm and check.

09 / Model and constructed response

Use long division: 2,868 ÷ 14. Show the algorithm and check.

10 / Constructed response

A club has 2447 cards. Each full packet holds 12. How many full packets can it make, and how many cards remain?

Day 007 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Divide by a Decimal

Standards: 6.NS.B.3

Learning target: I can divide by a decimal using an equivalent whole-number divisor.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 46.8 ÷ 0.9. Show your regrouping or division steps.

1. Make the divisor a whole number. Multiply BOTH numbers by 10: 46.8 ÷ 0.9 = 468 ÷ 9. This keeps the quotient unchanged.

2. 9 goes into 46 5 times. Write 5; multiply 5 × 9 = 45; subtract 46 − 45 = 1.

3. Bring down the next digit, 8, to make 18. 9 goes into 18 2 times. Write 2; multiply 2 × 9 = 18; subtract 18 − 18 = 0.

4. Check using multiplication: 0.9 × 52 = 46.8. The quotient is 52.

Answer: 52

Worked example 2

Use the standard algorithm: 1.752 ÷ 0.24. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 73
B. 83/10
C. 63/10
D. 73/10

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 100: 175.2 ÷ 24 = 7.3.

3. Check: 0.24 × 7.3 = 1.752.

Answer: D - 73/10

Day 007 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 55.3 ÷ 0.7. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 15.4 ÷ 0.2. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 21.5 ÷ 0.5. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 6.45 ÷ 1.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 20.4 ÷ 4. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 8.64 ÷ 1.2. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 2.16 ÷ 0.4. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 0.624 ÷ 0.24. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.

A. 8/5
B. 18/5
C. 13/5
D. 26

09 / Multiple choice

Use the standard algorithm: 3.456 ÷ 0.64. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.

A. 32/5
B. 22/5
C. 54
D. 27/5

10 / Constructed response

A container has 1.664 kg of seed. Each bag holds 0.32 kg. How many bags can be filled?

Day 008 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Explain Equivalent Decimal Division

Standards: 6.NS.B.3

Learning target: I can divide by a decimal using an equivalent whole-number divisor.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 16.8 ÷ 0.8. Show your regrouping or division steps.

1. Make the divisor a whole number. Multiply BOTH numbers by 10: 16.8 ÷ 0.8 = 168 ÷ 8. This keeps the quotient unchanged.

2. 8 goes into 16 2 times. Write 2; multiply 2 × 8 = 16; subtract 16 − 16 = 0.

3. Bring down the next digit, 8, to make 8. 8 goes into 8 1 time. Write 1; multiply 1 × 8 = 8; subtract 8 − 8 = 0.

4. Check using multiplication: 0.8 × 21 = 16.8. The quotient is 21.

Answer: 21

Worked example 2

Use the standard algorithm: 1.05 ÷ 0.25. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 26/5
B. 21/5
C. 16/5
D. 42

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 100: 105 ÷ 25 = 4.2.

3. Check: 0.25 × 4.2 = 1.05.

Answer: B - 21/5

Day 008 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 12.8 ÷ 0.4. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 35.2 ÷ 0.8. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 60 ÷ 0.8. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 8.52 ÷ 1.2. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 28.8 ÷ 6.4. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 16 ÷ 3.2. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 2.92 ÷ 0.4. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 0.744 ÷ 0.12. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 31/5
B. 62
C. 26/5
D. 36/5

09 / Multiple choice

Use the standard algorithm: 0.48 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 6/5
B. 1/5
C. 12
D. 11/5

10 / Constructed response

A container has 3.16 kg of seed. Each bag holds 0.4 kg. How many bags can be filled?

Day 009 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Apply Decimal Operations

Standards: 6.NS.B.3

Learning target: I can apply Decimal Operations, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Align decimal points and add placeholder zeros. Regroup one whole as ten tenths, or one tenth as ten hundredths. Subtract by place and check by adding the difference to the subtrahend. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 92.9 + 4.2. Show your regrouping or division steps.

1. Line up the decimal points. Add placeholder zeros so the numbers are 92.9 and 4.2.

2. Column 1 from the right: 9 + 2 = 11. Write 1 and carry 1 to the next column.

3. Column 2 from the right: 2 + 4 + 1 carried = 7. Write 7.

4. Column 3 from the right: 9 + 0 = 9. Write 9.

5. Bring the decimal point straight down, keeping 1 decimal place: 97.1. The simplified decimal answer is 97.1.

Answer: 97.1

Worked example 2

Use the standard algorithm: 2.112 ÷ 0.64. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 33/10
B. 43/10
C. 23/10
D. 33

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 100: 211.2 ÷ 64 = 3.3.

3. Check: 0.64 × 3.3 = 2.112.

Answer: A - 33/10

Day 009 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 47.3 + 6.8. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 36.3 - 2.4. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 90.3 × 6.5. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 12 ÷ 2.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 5.05 + 6.7. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 5.77 - 4.1. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 1.58 × 0.35. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 0.3 ÷ 0.25. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 11/5
B. 1/5
C. 12
D. 6/5

09 / Multiple choice

Use the standard algorithm: 5.31 + 0.24. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 111/2
B. 131/20
C. 111/20
D. 91/20

10 / Constructed response

A tank holds 1.26 L. A class uses 0.67 L. How much remains?

Day 010 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Model Multi-Step Decimal Decisions

Standards: 6.NS.B.3

Learning target: I can model Multi-Step Decimal Decisions, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Align decimal points and add placeholder zeros. Regroup one whole as ten tenths, or one tenth as ten hundredths. Subtract by place and check by adding the difference to the subtrahend. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 18.3 + 6.8. Show your regrouping or division steps.

1. Line up the decimal points. Add placeholder zeros so the numbers are 18.3 and 6.8.

2. Column 1 from the right: 3 + 8 = 11. Write 1 and carry 1 to the next column.

3. Column 2 from the right: 8 + 6 + 1 carried = 15. Write 5 and carry 1 to the next column.

4. Column 3 from the right: 1 + 0 + 1 carried = 2. Write 2.

5. Bring the decimal point straight down, keeping 1 decimal place: 25.1. The simplified decimal answer is 25.1.

Answer: 25.1

Worked example 2

Use the standard algorithm: 3.32 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 83
B. 93/10
C. 83/10
D. 73/10

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 10: 33.2 ÷ 4 = 8.3.

3. Check: 0.4 × 8.3 = 3.32.

Answer: C - 83/10

Day 010 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 36.5 + 3.2. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 53.1 - 8.5. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 70.1 × 4.3. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 11.85 ÷ 1.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 6.99 + 4.9. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 3.65 - 3.3. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 2.55 × 0.51. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 1.68 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 26/5
B. 21/5
C. 42
D. 16/5

09 / Multiple choice

Use the standard algorithm: 7.27 + 0.85. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 178/25
B. 228/25
C. 203/25
D. 406/5

10 / Constructed response

A tank holds 6.96 L. A class uses 0.26 L. How much remains?

Day 011 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Solve Decimal Measurement Problems

Standards: 6.NS.B.3

Learning target: I can solve Decimal Measurement Problems, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 43 × 8.8. Show your regrouping or division steps.

1. Estimate the size first. Treat the factors as whole numbers: 43 becomes 43, and 8.8 becomes 88.

2. Multiply the whole numbers: 43 × 88 = 3784.

3. Count decimal places: 0 in 43 plus 1 in 8.8 gives 1.

4. Place the decimal point 1 places from the right in 3784, adding leading zeros if necessary: 378.4. Remove only unnecessary trailing zeros.

Answer: 378.4

Worked example 2

Use the standard algorithm: 1.984 ÷ 0.64. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 21/10
B. 31
C. 41/10
D. 31/10

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 100: 198.4 ÷ 64 = 3.1.

3. Check: 0.64 × 3.1 = 1.984.

Answer: D - 31/10

Day 011 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 77.7 × 1.9. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 9.8 ÷ 0.2. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 32.2 × 6.8. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 5 ÷ 2.5. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 2.37 × 7.1. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 21 ÷ 2.5. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 4.3 × 0.4. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 3.136 ÷ 0.64. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 49
B. 39/10
C. 49/10
D. 59/10

09 / Multiple choice

Use the standard algorithm: 1.51 × 0.31. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 4681/1000
B. -5319/10000
C. 4681/10000
D. 14681/10000

10 / Constructed response

A container has 0.7 kg of seed. Each bag holds 0.25 kg. How many bags can be filled?

Day 012 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Interpret Whole-Number Remainders

Standards: 6.NS.B.2

Learning target: I can divide multi-digit whole numbers and check the quotient.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder.

Worked example 1

Use long division: 546 ÷ 7. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 54 ÷ 7: write 7, subtract 49, remainder 5

3. bring down the next digit: 56 ÷ 7: write 8, subtract 56, remainder 0.

4. Check: 7 × 78 = 546.

Answer: 78

Worked example 2

Use long division: 11,312 ÷ 33. Show the algorithm and check.

1. Estimate the quotient first.

2. 113 ÷ 33: write 3, subtract 99, remainder 14

3. bring down the next digit: 141 ÷ 33: write 4, subtract 132, remainder 9

4. bring down the next digit: 92 ÷ 33: write 2, subtract 66, remainder 26.

5. Check: 33 × 342 + 26 = 11312.

Answer: 342 R 26

Day 012 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 450 ÷ 6. Show the algorithm and check.

02 / Constructed response

Use long division: 448 ÷ 8. Show the algorithm and check.

03 / Constructed response

Use long division: 440 ÷ 5. Show the algorithm and check.

04 / Constructed response

Use long division: 6,647 ÷ 23. Show the algorithm and check.

05 / Constructed response

Use long division: 6,325 ÷ 23. Show the algorithm and check.

06 / Constructed response

Use long division: 2,327 ÷ 13. Show the algorithm and check.

07 / Constructed response

Use long division: 11,533 ÷ 30. Show the algorithm and check.

08 / Model and constructed response

Use long division: 13,555 ÷ 36. Show the algorithm and check.

09 / Model and constructed response

Use long division: 4,262 ÷ 34. Show the algorithm and check.

10 / Constructed response

A club has 6605 cards. Each full packet holds 25. How many full packets can it make, and how many cards remain?

Day 013 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Critique Decimal Calculations

Standards: 6.NS.B.3

Learning target: I can critique Decimal Calculations, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Align decimal points and add placeholder zeros. Regroup one whole as ten tenths, or one tenth as ten hundredths. Subtract by place and check by adding the difference to the subtrahend. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use the standard algorithm: 65.9 - 5. Show your regrouping or division steps.

1. Line up the decimal points. Add placeholder zeros so the numbers are 65.9 and 5.0.

2. Column 1 from the right: 9 − 0 = 9. Write 9.

3. Column 2 from the right: 5 − 5 = 0. Write 0.

4. Column 3 from the right: 6 − 0 = 6. Write 6.

5. Bring the decimal point straight down, keeping 1 decimal place: 60.9. The simplified decimal answer is 60.9.

6. Check: 60.9 + 5 = 65.9.

Answer: 60.9

Worked example 2

Use the standard algorithm: 1.4 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 7/2
B. 5/2
C. 9/2
D. 35

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 10: 14 ÷ 4 = 3.5.

3. Check: 0.4 × 3.5 = 1.4.

Answer: A - 7/2

Day 013 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use the standard algorithm: 12.2 - 3.6. Show your regrouping or division steps.

02 / Constructed response

Use the standard algorithm: 31 × 8.6. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 52.2 ÷ 0.9. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 3.94 - 2. Show your regrouping or division steps.

05 / Constructed response

Use the standard algorithm: 3.9 × 2.5. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 5.4 ÷ 1.2. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 1.69 - 0.2. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 1.33 × 0.51. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 6783/1000
B. 16783/10000
C. 6783/10000
D. -3217/10000

09 / Multiple choice

Use the standard algorithm: 0.396 ÷ 0.12. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 23/10
B. 33
C. 43/10
D. 33/10

10 / Constructed response

A tank holds 4.46 L. A class uses 0.83 L. How much remains?

Day 014 | Unit 01 | Lesson

Positive Rational Numbers: Decimals and Division

Model a Rational-Number Situation

Standards: 6.NS.B.2, 6.NS.B.3

Learning target: I can model a Rational-Number Situation, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

01 / Learn

Estimate the quotient first. Divide, multiply, subtract, and bring down one digit at a time. Keep each quotient digit in its place; write zero when needed. Check dividend = divisor × quotient + remainder. Line up decimal points so tenths are under tenths and hundredths under hundredths. Add zeros if helpful. Add from right to left, regroup, and bring the decimal point straight down. Multiply as whole numbers using the standard algorithm. The product has the total number of decimal places in the two factors before trailing zeros are removed. Check its size against an estimate. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number. Then use long division, adding zeros to the dividend as needed. Check by multiplication.

Worked example 1

Use long division: 648 ÷ 9. Show the algorithm and check.

1. Estimate how many digits the quotient should have. Work from the leftmost digits of the dividend.

2. 64 ÷ 9: write 7, subtract 63, remainder 1

3. bring down the next digit: 18 ÷ 9: write 2, subtract 18, remainder 0.

4. Check: 9 × 72 = 648.

Answer: 72

Worked example 2

Use the standard algorithm: 2.56 ÷ 0.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 32/5
B. 64
C. 27/5
D. 37/5

1. Multiply both dividend and divisor by the same power of 10 so the divisor is a whole number.

2. Multiply both numbers by 10: 25.6 ÷ 4 = 6.4.

3. Check: 0.4 × 6.4 = 2.56.

Answer: A - 32/5

Day 014 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Use long division: 129 ÷ 3. Show the algorithm and check.

02 / Constructed response

Use the standard algorithm: 47.4 + 6.7. Show your regrouping or division steps.

03 / Constructed response

Use the standard algorithm: 65.4 × 3.5. Show your regrouping or division steps.

04 / Constructed response

Use the standard algorithm: 9.36 ÷ 1.2. Show your regrouping or division steps.

05 / Constructed response

Use long division: 8,178 ÷ 29. Show the algorithm and check.

06 / Constructed response

Use the standard algorithm: 4.02 + 4. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 5.57 × 0.15. Show your regrouping or division steps.

08 / Multiple choice

Use the standard algorithm: 0.825 ÷ 0.15. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 13/2
B. 11/2
C. 9/2
D. 55

09 / Model and constructed response

Use long division: 8,443 ÷ 23. Show the algorithm and check.

10 / Constructed response

A runner walks 1.31 km in the morning and 0.27 km later. What is the total distance?

Day 015 | Unit 01 | Unit Test

Positive Rational Numbers: Decimals and Division

Unit 1 Test

Standards: Grade 5 prerequisite, 6.NS.B.3, 6.NS.B.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 1

Directions

The first 4 questions (20%) review Grade 5 readiness skills. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Use long division: 522 ÷ 6. Show the algorithm and check.

02 / Previous unit review | Constructed response

Use the standard algorithm: 66.2 + 5.7. Show your regrouping or division steps.

03 / Previous unit review | Constructed response

Use the standard algorithm: 68.2 - 7.4. Show your regrouping or division steps.

04 / Previous unit review | Constructed response

Calculate 1 × 1/2. Simplify.

05 / Constructed response

Use the standard algorithm: 18.7 + 1.5. Show your regrouping or division steps.

06 / Constructed response

Use the standard algorithm: 31.6 - 3.4. Show your regrouping or division steps.

07 / Constructed response

Use the standard algorithm: 26.6 × 1.5. Show your regrouping or division steps.

08 / Constructed response

Use long division: 7,025 ÷ 25. Show the algorithm and check.

09 / Constructed response

Use long division: 6,048 ÷ 36. Show the algorithm and check.

10 / Constructed response

Use long division: 3,815 ÷ 35. Show the algorithm and check.

11 / Constructed response

Use the standard algorithm: 17.92 ÷ 3.2. Show your regrouping or division steps.

12 / Multiple choice

Use the standard algorithm: 2.55 ÷ 1.5. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 27/10
B. 17/10
C. 17
D. 7/10

13 / Multiple choice

Use the standard algorithm: 7.3 + 2.1. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 94
B. 42/5
C. 47/5
D. 52/5

14 / Multiple choice

Use the standard algorithm: 5.58 + 0.68. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 313/5
B. 363/50
C. 313/50
D. 263/50

15 / Multiple choice

Use the standard algorithm: 1.968 ÷ 0.24. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 41/5
B. 82
C. 46/5
D. 36/5

16 / Model and constructed response

Use long division: 7,129 ÷ 27. Show the algorithm and check.

17 / Multiple choice

Use the standard algorithm: 4.3 × 0.73. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 3139/100
B. 2139/1000
C. 3139/1000
D. 4139/1000

18 / Multiple choice

Use the standard algorithm: 7.79 + 0.25. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 201/25
B. 226/25
C. 176/25
D. 402/5

19 / Multiple choice

Use the standard algorithm: 6.12 + 0.59. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 771/100
B. 571/100
C. 671/100
D. 671/10

20 / Multiple choice

Use the standard algorithm: 3.69 - 0.82. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.

A. 287/10
B. 287/100
C. 387/100
D. 187/100

Day 016 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Multiply Fractions: Readiness

Standards: 6.NS.A.1

Learning target: I can use fraction multiplication to prepare for fraction division.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

For a product of fractions, multiply numerators and multiply denominators, then simplify. Convert mixed numbers to improper fractions first. Multiplying by a fraction less than 1 makes a positive quantity smaller.

Worked example 1

Calculate 1/4 × 1/2. Simplify.

1. Rename the numbers as fractions: 1/4 = 1/4 and 1/2 = 1/2.

2. Keep multiplication and multiply the fractions: 1/4 × 1/2.

3. Multiply numerators and denominators: (1 × 1) / (4 × 2) = 1/8.

4. Simplify by dividing top and bottom by 1: 1/8.

5. Check the result: The product of 1/4 and 1/2 is 1/8. A positive factor below 1 makes the other positive factor smaller.

Answer: 1/8

Worked example 2

Calculate 3 × 5/8. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 23/8
B. 8/15
C. 7/8
D. 15/8

1. For a product of fractions, multiply numerators and multiply denominators, then simplify.

2. 3 × 5/8 = 15/8.

Answer: D - 15/8

Day 016 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1 × 1/2. Simplify.

02 / Constructed response

Calculate 1/4 × 1/4. Simplify.

03 / Constructed response

Calculate 1/2 × 1/2. Simplify.

04 / Constructed response

Calculate 2 × 1/4. Simplify.

05 / Constructed response

Calculate 8/5 × 5/3. Simplify.

06 / Constructed response

Calculate 1/2 × 1/5. Simplify.

07 / Constructed response

Calculate 1/2 × 1/4. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 1/2 × 2/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 6/5
B. -4/5
C. 5
D. 1/5

09 / Multiple choice

Calculate 7/2 × 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 3/4
B. 11/4
C. 4/7
D. 7/4

10 / Constructed response

Calculate 3/4 × 3/5. Simplify and explain why your answer is reasonable.

Day 017 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Multiply Mixed Numbers: Readiness

Standards: 6.NS.A.1

Learning target: I can use fraction multiplication to prepare for fraction division.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

For a product of fractions, multiply numerators and multiply denominators, then simplify. Convert mixed numbers to improper fractions first. Multiplying by a fraction less than 1 makes a positive quantity smaller.

Worked example 1

Calculate 2 1/4 × 2 1/4. Simplify.

1. Rename the numbers as fractions: 2 1/4 = 9/4 and 2 1/4 = 9/4.

2. Keep multiplication and multiply the fractions: 9/4 × 9/4.

3. Multiply numerators and denominators: (9 × 9) / (4 × 4) = 81/16.

4. Simplify by dividing top and bottom by 1: 81/16.

5. Check the result: The product of 9/4 and 9/4 is 81/16. A positive factor below 1 makes the other positive factor smaller.

Answer: 81/16

Worked example 2

Calculate 3 1/4 × 1 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 31/8
B. 39/8
C. 8/39
D. 47/8

1. For a product of fractions, multiply numerators and multiply denominators, then simplify.

2. 13/4 × 3/2 = 39/8.

Answer: B - 39/8

Day 017 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1 1/4 × 2 1/2. Simplify.

02 / Constructed response

Calculate 2 1/2 × 1 1/3. Simplify.

03 / Constructed response

Calculate 3 1/2 × 1 1/2. Simplify.

04 / Constructed response

Calculate 3 3/4 × 1 1/3. Simplify.

05 / Constructed response

Calculate 3 1/2 × 1 1/3. Simplify.

06 / Constructed response

Calculate 1 1/2 × 1 1/2. Simplify.

07 / Constructed response

Calculate 2 1/2 × 1 1/3. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 3 3/4 × 1 1/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 6
B. 4
C. 1/5
D. 5

09 / Multiple choice

Calculate 1 1/2 × 1 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 4/9
B. 9/4
C. 5/4
D. 13/4

10 / Constructed response

Calculate 3 1/4 × 1 1/3. Simplify and explain why your answer is reasonable.

Day 018 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Model Fraction Division with Bars

Standards: 6.NS.A.1

Learning target: I can model fraction division as equal-sized groups.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole.

Worked example 1

Draw a bar partitioned into 6 equal parts. Shade 3 parts. Circle groups of 3 parts. Use the model to find 1/2 ÷ 1/2.

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Rename both fractions with a common denominator, draw equal parts, and count groups.

3. Each small part is 1/6

4. 3 shaded parts divided into groups of 3 gives 3/3 = 1 groups.

5. State the result in the form requested: 1.

Answer: 1

Worked example 2

Draw a bar partitioned into 10 equal parts. Shade 6 parts. Circle groups of 2 parts. Use the model to find 3/5 ÷ 1/5. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 3
B. 30
C. 4
D. 2

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Each small part is 1/10

3. 6 shaded parts divided into groups of 2 gives 6/2 = 3 groups.

Answer: A - 3

Day 018 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 9 parts. Circle groups of 6 parts. Use the model to find 3/4 ÷ 1/2.

02 / Constructed response

Draw a bar partitioned into 4 equal parts. Shade 4 parts. Circle groups of 4 parts. Use the model to find 1 ÷ 1.

03 / Constructed response

Draw a bar partitioned into 6 equal parts. Shade 3 parts. Circle groups of 2 parts. Use the model to find 1/2 ÷ 1/3.

04 / Constructed response

Draw a bar partitioned into 6 equal parts. Shade 4 parts. Circle groups of 4 parts. Use the model to find 2/3 ÷ 2/3.

05 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 11 parts. Circle groups of 2 parts. Use the model to find 11/12 ÷ 1/6.

06 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 9 parts. Circle groups of 5 parts. Use the model to find 3/4 ÷ 5/12.

07 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 4 parts. Circle groups of 4 parts. Use the model to find 2/5 ÷ 2/5. Explain the partial group.

08 / Multiple choice

Draw a bar partitioned into 6 equal parts. Shade 4 parts. Circle groups of 1 parts. Use the model to find 2/3 ÷ 1/6. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 4
B. 40
C. 5
D. 3

09 / Multiple choice

Draw a bar partitioned into 8 equal parts. Shade 8 parts. Circle groups of 1 parts. Use the model to find 1 ÷ 1/8. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 80
B. 7
C. 9
D. 8

10 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 6 parts. Circle groups of 1 parts. Use the model to find 3/4 ÷ 1/8. Explain the partial group.

Day 019 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Interpret a Partial Fraction Group

Standards: 6.NS.A.1

Learning target: I can model fraction division as equal-sized groups.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole.

Worked example 1

Draw a bar partitioned into 10 equal parts. Shade 4 parts. Circle groups of 3 parts. Use the model to find 2/5 ÷ 3/10.

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Rename both fractions with a common denominator, draw equal parts, and count groups.

3. Each small part is 1/10

4. 4 shaded parts divided into groups of 3 gives 4/3 = 4/3 groups.

5. State the result in the form requested: 4/3.

Answer: 4/3

Worked example 2

Draw a bar partitioned into 6 equal parts. Shade 4 parts. Circle groups of 2 parts. Use the model to find 2/3 ÷ 1/3. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 2
B. 20
C. 3
D. 1

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Each small part is 1/6

3. 4 shaded parts divided into groups of 2 gives 4/2 = 2 groups.

Answer: A - 2

Day 019 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 10 parts. Circle groups of 5 parts. Use the model to find 5/6 ÷ 5/12.

02 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 9 parts. Circle groups of 2 parts. Use the model to find 9/10 ÷ 1/5.

03 / Constructed response

Draw a bar partitioned into 4 equal parts. Shade 2 parts. Circle groups of 1 parts. Use the model to find 1/2 ÷ 1/4.

04 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 7 parts. Circle groups of 7 parts. Use the model to find 7/10 ÷ 7/10.

05 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 2 parts. Circle groups of 1 parts. Use the model to find 1/6 ÷ 1/12.

06 / Constructed response

Draw a bar partitioned into 4 equal parts. Shade 4 parts. Circle groups of 1 parts. Use the model to find 1 ÷ 1/4.

07 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 8 parts. Circle groups of 3 parts. Use the model to find 4/5 ÷ 3/10. Explain the partial group.

08 / Multiple choice

Draw a bar partitioned into 4 equal parts. Shade 4 parts. Circle groups of 4 parts. Use the model to find 1 ÷ 1. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 0
B. 10
C. 2
D. 1

09 / Multiple choice

Draw a bar partitioned into 10 equal parts. Shade 9 parts. Circle groups of 4 parts. Use the model to find 9/10 ÷ 2/5. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 13/4
B. 9/4
C. 45/2
D. 5/4

10 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 6 parts. Circle groups of 4 parts. Use the model to find 3/5 ÷ 2/5. Explain the partial group.

Day 020 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Connect Fraction Bars and Related Equations

Standards: 6.NS.A.1

Learning target: I can connect Fraction Bars and Related Equations, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole. Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

Worked example 1

Draw a bar partitioned into 6 equal parts. Shade 5 parts. Circle groups of 4 parts. Use the model to find 5/6 ÷ 2/3.

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Rename both fractions with a common denominator, draw equal parts, and count groups.

3. Each small part is 1/6

4. 5 shaded parts divided into groups of 4 gives 5/4 = 5/4 groups.

5. State the result in the form requested: 5/4.

Answer: 5/4

Worked example 2

The equation 1/5 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

1. Division and multiplication are inverse operations.

2. 1 ÷ 1/5 = 1 × 5 = 5.

3. Check: 1/5 × 5 = 1.

Answer: n = 5; 1 ÷ 1/5 = 5

Day 020 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a bar partitioned into 12 equal parts. Shade 11 parts. Circle groups of 8 parts. Use the model to find 11/12 ÷ 2/3.

02 / Constructed response

The equation 1/4 × n = 1/2 represents equally sized portions. Find n, write the related division equation, and check.

03 / Constructed response

Draw a bar partitioned into 10 equal parts. Shade 10 parts. Circle groups of 6 parts. Use the model to find 1 ÷ 3/5.

04 / Constructed response

The equation 1/8 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

05 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 3 parts. Circle groups of 3 parts. Use the model to find 3/8 ÷ 3/8.

06 / Constructed response

The equation 1/2 × n = 1/3 represents equally sized portions. Find n, write the related division equation, and check.

07 / Constructed response

Draw a bar partitioned into 6 equal parts. Shade 5 parts. Circle groups of 4 parts. Use the model to find 5/6 ÷ 2/3. Explain the partial group.

08 / Model and constructed response

The equation 2/5 × n = 4/5 represents equally sized portions. Find n, write the related division equation, and check.

09 / Multiple choice

Draw a bar partitioned into 8 equal parts. Shade 7 parts. Circle groups of 4 parts. Use the model to find 7/8 ÷ 1/2. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 7/4
B. 11/4
C. 3/4
D. 35/2

10 / Constructed response

The equation 5/3 × n = 2 represents equally sized portions. Find n, write the related division equation, and check.

Day 021 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Divide Fractions Using Reciprocals

Standards: 6.NS.A.1

Learning target: I can divide fractions and check using multiplication.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product.

Worked example 1

Calculate 1 ÷ 1/4. Simplify.

1. Rename the numbers as fractions: 1 = 1 and 1/4 = 1/4.

2. Keep the first fraction and multiply by the reciprocal of the second: 1 × 4.

3. Multiply numerators and denominators: (1 × 4) / (1 × 1) = 4/1.

4. Simplify by dividing top and bottom by 1: 4.

5. Check the result: 1/4 × 4 = 1, so the quotient is correct.

Answer: 4

Worked example 2

Calculate 2 ÷ 4/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 1/2
B. 5/2
C. 2/3
D. 3/2

1. If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b.

2. 2 ÷ 4/3 = 2 × 3/4 = 3/2.

Answer: D - 3/2

Day 021 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 3/4 ÷ 1/4. Simplify.

02 / Constructed response

Calculate 1/2 ÷ 1/4. Simplify.

03 / Constructed response

Calculate 1 ÷ 1/2. Simplify.

04 / Constructed response

Calculate 5/2 ÷ 1/4. Simplify.

05 / Constructed response

Calculate 4/3 ÷ 1/4. Simplify.

06 / Constructed response

Calculate 2 ÷ 4/3. Simplify.

07 / Constructed response

Calculate 1/2 ÷ 1/2. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 1 ÷ 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 3
B. 1/2
C. 2
D. 1

09 / Multiple choice

Calculate 3/4 ÷ 1/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 15/4
B. 4/15
C. 11/4
D. 19/4

10 / Constructed response

Calculate 2/3 ÷ 1/2. Simplify and explain why your answer is reasonable.

Day 022 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Check a Fraction Quotient

Standards: 6.NS.A.1

Learning target: I can check a Fraction Quotient, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

Worked example 1

Calculate 1/4 ÷ 1/4. Simplify.

1. Rename the numbers as fractions: 1/4 = 1/4 and 1/4 = 1/4.

2. Keep the first fraction and multiply by the reciprocal of the second: 1/4 × 4.

3. Multiply numerators and denominators: (1 × 4) / (4 × 1) = 4/4.

4. Simplify by dividing top and bottom by 4: 1.

5. Check the result: 1/4 × 1 = 1/4, so the quotient is correct.

Answer: 1

Worked example 2

The equation 1 × n = 3/2 represents equally sized portions. Find n, write the related division equation, and check.

1. Division and multiplication are inverse operations.

2. 3/2 ÷ 1 = 3/2 × 1 = 3/2.

3. Check: 1 × 3/2 = 3/2.

Answer: n = 3/2; 3/2 ÷ 1 = 3/2

Day 022 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1 ÷ 1/2. Simplify.

02 / Constructed response

The equation 1/2 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

03 / Constructed response

Calculate 1/2 ÷ 1/4. Simplify.

04 / Constructed response

The equation 1/4 × n = 1/2 represents equally sized portions. Find n, write the related division equation, and check.

05 / Constructed response

Calculate 1/6 ÷ 5/4. Simplify.

06 / Constructed response

The equation 2/3 × n = 3/2 represents equally sized portions. Find n, write the related division equation, and check.

07 / Constructed response

Calculate 4/3 ÷ 1. Simplify and explain why your answer is reasonable.

08 / Model and constructed response

The equation 5/4 × n = 2 represents equally sized portions. Find n, write the related division equation, and check.

09 / Multiple choice

Calculate 3/2 ÷ 3/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 7/2
B. 5/2
C. 3/2
D. 2/5

10 / Constructed response

The equation 4/3 × n = 1/3 represents equally sized portions. Find n, write the related division equation, and check.

Day 023 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Divide Mixed Numbers

Standards: 6.NS.A.1

Learning target: I can divide mixed numbers after renaming them as fractions.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

Rename a mixed number as (whole × denominator + numerator)/denominator. Keep the dividend, change division to multiplication, and use the reciprocal of the divisor. Simplify and interpret the result.

Worked example 1

Calculate 2 ÷ 1/2. Simplify.

1. Rename the numbers as fractions: 2 = 2 and 1/2 = 1/2.

2. Keep the first fraction and multiply by the reciprocal of the second: 2 × 2.

3. Multiply numerators and denominators: (2 × 2) / (1 × 1) = 4/1.

4. Simplify by dividing top and bottom by 1: 4.

5. Check the result: 1/2 × 4 = 2, so the quotient is correct.

Answer: 4

Worked example 2

Calculate 3 1/3 ÷ 4 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. -7/27
B. 20/27
C. 47/27
D. 27/20

1. Rename a mixed number as (whole × denominator + numerator)/denominator.

2. 10/3 ÷ 9/2 = 10/3 × 2/9 = 20/27.

Answer: B - 20/27

Day 023 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1 1/2 ÷ 1/2. Simplify.

02 / Constructed response

Calculate 1 1/4 ÷ 1/2. Simplify.

03 / Constructed response

Calculate 2 1/2 ÷ 1/4. Simplify.

04 / Constructed response

Calculate 2 1/2 ÷ 2 3/8. Simplify.

05 / Constructed response

Calculate 1 5/6 ÷ 3. Simplify.

06 / Constructed response

Calculate 3 1/4 ÷ 1 3/8. Simplify.

07 / Constructed response

Calculate 1 1/2 ÷ 3 1/3. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 3 3/5 ÷ 2 3/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 31/13
B. 18/13
C. 13/18
D. 5/13

09 / Multiple choice

Calculate 1 3/4 ÷ 3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 7/12
B. -5/12
C. 12/7
D. 19/12

10 / Constructed response

Calculate 3 7/8 ÷ 3. Simplify and explain why your answer is reasonable.

Day 024 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Interpret Mixed-Number Quotients

Standards: 6.NS.A.1

Learning target: I can divide mixed numbers after renaming them as fractions.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

Rename a mixed number as (whole × denominator + numerator)/denominator. Keep the dividend, change division to multiplication, and use the reciprocal of the divisor. Simplify and interpret the result.

Worked example 1

Calculate 3 ÷ 1/2. Simplify.

1. Rename the numbers as fractions: 3 = 3 and 1/2 = 1/2.

2. Keep the first fraction and multiply by the reciprocal of the second: 3 × 2.

3. Multiply numerators and denominators: (3 × 2) / (1 × 1) = 6/1.

4. Simplify by dividing top and bottom by 1: 6.

5. Check the result: 1/2 × 6 = 3, so the quotient is correct.

Answer: 6

Worked example 2

Calculate 5 1/2 ÷ 2 1/4. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 31/9
B. 13/9
C. 22/9
D. 9/22

1. Rename a mixed number as (whole × denominator + numerator)/denominator.

2. 11/2 ÷ 9/4 = 11/2 × 4/9 = 22/9.

Answer: C - 22/9

Day 024 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 2 1/4 ÷ 1/2. Simplify.

02 / Constructed response

Calculate 2 ÷ 1/2. Simplify.

03 / Constructed response

Calculate 2 1/2 ÷ 1/2. Simplify.

04 / Constructed response

Calculate 2 1/8 ÷ 1 4/5. Simplify.

05 / Constructed response

Calculate 3 ÷ 2. Simplify.

06 / Constructed response

Calculate 2 ÷ 1 1/4. Simplify.

07 / Constructed response

Calculate 1 2/3 ÷ 4 1/2. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 6 1/2 ÷ 1 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 3/13
B. 10/3
C. 16/3
D. 13/3

09 / Multiple choice

Calculate 4 ÷ 2 1/4. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 16/9
B. 7/9
C. 25/9
D. 9/16

10 / Constructed response

Calculate 4 ÷ 3. Simplify and explain why your answer is reasonable.

Day 025 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Explain Fraction Quotients in Context

Standards: 6.NS.A.1

Learning target: I can explain Fraction Quotients in Context, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Rename a mixed number as (whole × denominator + numerator)/denominator. Keep the dividend, change division to multiplication, and use the reciprocal of the divisor. Simplify and interpret the result.

Worked example 1

Calculate 3/4 ÷ 1/2. Simplify.

1. Rename the numbers as fractions: 3/4 = 3/4 and 1/2 = 1/2.

2. Keep the first fraction and multiply by the reciprocal of the second: 3/4 × 2.

3. Multiply numerators and denominators: (3 × 2) / (4 × 1) = 6/4.

4. Simplify by dividing top and bottom by 2: 3/2.

5. Check the result: 1/2 × 3/2 = 3/4, so the quotient is correct.

Answer: 3/2

Worked example 2

Calculate 6 1/2 ÷ 3 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 7/13
B. 13/7
C. 6/7
D. 20/7

1. Rename a mixed number as (whole × denominator + numerator)/denominator.

2. 13/2 ÷ 7/2 = 13/2 × 2/7 = 13/7.

Answer: B - 13/7

Day 025 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1/4 ÷ 1/2. Simplify.

02 / Constructed response

Calculate 1 1/4 ÷ 1/2. Simplify.

03 / Constructed response

Calculate 1/2 ÷ 1/4. Simplify.

04 / Constructed response

Calculate 3 ÷ 1 2/3. Simplify.

05 / Constructed response

Calculate 1/5 ÷ 2/3. Simplify.

06 / Constructed response

Calculate 2 1/6 ÷ 2. Simplify.

07 / Constructed response

Calculate 1/2 ÷ 1/8. Simplify and explain why your answer is reasonable.

08 / Multiple choice

Calculate 3 3/8 ÷ 2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 27/16
B. 11/16
C. 43/16
D. 16/27

09 / Multiple choice

Calculate 1 ÷ 1/8. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 7
B. 9
C. 1/8
D. 8

10 / Constructed response

Calculate 4 ÷ 1 2/3. Simplify and explain why your answer is reasonable.

Day 026 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Write Fraction-Division Equations

Standards: 6.NS.A.1

Learning target: I can connect a fraction quotient to a missing factor.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

Division and multiplication are inverse operations. For T ÷ s = n, the related equation is s × n = T. Label whether the unknown is the total, the group size, or the number of groups.

Worked example 1

The equation 1/4 × n = 3/4 represents equally sized portions. Find n, write the related division equation, and check.

1. Division and multiplication are inverse operations.

2. For T ÷ s = n, the related equation is s × n = T.

3. 3/4 ÷ 1/4 = 3/4 × 4 = 3.

4. Check: 1/4 × 3 = 3/4.

5. State the result in the form requested: n = 3; 3/4 ÷ 1/4 = 3.

Answer: n = 3; 3/4 ÷ 1/4 = 3

Worked example 2

The equation 1/5 × n = 3/8 represents equally sized portions. Find n, write the related division equation, and check.

1. Division and multiplication are inverse operations.

2. 3/8 ÷ 1/5 = 3/8 × 5 = 15/8.

3. Check: 1/5 × 15/8 = 3/8.

Answer: n = 15/8; 3/8 ÷ 1/5 = 15/8

Day 026 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

The equation 1/2 × n = 3/4 represents equally sized portions. Find n, write the related division equation, and check.

02 / Constructed response

The equation 1/4 × n = 1/4 represents equally sized portions. Find n, write the related division equation, and check.

03 / Constructed response

The equation 1/4 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

04 / Constructed response

The equation 3/5 × n = 1/2 represents equally sized portions. Find n, write the related division equation, and check.

05 / Constructed response

The equation 1 × n = 7/8 represents equally sized portions. Find n, write the related division equation, and check.

06 / Constructed response

The equation 2 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

07 / Constructed response

The equation 1 × n = 1 represents equally sized portions. Find n, write the related division equation, and check.

08 / Model and constructed response

The equation 1/3 × n = 2/5 represents equally sized portions. Find n, write the related division equation, and check.

09 / Model and constructed response

The equation 5/4 × n = 2/3 represents equally sized portions. Find n, write the related division equation, and check.

10 / Constructed response

The equation 4/5 × n = 6/5 represents equally sized portions. Find n, write the related division equation, and check.

Day 027 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Solve Fraction-Division Stories

Standards: 6.NS.A.1

Learning target: I can write and solve a fraction-division equation for a situation.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

Worked example 1

A ribbon is 1/2 m long. Each piece is 1/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

1. Identify the total amount and the size of one group.

2. Use total ÷ group size for the number of groups.

3. 1/2 ÷ 1/4 = 1/2 × 4 = 2.

4. State the result in the form requested: 2 piece-lengths.

Answer: 2 piece-lengths

Worked example 2

A jug contains 2/3 L. Each full bottle needs 1/4 L. How many full bottles can be filled, and how much liquid remains?

1. Identify the total amount and the size of one group.

2. 2/3 ÷ 1/4 = 2/3 × 4 = 8/3.

3. Only whole bottles count

4. subtract the liquid used.

Answer: 2 bottles; 1/6 L left

Day 027 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A ribbon is 1/2 m long. Each piece is 1/2 m. How many piece-lengths are in the ribbon? Include any fractional piece.

02 / Constructed response

A ribbon is 1 m long. Each piece is 1/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

03 / Constructed response

A ribbon is 1/4 m long. Each piece is 1/2 m. How many piece-lengths are in the ribbon? Include any fractional piece.

04 / Constructed response

A ribbon is 1 1/3 m long. Each piece is 5/8 m. How many piece-lengths are in the ribbon? Include any fractional piece.

05 / Constructed response

A ribbon is 1 3/4 m long. Each piece is 2/5 m. How many piece-lengths are in the ribbon? Include any fractional piece.

06 / Constructed response

A ribbon is 1/4 m long. Each piece is 1/3 m. How many piece-lengths are in the ribbon? Include any fractional piece.

07 / Constructed response

A jug contains 2 2/3 L. Each full bottle needs 1/5 L. How many full bottles can be filled, and how much liquid remains?

08 / Model and constructed response

A jug contains 1/2 L. Each full bottle needs 4/3 L. How many full bottles can be filled, and how much liquid remains?

09 / Model and constructed response

A jug contains 2 1/2 L. Each full bottle needs 4/5 L. How many full bottles can be filled, and how much liquid remains?

10 / Constructed response

A jug contains 1 L. Each full bottle needs 2/5 L. How many full bottles can be filled, and how much liquid remains?

Day 028 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Find Full Containers and Leftovers

Standards: 6.NS.A.1

Learning target: I can write and solve a fraction-division equation for a situation.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

Worked example 1

A ribbon is 1/2 m long. Each piece is 1/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

1. Identify the total amount and the size of one group.

2. Use total ÷ group size for the number of groups.

3. 1/2 ÷ 1/4 = 1/2 × 4 = 2.

4. State the result in the form requested: 2 piece-lengths.

Answer: 2 piece-lengths

Worked example 2

A jug contains 1 L. Each full bottle needs 1 L. How many full bottles can be filled, and how much liquid remains?

1. Identify the total amount and the size of one group.

2. 1 ÷ 1 = 1 × 1 = 1.

3. Only whole bottles count

4. subtract the liquid used.

Answer: 1 bottles; 0 L left

Day 028 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A ribbon is 1/4 m long. Each piece is 1/2 m. How many piece-lengths are in the ribbon? Include any fractional piece.

02 / Constructed response

A ribbon is 3/4 m long. Each piece is 1/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

03 / Constructed response

A ribbon is 1 m long. Each piece is 1/2 m. How many piece-lengths are in the ribbon? Include any fractional piece.

04 / Constructed response

A ribbon is 2/3 m long. Each piece is 2/5 m. How many piece-lengths are in the ribbon? Include any fractional piece.

05 / Constructed response

A ribbon is 1/2 m long. Each piece is 5/3 m. How many piece-lengths are in the ribbon? Include any fractional piece.

06 / Constructed response

A ribbon is 3/4 m long. Each piece is 3/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

07 / Constructed response

A jug contains 1 1/2 L. Each full bottle needs 1 L. How many full bottles can be filled, and how much liquid remains?

08 / Model and constructed response

A jug contains 3 1/2 L. Each full bottle needs 3/2 L. How many full bottles can be filled, and how much liquid remains?

09 / Model and constructed response

A jug contains 4 L. Each full bottle needs 1/3 L. How many full bottles can be filled, and how much liquid remains?

10 / Constructed response

A jug contains 3/4 L. Each full bottle needs 1/3 L. How many full bottles can be filled, and how much liquid remains?

Day 029 | Unit 02 | Lesson

Positive Rational Numbers: Fractions and Mixed Numbers

Explain Fraction Models and Algorithms

Standards: 6.NS.A.1

Learning target: I can explain Fraction Models and Algorithms, and explain and check my reasoning.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

01 / Learn

A quotient asks how many groups of the divisor fit in the dividend. Rename both fractions with a common denominator, draw equal parts, and count groups. A partial group is a fraction of the divisor, not of the original whole. If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. Rename a mixed number as (whole × denominator + numerator)/denominator. Keep the dividend, change division to multiplication, and use the reciprocal of the divisor. Simplify and interpret the result. Identify the total amount and the size of one group. Use total ÷ group size for the number of groups. A quotient may include a partial group; if the question asks for full servings, use only the whole-number part.

Worked example 1

Draw a bar partitioned into 12 equal parts. Shade 6 parts. Circle groups of 3 parts. Use the model to find 1/2 ÷ 1/4.

1. A quotient asks how many groups of the divisor fit in the dividend.

2. Rename both fractions with a common denominator, draw equal parts, and count groups.

3. Each small part is 1/12

4. 6 shaded parts divided into groups of 3 gives 6/3 = 2 groups.

5. State the result in the form requested: 2.

Answer: 2

Worked example 2

A jug contains 1/3 L. Each full bottle needs 1 L. How many full bottles can be filled, and how much liquid remains?

1. Identify the total amount and the size of one group.

2. 1/3 ÷ 1 = 1/3 × 1 = 1/3.

3. Only whole bottles count

4. subtract the liquid used.

Answer: 0 bottles; 1/3 L left

Day 029 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a bar partitioned into 4 equal parts. Shade 3 parts. Circle groups of 2 parts. Use the model to find 3/4 ÷ 1/2.

02 / Constructed response

Calculate 3/4 ÷ 1/2. Simplify.

03 / Constructed response

Calculate 1 3/4 ÷ 1/2. Simplify.

04 / Constructed response

A ribbon is 3/4 m long. Each piece is 5/4 m. How many piece-lengths are in the ribbon? Include any fractional piece.

05 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 2 parts. Circle groups of 2 parts. Use the model to find 1/4 ÷ 1/4.

06 / Constructed response

Calculate 1/4 ÷ 5/4. Simplify.

07 / Constructed response

Calculate 1 3/8 ÷ 2 2/3. Simplify and explain why your answer is reasonable.

08 / Model and constructed response

A jug contains 3/8 L. Each full bottle needs 1 L. How many full bottles can be filled, and how much liquid remains?

09 / Multiple choice

Draw a bar partitioned into 8 equal parts. Shade 5 parts. Circle groups of 4 parts. Use the model to find 5/8 ÷ 1/2. Explain the partial group. Select one answer, then justify it with a model or calculation.
A. 1/4
B. 25/2
C. 9/4
D. 5/4

10 / Constructed response

Calculate 2 ÷ 5/8. Simplify and explain why your answer is reasonable.

Day 030 | Unit 02 | Unit Test

Positive Rational Numbers: Fractions and Mixed Numbers

Unit 2 Test

Standards: 6.NS.B.3, 6.NS.B.2, 6.NS.A.1

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

Directions

The first 4 questions (20%) review Unit 1. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Use the standard algorithm: 20.8 + 8.7. Show your regrouping or division steps.

02 / Previous unit review | Constructed response

Use the standard algorithm: 35.3 - 1.5. Show your regrouping or division steps.

03 / Previous unit review | Constructed response

Use the standard algorithm: 93.3 × 8.4. Show your regrouping or division steps.

04 / Previous unit review | Constructed response

Use long division: 350 ÷ 7. Show the algorithm and check.

05 / Constructed response

Calculate 1/4 × 1/4. Simplify.

06 / Constructed response

Calculate 2 1/4 × 2 1/2. Simplify.

07 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 8 parts. Circle groups of 1 parts. Use the model to find 1 ÷ 1/8.

08 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 3 parts. Circle groups of 2 parts. Use the model to find 3/8 ÷ 1/4.

09 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 4 parts. Circle groups of 3 parts. Use the model to find 1/2 ÷ 3/8.

10 / Constructed response

Calculate 8/5 ÷ 1/4. Simplify.

11 / Constructed response

Calculate 3/4 ÷ 1/5. Simplify.

12 / Multiple choice

Calculate 3 1/6 ÷ 1 1/4. Simplify. Select one answer, then justify it with a model or calculation.
A. 15/38
B. 23/15
C. 38/15
D. 53/15

13 / Multiple choice

Calculate 4 1/3 ÷ 2 3/4. Simplify. Select one answer, then justify it with a model or calculation.
A. 85/33
B. 52/33
C. 19/33
D. 33/52

14 / Multiple choice

Calculate 1/2 ÷ 2/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 4/3
B. 7/4
C. -1/4
D. 3/4

15 / Model and constructed response

The equation 3/5 × n = 7/3 represents equally sized portions. Find n, write the related division equation, and check.

16 / Model and constructed response

A jug contains 1/3 L. Each full bottle needs 2 L. How many full bottles can be filled, and how much liquid remains?

17 / Model and constructed response

A jug contains 1 L. Each full bottle needs 1/2 L. How many full bottles can be filled, and how much liquid remains?

18 / Multiple choice

Calculate 2/3 ÷ 2/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 2
B. 1
C. 0

19 / Multiple choice

Calculate 1/2 × 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 1/4
B. -3/4
C. 4
D. 5/4

20 / Multiple choice

Calculate 2 3/4 × 1 1/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 3/11
B. 11/3
C. 14/3
D. 8/3

Day 031 | Unit 03 | Lesson

Integers and Rational Numbers

Represent Signed Quantities and Zero

Standards: 6.NS.C.5

Learning target: I can use positive numbers, negative numbers, and zero in context.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Choose a reference point for zero. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative. Zero means at the chosen reference, not always nothing.

Worked example 1

A marker is 32 m below sea level. Write its signed elevation and explain what elevation 0 means.

1. Choose a reference point for zero.

2. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative.

3. Sea level is the reference.

4. Below is negative and above is positive.

5. State the result in the form requested: -32 m; 0 means at sea level.

Answer: -32 m; 0 means at sea level

Worked example 2

A sensor is 7 meters below a reference point. Select all correct interpretations.
A. Its signed position is -7 m.
B. Its signed position is +7 m.
C. The opposite position is 7 m below the reference.
D. A reading of 0 m means at the reference point.

1. Choose a reference point for zero.

2. Below the chosen zero is negative, so the position is -7.

3. Zero identifies the reference

4. the opposite is 7 m above it.

Answer: A, D - Its signed position is -7 m.; A reading of 0 m means at the reference point.

Day 031 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A marker is 14 m above sea level. Write its signed elevation and explain what elevation 0 means.

02 / Constructed response

A marker is 9 m below sea level. Write its signed elevation and explain what elevation 0 means.

03 / Constructed response

A marker is 10 m above sea level. Write its signed elevation and explain what elevation 0 means.

04 / Constructed response

A marker is 10 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

A marker is 35 m above sea level. Write its signed elevation and explain what elevation 0 means.

06 / Constructed response

A marker is 28 m below sea level. Write its signed elevation and explain what elevation 0 means.

07 / Constructed response

A marker is 15 m above sea level. Write its signed elevation and explain what elevation 0 means. Write the elevation of the point equally far on the opposite side of sea level.

08 / Multiple select

A sensor is 2 meters below a reference point. Select all correct interpretations.
A. Its signed position is -2 m.
B. The opposite position is 2 m below the reference.
C. A reading of 0 m means at the reference point.
D. Its signed position is +2 m.

09 / Multiple select

A sensor is 5 meters below a reference point. Select all correct interpretations.
A. Its signed position is +5 m.
B. A reading of 0 m means at the reference point.
C. The opposite position is 5 m below the reference.
D. Its signed position is -5 m.

10 / Constructed response

A marker is 15 m below sea level. Write its signed elevation and explain what elevation 0 means. Write the elevation of the point equally far on the opposite side of sea level.

Day 032 | Unit 03 | Lesson

Integers and Rational Numbers

Recognize Opposites

Standards: 6.NS.C.6a

Learning target: I can identify opposites and the opposite of an opposite.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Opposites are equally far from zero on opposite sides of a number line. The opposite of 0 is 0. Reversing direction twice returns to the original number.

Worked example 1

Find the opposite of -2. Show both values on a number line.

1. Opposites are equally far from zero on opposite sides of a number line.

2. The opposite of 0 is 0.

3. -2 and 2 are each 2 units from 0 on opposite sides.

4. State the result in the form requested: 2.

Answer: 2

Worked example 2

Select all true statements about -15/2 and its opposite. Explain the role of zero.
A. -15/2 and 15/2 are on the same side of zero.
B. The opposite of the opposite of -15/2 is -15/2.
C. The opposite of 0 is 1.
D. The opposite of -15/2 is 15/2.

1. Opposites are equally far from zero on opposite sides of a number line.

2. An opposite reverses direction without changing distance.

3. Reversing twice returns to -15/2.

4. Zero is its own opposite.

Answer: B, D - The opposite of the opposite of -15/2 is -15/2.; The opposite of -15/2 is 15/2.

Day 032 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the opposite of -3. Show both values on a number line.

02 / Constructed response

Find the opposite of -6. Show both values on a number line.

03 / Constructed response

Find the opposite of -5. Show both values on a number line.

04 / Constructed response

Find the opposite of -4/5 and the opposite of that result. Locate all values on a number line.

05 / Constructed response

Find the opposite of -3/4 and the opposite of that result. Locate all values on a number line.

06 / Constructed response

Find the opposite of -2 and the opposite of that result. Locate all values on a number line.

07 / Constructed response

Find the opposite of -7 and the opposite of that result. Locate all values on a number line.

08 / Multiple select

Select all true statements about -5/2 and its opposite. Explain the role of zero.
A. The opposite of the opposite of -5/2 is -5/2.
B. The opposite of 0 is 1.
C. -5/2 and 5/2 are on the same side of zero.
D. The opposite of -5/2 is 5/2.

09 / Multiple select

Select all true statements about -7/2 and its opposite. Explain the role of zero.
A. The opposite of the opposite of -7/2 is -7/2.
B. The opposite of 0 is 1.
C. -7/2 and 7/2 are on the same side of zero.
D. The opposite of -7/2 is 7/2.

10 / Constructed response

Find the opposite of -7/5 and the opposite of that result. Locate all values on a number line.

Day 033 | Unit 03 | Lesson

Integers and Rational Numbers

Locate Integers on Number Lines

Standards: 6.NS.C.6c

Learning target: I can locate integers and negative fractions on a number line.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Mark zero and equally spaced whole-number ticks. Partition each interval into equal parts for the denominator. Negative values are left of zero on a horizontal line and below zero on a vertical line.

Worked example 1

Draw a horizontal or vertical number line. Plot and label -4, 0, and 7.

1. Mark zero and equally spaced whole-number ticks.

2. Partition each interval into equal parts for the denominator.

3. Negative -4 is left of or below 0

4. 7 is right of or above 0.

5. Use equal intervals.

6. State the result in the form requested: -4 < 0 < 7; unit steps.

Answer: -4 < 0 < 7; unit steps

Worked example 2

On a number line, consecutive integer ticks are 1 unit apart. Plot -5/2, -2, and 0. Part A: name the leftmost point. Part B: give the midpoint between -3 and -2.

1. Mark zero and equally spaced whole-number ticks.

2. -5/2 = -2 - 1/2 lies halfway between -3 and -2, left of both -2 and 0.

Answer: Leftmost -5/2; midpoint -5/2

Day 033 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a horizontal or vertical number line. Plot and label -8, 0, and 7.

02 / Constructed response

Draw a horizontal or vertical number line. Plot and label -5, 0, and 5.

03 / Constructed response

Draw a horizontal or vertical number line. Plot and label -3, 0, and 2.

04 / Constructed response

Draw a horizontal or vertical number line. Plot and label -2, 0, and 3/4.

05 / Constructed response

Draw a horizontal or vertical number line. Plot and label -1/2, 0, and 3/4.

06 / Constructed response

Draw a horizontal or vertical number line. Plot and label -7/10, 0, and 3/10.

07 / Constructed response

Draw a horizontal or vertical number line. Plot and label -5/4, 0, and 1/2. Name the step size of your scale and explain the placement.

08 / Graphing and two-part response

On a number line, consecutive integer ticks are 1 unit apart. Plot -11/2, -5, and 0. Part A: name the leftmost point. Part B: give the midpoint between -6 and -5.



Choose and label an equal-interval scale.

09 / Graphing and two-part response

On a number line, consecutive integer ticks are 1 unit apart. Plot -15/2, -7, and 0. Part A: name the leftmost point. Part B: give the midpoint between -8 and -7.



Choose and label an equal-interval scale.

10 / Constructed response

Draw a horizontal or vertical number line. Plot and label -3/4, 0, and 1. Name the step size of your scale and explain the placement.

Day 034 | Unit 03 | Lesson

Integers and Rational Numbers

Locate Negative Fractions

Standards: 6.NS.C.6c

Learning target: I can locate integers and negative fractions on a number line.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Mark zero and equally spaced whole-number ticks. Partition each interval into equal parts for the denominator. Negative values are left of zero on a horizontal line and below zero on a vertical line.

Worked example 1

Draw a horizontal or vertical number line. Plot and label -5, 0, and 7.

1. Mark zero and equally spaced whole-number ticks.

2. Partition each interval into equal parts for the denominator.

3. Negative -5 is left of or below 0

4. 7 is right of or above 0.

5. Use equal intervals.

6. State the result in the form requested: -5 < 0 < 7; unit steps.

Answer: -5 < 0 < 7; unit steps

Worked example 2

On a number line, consecutive integer ticks are 1 unit apart. Plot -11/2, -5, and 0. Part A: name the leftmost point. Part B: give the midpoint between -6 and -5.

1. Mark zero and equally spaced whole-number ticks.

2. -11/2 = -5 - 1/2 lies halfway between -6 and -5, left of both -5 and 0.

Answer: Leftmost -11/2; midpoint -11/2

Day 034 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a horizontal or vertical number line. Plot and label -3, 0, and 2.

02 / Constructed response

Draw a horizontal or vertical number line. Plot and label -8, 0, and 3.

03 / Constructed response

Draw a horizontal or vertical number line. Plot and label -2, 0, and 7.

04 / Constructed response

Draw a horizontal or vertical number line. Plot and label -1, 0, and 2/5.

05 / Constructed response

Draw a horizontal or vertical number line. Plot and label -2, 0, and 7/2.

06 / Constructed response

Draw a horizontal or vertical number line. Plot and label -7/5, 0, and 3/5.

07 / Constructed response

Draw a horizontal or vertical number line. Plot and label -7/4, 0, and 7/4. Name the step size of your scale and explain the placement.

08 / Graphing and two-part response

On a number line, consecutive integer ticks are 1 unit apart. Plot -13/2, -6, and 0. Part A: name the leftmost point. Part B: give the midpoint between -7 and -6.



Choose and label an equal-interval scale.

09 / Graphing and two-part response

On a number line, consecutive integer ticks are 1 unit apart. Plot -7/2, -3, and 0. Part A: name the leftmost point. Part B: give the midpoint between -4 and -3.



Choose and label an equal-interval scale.

10 / Constructed response

Draw a horizontal or vertical number line. Plot and label -2/5, 0, and 2/5. Name the step size of your scale and explain the placement.

Day 035 | Unit 03 | Lesson

Integers and Rational Numbers

Order Negative Fractions with Benchmarks

Standards: 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b

Learning target: I can order Negative Fractions with Benchmarks, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Mark zero and equally spaced whole-number ticks. Partition each interval into equal parts for the denominator. Negative values are left of zero on a horizontal line and below zero on a vertical line. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

Worked example 1

Draw a horizontal or vertical number line. Plot and label -8, 0, and 6.

1. Mark zero and equally spaced whole-number ticks.

2. Partition each interval into equal parts for the denominator.

3. Negative -8 is left of or below 0

4. 6 is right of or above 0.

5. Use equal intervals.

6. State the result in the form requested: -8 < 0 < 6; unit steps.

Answer: -8 < 0 < 6; unit steps

Worked example 2

Temperatures are -5 °C and -3 °C. Select all true comparisons.
A. -5 is warmer than -3.
B. -5 < -3
C. -3 lies to the right of -5 on a number line.
D. -5 > 0

1. A number farther right is greater.

2. Among negatives, the number closer to zero is greater.

3. Thus -3 is warmer and is farther right.

Answer: B, C - -5 < -3; -3 lies to the right of -5 on a number line.

Day 035 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a horizontal or vertical number line. Plot and label -8, 0, and 4.

02 / Constructed response

Compare -3 ___ -3 using <, >, or =. Sketch a number line to justify.

03 / Constructed response

Draw a horizontal or vertical number line. Plot and label -6, 0, and 2.

04 / Constructed response

Compare -2 ___ -1/2 using <, >, or =. Sketch a number line to justify.

05 / Constructed response

Draw a horizontal or vertical number line. Plot and label -2/3, 0, and 4/3.

06 / Constructed response

Compare -3/5 ___ -4/3 using <, >, or =. Sketch a number line to justify.

07 / Constructed response

Draw a horizontal or vertical number line. Plot and label -5/4, 0, and 3/4. Name the step size of your scale and explain the placement.

08 / Multiple select

Temperatures are -6 °C and -4 °C. Select all true comparisons.
A. -6 is warmer than -4.
B. -6 < -4
C. -4 lies to the right of -6 on a number line.
D. -6 > 0

09 / Graphing and two-part response

On a number line, consecutive integer ticks are 1 unit apart. Plot -13/2, -6, and 0. Part A: name the leftmost point. Part B: give the midpoint between -7 and -6.



Choose and label an equal-interval scale.

10 / Constructed response

Temperatures are -7/5°C and -2/5°C. Write an inequality comparing them and explain which is warmer.

Day 036 | Unit 03 | Lesson

Integers and Rational Numbers

Compare and Order Rational Numbers

Standards: 6.NS.C.7a, 6.NS.C.7b

Learning target: I can order rational numbers using their positions.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

Worked example 1

Compare -2 ___ -6 using <, >, or =. Sketch a number line to justify.

1. A number farther right is greater.

2. Among negative numbers, the value closer to zero is greater.

3. -2 > -6

4. greater numbers lie farther right.

5. State the result in the form requested: >.

Answer: >

Worked example 2

Temperatures are -8 °C and -6 °C. Select all true comparisons.
A. -8 is warmer than -6.
B. -6 lies to the right of -8 on a number line.
C. -8 < -6
D. -8 > 0

1. A number farther right is greater.

2. Among negatives, the number closer to zero is greater.

3. Thus -6 is warmer and is farther right.

Answer: B, C - -6 lies to the right of -8 on a number line.; -8 < -6

Day 036 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Compare -8 ___ -6 using <, >, or =. Sketch a number line to justify.

02 / Constructed response

Compare -8 ___ -2 using <, >, or =. Sketch a number line to justify.

03 / Constructed response

Compare -7 ___ -7 using <, >, or =. Sketch a number line to justify.

04 / Constructed response

Compare -3/10 ___ -5/2 using <, >, or =. Sketch a number line to justify.

05 / Constructed response

Compare -7/2 ___ -7/5 using <, >, or =. Sketch a number line to justify.

06 / Constructed response

Compare -1/5 ___ -2 using <, >, or =. Sketch a number line to justify.

07 / Constructed response

Temperatures are -3/10°C and -3/10°C. Write an inequality comparing them and explain which is warmer.

08 / Multiple select

Temperatures are -4 °C and -2 °C. Select all true comparisons.
A. -4 > 0
B. -4 is warmer than -2.
C. -4 < -2
D. -2 lies to the right of -4 on a number line.

09 / Multiple select

Temperatures are -5 °C and -3 °C. Select all true comparisons.
A. -5 is warmer than -3.
B. -5 < -3
C. -3 lies to the right of -5 on a number line.
D. -5 > 0

10 / Constructed response

Temperatures are -7/4°C and -5/2°C. Write an inequality comparing them and explain which is warmer.

Day 037 | Unit 03 | Lesson

Integers and Rational Numbers

Interpret Order in Context

Standards: 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.5

Learning target: I can interpret Order in Context, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions. Choose a reference point for zero. Opposite directions or values use opposite signs: above sea level is positive and below sea level is negative. Zero means at the chosen reference, not always nothing.

Worked example 1

Compare -7 ___ -6 using <, >, or =. Sketch a number line to justify.

1. A number farther right is greater.

2. Among negative numbers, the value closer to zero is greater.

3. -7 < -6

4. greater numbers lie farther right.

5. State the result in the form requested: <.

Answer: <

Worked example 2

A sensor is 5 meters below a reference point. Select all correct interpretations.
A. Its signed position is +5 m.
B. A reading of 0 m means at the reference point.
C. The opposite position is 5 m below the reference.
D. Its signed position is -5 m.

1. Choose a reference point for zero.

2. Below the chosen zero is negative, so the position is -5.

3. Zero identifies the reference

4. the opposite is 5 m above it.

Answer: B, D - A reading of 0 m means at the reference point.; Its signed position is -5 m.

Day 037 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Compare -5 ___ -2 using <, >, or =. Sketch a number line to justify.

02 / Constructed response

A marker is 12 m below sea level. Write its signed elevation and explain what elevation 0 means.

03 / Constructed response

Compare -3 ___ -4 using <, >, or =. Sketch a number line to justify.

04 / Constructed response

A marker is 32 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

Compare -4 ___ -1 using <, >, or =. Sketch a number line to justify.

06 / Constructed response

A marker is 8 m below sea level. Write its signed elevation and explain what elevation 0 means.

07 / Constructed response

Temperatures are -1/2°C and -1/5°C. Write an inequality comparing them and explain which is warmer.

08 / Multiple select

A sensor is 3 meters below a reference point. Select all correct interpretations.
A. Its signed position is -3 m.
B. A reading of 0 m means at the reference point.
C. The opposite position is 3 m below the reference.
D. Its signed position is +3 m.

09 / Multiple select

Temperatures are -6 °C and -4 °C. Select all true comparisons.
A. -6 is warmer than -4.
B. -6 < -4
C. -4 lies to the right of -6 on a number line.
D. -6 > 0

10 / Constructed response

A marker is 42 m below sea level. Write its signed elevation and explain what elevation 0 means. Write the elevation of the point equally far on the opposite side of sea level.

Day 038 | Unit 03 | Lesson

Integers and Rational Numbers

Understand Absolute Value

Standards: 6.NS.C.7c, 6.NS.C.7d

Learning target: I can distinguish a number from its distance to zero.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

The absolute value |x| is the distance from x to zero, so it cannot be negative. A larger absolute value means farther from zero, but does not necessarily mean a greater number.

Worked example 1

Find |-36|. Explain the distance meaning.

1. The absolute value |x| is the distance from x to zero, so it cannot be negative.

2. A larger absolute value means farther from zero, but does not necessarily mean a greater number.

3. -36 is 36 units from 0

4. distance is nonnegative.

5. State the result in the form requested: 36.

Answer: 36

Worked example 2

Consider -5 and 3. Select all true statements.
A. |-5| = -5
B. |-5| = 5
C. -5 is farther from zero than 3.
D. -5 > 3

1. The absolute value |x| is the distance from x to zero, so it cannot be negative.

2. Absolute value measures distance: 5 versus 3.

3. A larger distance does not make the negative number greater.

Answer: B, C - |-5| = 5; -5 is farther from zero than 3.

Day 038 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find |-21|. Explain the distance meaning.

02 / Constructed response

Find |-14|. Explain the distance meaning.

03 / Constructed response

Find |-24|. Explain the distance meaning.

04 / Constructed response

Find |-9/2| and compare it with |3/2|.

05 / Constructed response

Find |-5| and compare it with |5/2|.

06 / Constructed response

Find |-3| and compare it with |1|.

07 / Constructed response

A diver is at -6 m and a bird is at 2 m relative to sea level. Who is farther from sea level? Who has greater elevation? Use absolute values.

08 / Multiple select

Consider -4 and 2. Select all true statements.
A. |-4| = 4
B. -4 is farther from zero than 2.
C. |-4| = -4
D. -4 > 2

09 / Multiple select

Consider -6 and 4. Select all true statements.
A. -6 is farther from zero than 4.
B. |-6| = -6
C. |-6| = 6
D. -6 > 4

10 / Constructed response

A diver is at -8 m and a bird is at 4 m relative to sea level. Who is farther from sea level? Who has greater elevation? Use absolute values.

Day 039 | Unit 03 | Lesson

Integers and Rational Numbers

Compare Order and Distance from Zero

Standards: 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.7a, 6.NS.C.7b

Learning target: I can compare Order and Distance from Zero, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

The absolute value |x| is the distance from x to zero, so it cannot be negative. A larger absolute value means farther from zero, but does not necessarily mean a greater number. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions.

Worked example 1

Find |-10|. Explain the distance meaning.

1. The absolute value |x| is the distance from x to zero, so it cannot be negative.

2. A larger absolute value means farther from zero, but does not necessarily mean a greater number.

3. -10 is 10 units from 0

4. distance is nonnegative.

5. State the result in the form requested: 10.

Answer: 10

Worked example 2

Temperatures are -9 °C and -7 °C. Select all true comparisons.
A. -9 is warmer than -7.
B. -9 < -7
C. -7 lies to the right of -9 on a number line.
D. -9 > 0

1. A number farther right is greater.

2. Among negatives, the number closer to zero is greater.

3. Thus -7 is warmer and is farther right.

Answer: B, C - -9 < -7; -7 lies to the right of -9 on a number line.

Day 039 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find |-40|. Explain the distance meaning.

02 / Constructed response

Compare -5 ___ -4 using <, >, or =. Sketch a number line to justify.

03 / Constructed response

Find |-9|. Explain the distance meaning.

04 / Constructed response

Compare -3/5 ___ -6/5 using <, >, or =. Sketch a number line to justify.

05 / Constructed response

Find |-9/2| and compare it with |3/2|.

06 / Constructed response

Compare -1 ___ -4/3 using <, >, or =. Sketch a number line to justify.

07 / Constructed response

A diver is at -15 m and a bird is at 5/2 m relative to sea level. Who is farther from sea level? Who has greater elevation? Use absolute values.

08 / Multiple select

Temperatures are -6 °C and -4 °C. Select all true comparisons.
A. -6 is warmer than -4.
B. -6 < -4
C. -4 lies to the right of -6 on a number line.
D. -6 > 0

09 / Multiple select

Consider -6 and 4. Select all true statements.
A. -6 is farther from zero than 4.
B. |-6| = -6
C. |-6| = 6
D. -6 > 4

10 / Constructed response

Temperatures are -2°C and -2/5°C. Write an inequality comparing them and explain which is warmer.

Day 040 | Unit 03 | Lesson

Integers and Rational Numbers

Interpret Points on Axes and in Quadrants

Standards: 6.NS.C.6b, 6.NS.C.6c

Learning target: I can plot ordered pairs in all four quadrants.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant.

Worked example 1

Plot P(6, -4) on a coordinate plane and name its quadrant.

1. Start at the origin.

2. Move horizontally by x, then vertically by y.

3. Move 6 right, then 4 down.

4. State the result in the form requested: Quadrant IV.

Answer: Quadrant IV

Worked example 2

Plot P(-5, 4), Q(5, -4), and R(0, -4). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.

1. Start at the origin.

2. P has signs (-,+), so II.

3. Q has (+,-), so IV.

4. R has x = 0 and is on an axis.

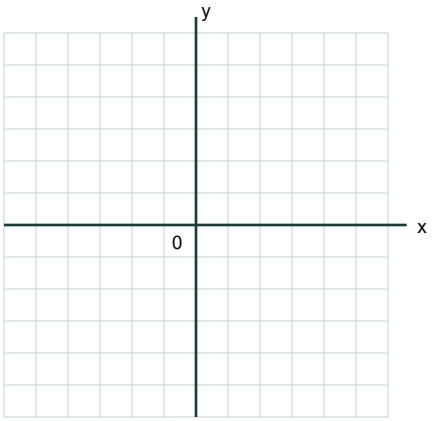
Answer: P: II; Q: IV; R is on the y-axis.

Day 040 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

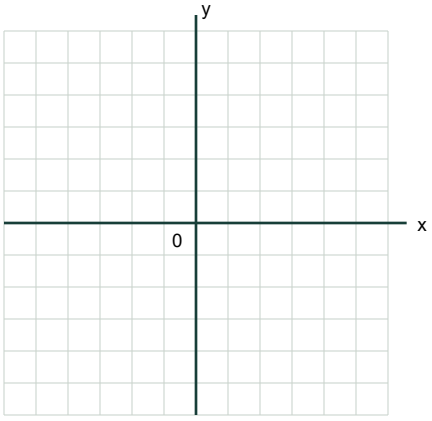
Plot P(-7, -5) on a coordinate plane and name its quadrant.



Label the scale and units.

02 / Constructed response

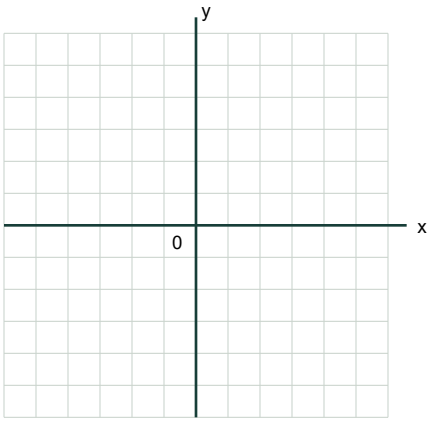
Plot P(5, -3) on a coordinate plane and name its quadrant.



Label the scale and units.

03 / Constructed response

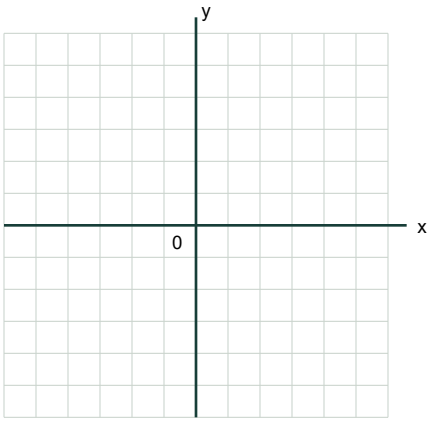
Plot P(-8, 2) on a coordinate plane and name its quadrant.



Label the scale and units.

04 / Constructed response

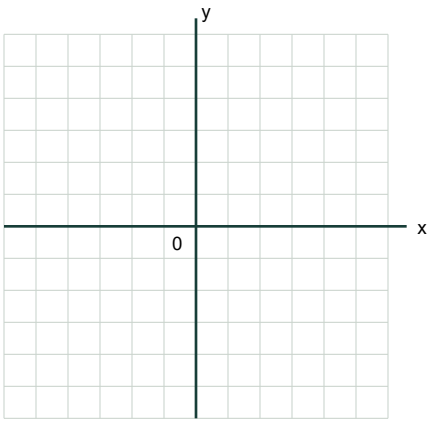
Plot P(5, -6) on a coordinate plane and name its quadrant.



Label the scale and units.

05 / Constructed response

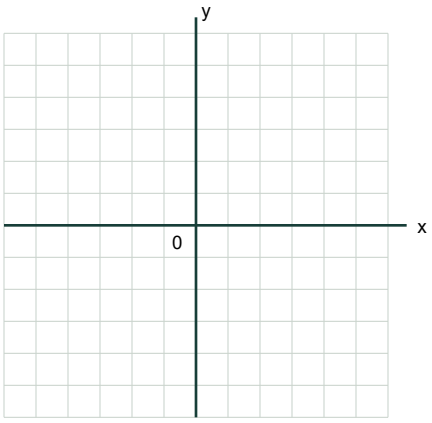
Plot P(-8, -4) on a coordinate plane and name its quadrant.



Label the scale and units.

06 / Constructed response

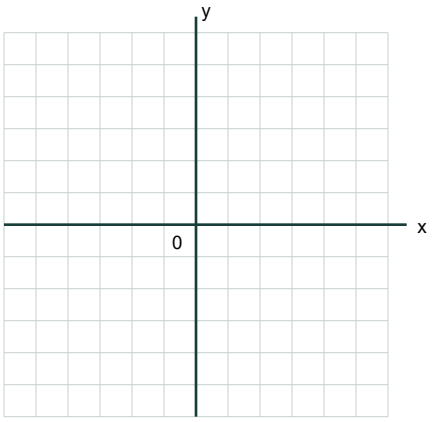
Plot P(3, 4) on a coordinate plane and name its quadrant.



Label the scale and units.

07 / Constructed response

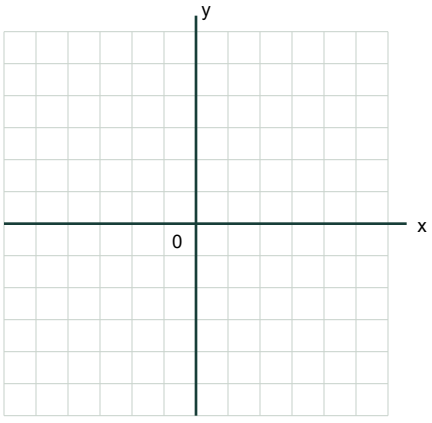
Plot P(-3/2, -1) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

08 / Coordinate plotting and explanation

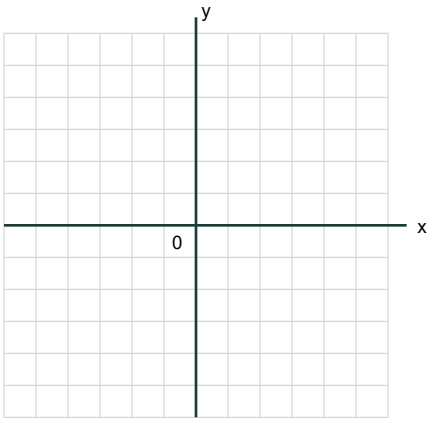
Plot P(-4, 5), Q(4, -5), and R(0, -5). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

09 / Coordinate plotting and explanation

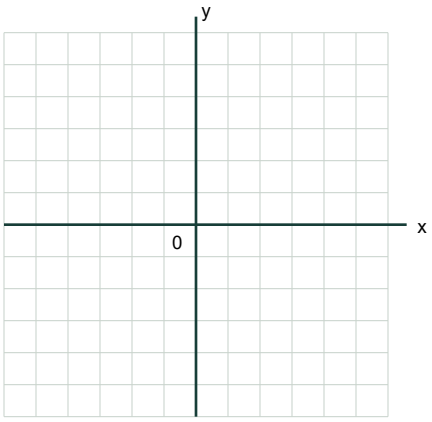
Plot P(-4, 6), Q(4, -6), and R(0, -6). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

10 / Constructed response

Plot P(1, -3/2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

Day 041 | Unit 03 | Lesson

Integers and Rational Numbers

Plot Points in Four Quadrants

Standards: 6.NS.C.6b, 6.NS.C.6c

Learning target: I can plot ordered pairs in all four quadrants.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant.

Worked example 1

Plot P(5, -7) on a coordinate plane and name its quadrant.

1. Start at the origin.

2. Move horizontally by x, then vertically by y.

3. Move 5 right, then 7 down.

4. State the result in the form requested: Quadrant IV.

Answer: Quadrant IV

Worked example 2

Plot P(-3, 5), Q(3, -5), and R(0, -5). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.

1. Start at the origin.

2. P has signs (-,+), so II.

3. Q has (+,-), so IV.

4. R has x = 0 and is on an axis.

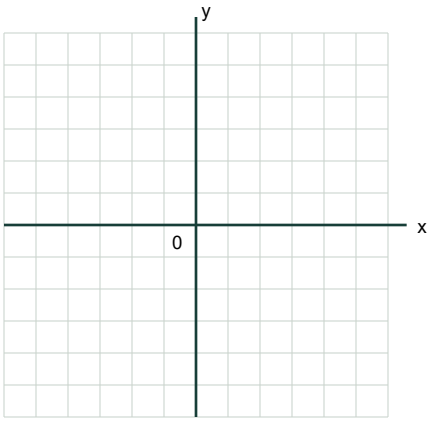
Answer: P: II; Q: IV; R is on the y-axis.

Day 041 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

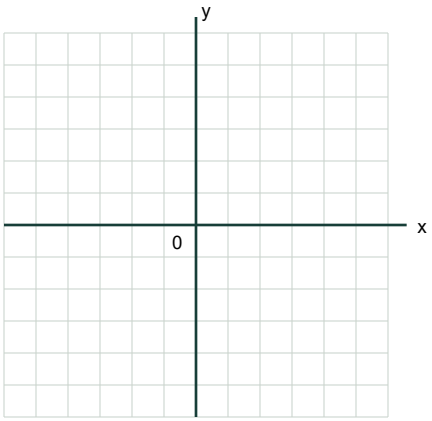
Plot P(-8, -3) on a coordinate plane and name its quadrant.



Label the scale and units.

02 / Constructed response

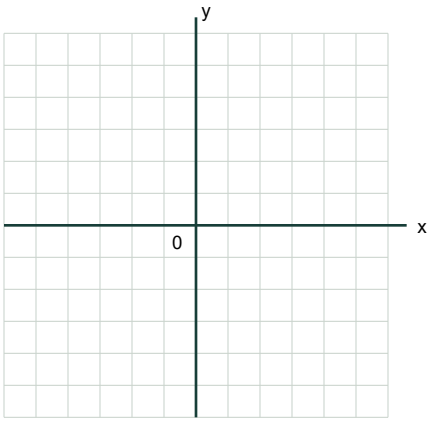
Plot P(2, -5) on a coordinate plane and name its quadrant.



Label the scale and units.

03 / Constructed response

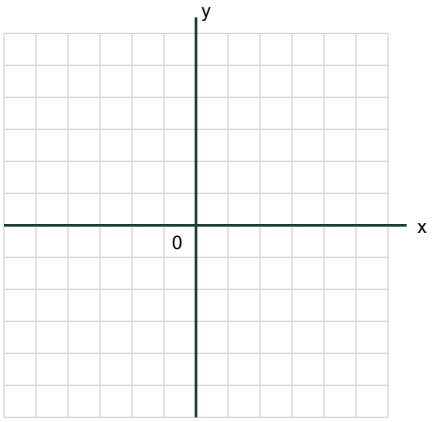
Plot P(-3, 2) on a coordinate plane and name its quadrant.



Label the scale and units.

04 / Constructed response

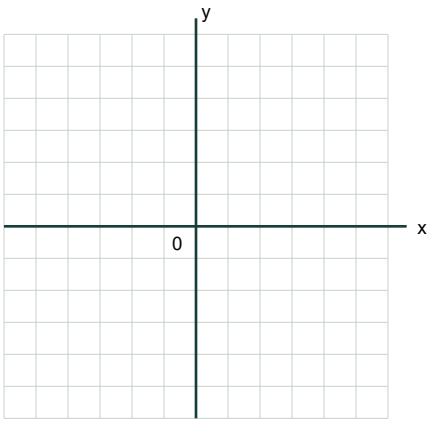
Plot P(6, -3) on a coordinate plane and name its quadrant.



Label the scale and units.

05 / Constructed response

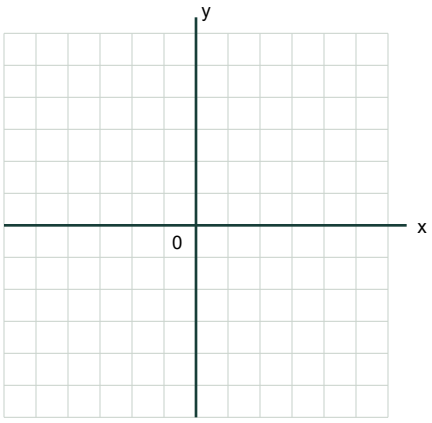
Plot P(-6, -7) on a coordinate plane and name its quadrant.



Label the scale and units.

06 / Constructed response

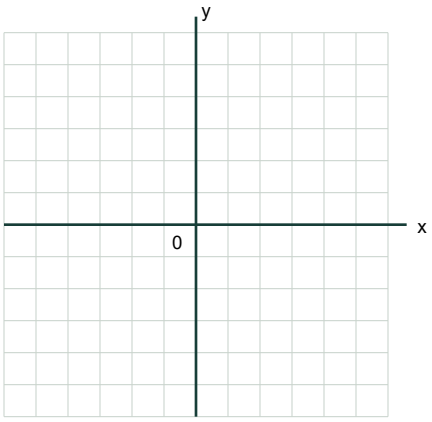
Plot P(2, 3) on a coordinate plane and name its quadrant.



Label the scale and units.

07 / Constructed response

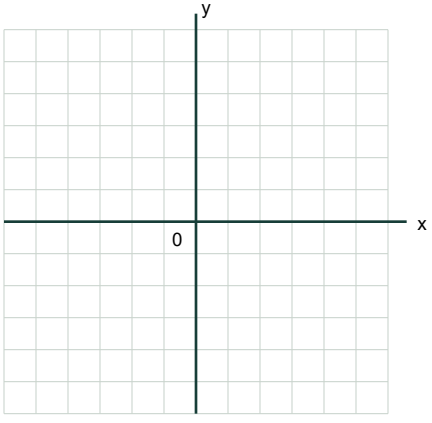
Plot P(-5/2, -3/2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

08 / Coordinate plotting and explanation

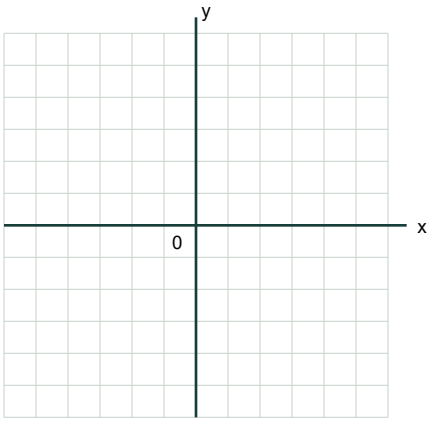
Plot P(-3, 3), Q(3, -3), and R(0, -3). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

09 / Coordinate plotting and explanation

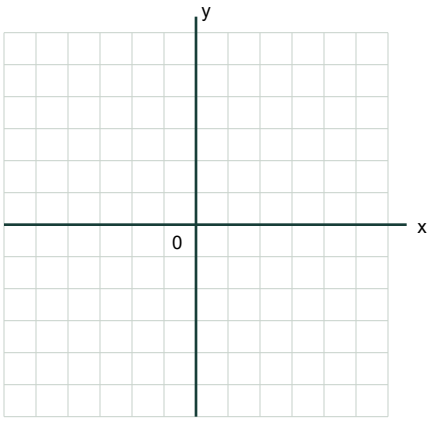
Plot P(-7, 6), Q(7, -6), and R(0, -6). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

10 / Constructed response

Plot P(3, -3/2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

Day 042 | Unit 03 | Lesson

Integers and Rational Numbers

Reflect Points Across Axes

Standards: 6.NS.C.6b

Learning target: I can reflect points across an axis using signs.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis.

Worked example 1

Plot (-2, -7), reflect it across y-axis, and give the reflected coordinates.

1. Across the x-axis, keep x and reverse the sign of y.

2. Across the y-axis, reverse x and keep y.

3. Reverse the x sign: (2, -7).

4. State the result in the form requested: (2, -7).

Answer: (2, -7)

Worked example 2

Point P is (-4, 3). Select all correct reflections.
A. Across both axes: (-4, 3).
B. Across the x-axis: (-4, -3).
C. Across the x-axis: (4, 3).
D. Across the y-axis: (4, 3).

1. Across the x-axis, keep x and reverse the sign of y.

2. The x-axis reflection changes y only

3. the y-axis reflection changes x only.

4. Across both gives (4, -3).

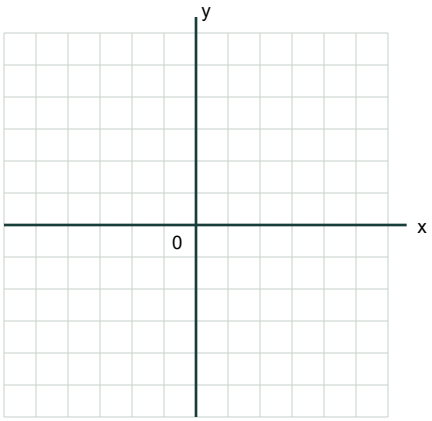
Answer: B, D - Across the x-axis: (-4, -3).; Across the y-axis: (4, 3).

Day 042 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

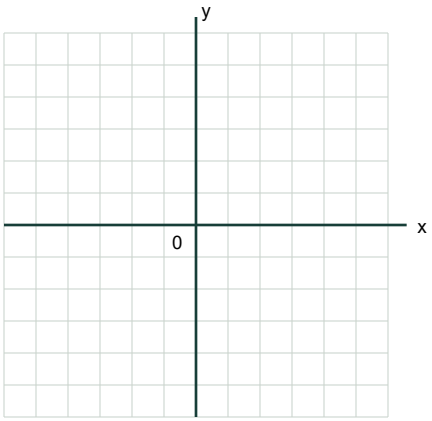
Plot (-3, 4), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

02 / Constructed response

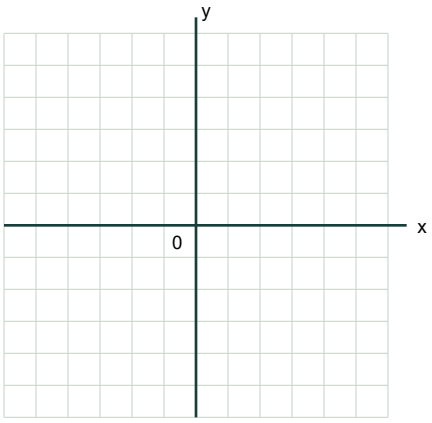
Plot (-4, -7), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

03 / Constructed response

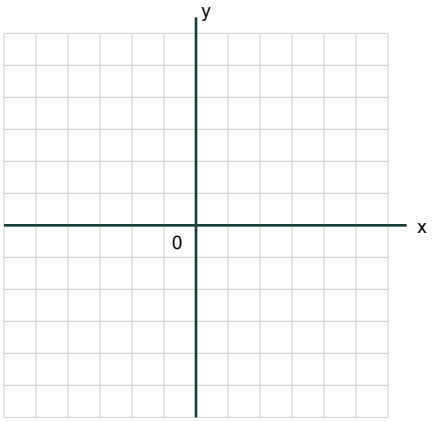
Plot (-8, 6), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

04 / Constructed response

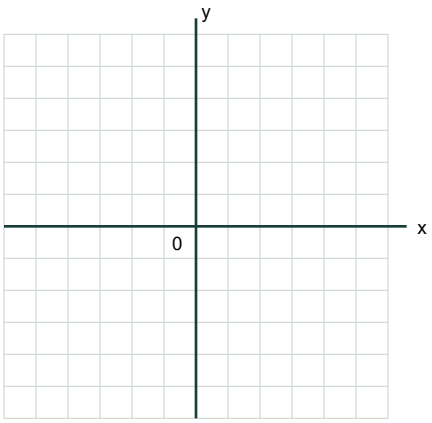
Plot (-7, -7), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

05 / Constructed response

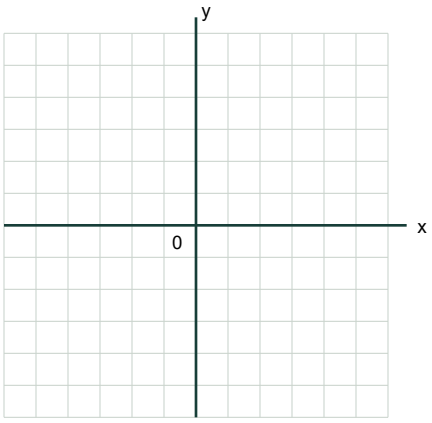
Plot (-4, 3), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

06 / Constructed response

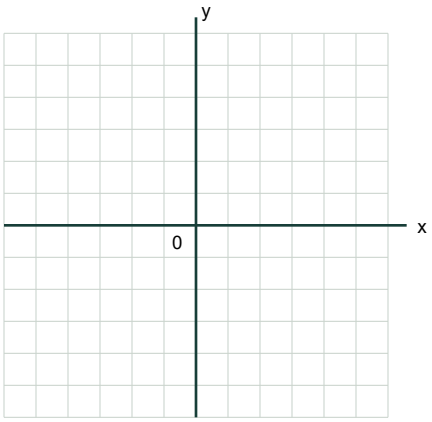
Plot (-7, -3), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

07 / Constructed response

Plot (-8, 4), reflect it across both axes, and give the reflected coordinates.



Label the scale and units.

08 / Multiple select

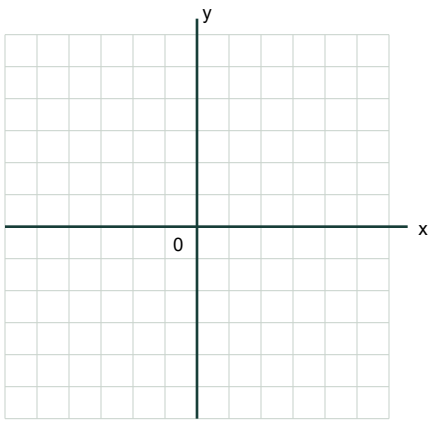
Point P is (-6, 4). Select all correct reflections.
A. Across the y-axis: (6, 4).
B. Across the x-axis: (6, 4).
C. Across the x-axis: (-6, -4).
D. Across both axes: (-6, 4).

09 / Multiple select

Point P is (-2, 2). Select all correct reflections.
A. Across the x-axis: (2, 2).
B. Across the x-axis: (-2, -2).
C. Across the y-axis: (2, 2).
D. Across both axes: (-2, 2).

10 / Constructed response

Plot (-8, -4), reflect it across both axes, and give the reflected coordinates.



Label the scale and units.

Day 043 | Unit 03 | Lesson

Integers and Rational Numbers

Find Horizontal and Vertical Distance

Standards: 6.NS.C.8

Learning target: I can graph points and find horizontal or vertical distance.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes.

Worked example 1

Plot (-4, 4) and (-8, 4). Find their distance.

1. Plot both points.

2. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units.

3. Only one coordinate changes.

4. Same side of 0: subtract magnitudes to get 4.

5. State the result in the form requested: 4 units.

Answer: 4 units

Worked example 2

Plot A(-7, -4), B(2, -4), and C(-7, 4). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.

1. Plot both points.

2. AB is horizontal: 7 + 2 = 9.

3. AC is vertical: 4 + 4 = 8.

4. Each segment crosses an axis.

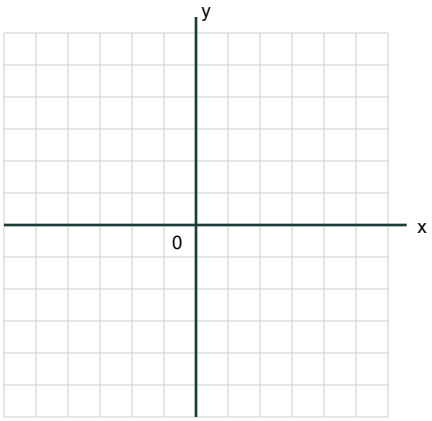
Answer: AB = 9 units; AC = 8 units

Day 043 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

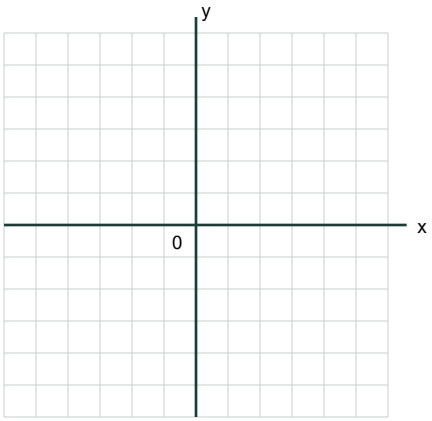
Plot (4, -5) and (4, -9). Find their distance.



Label the scale and units.

02 / Constructed response

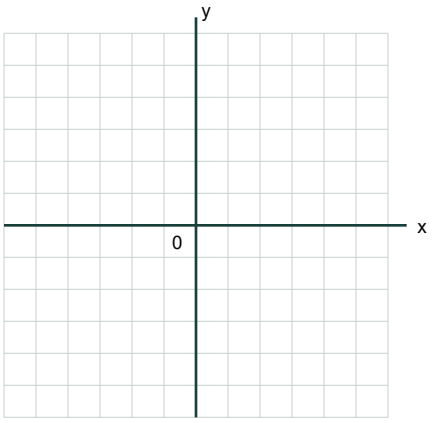
Plot (-8, 2) and (-10, 2). Find their distance.



Label the scale and units.

03 / Constructed response

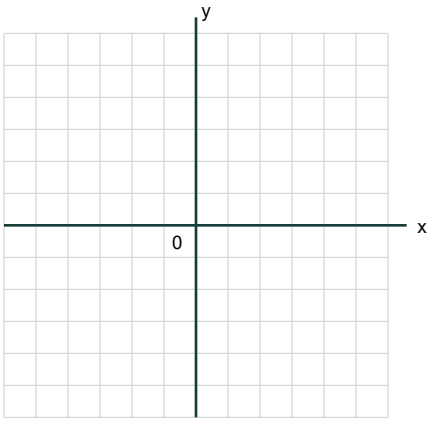
Plot (4, -2) and (4, -6). Find their distance.



Label the scale and units.

04 / Constructed response

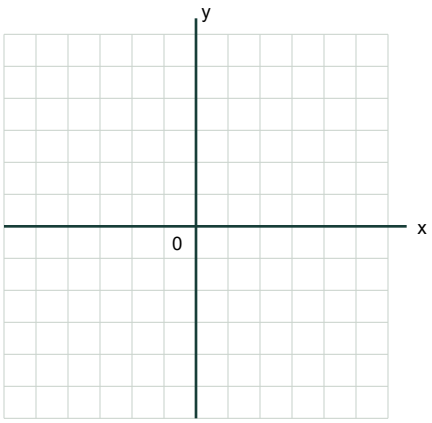
Plot (-5, 5) and (6, 5). Find their distance.



Label the scale and units.

05 / Constructed response

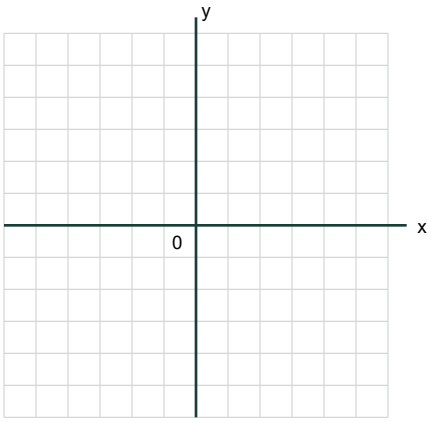
Plot (4, -7) and (4, 2). Find their distance.



Label the scale and units.

06 / Constructed response

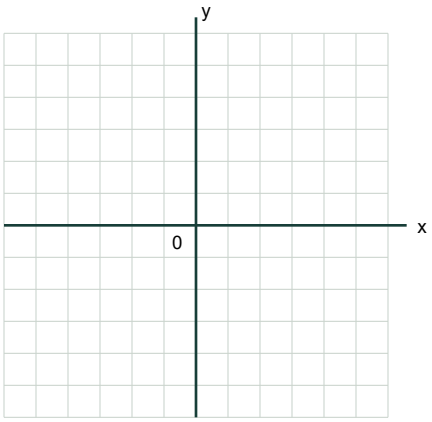
Plot (-7, 6) and (5, 6). Find their distance.



Label the scale and units.

07 / Constructed response

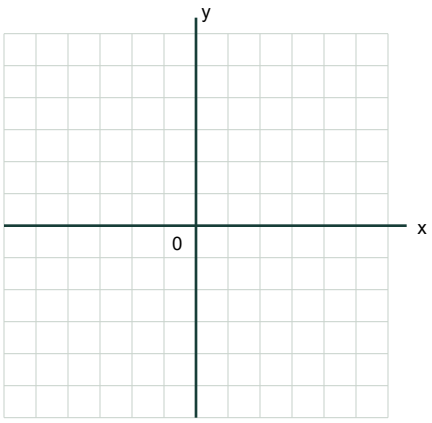
Plot (5, -3) and (5, 2). Find their distance. Explain why adding magnitudes works or does not work for these points.



Label the scale and units.

08 / Coordinate plotting and numeric entry

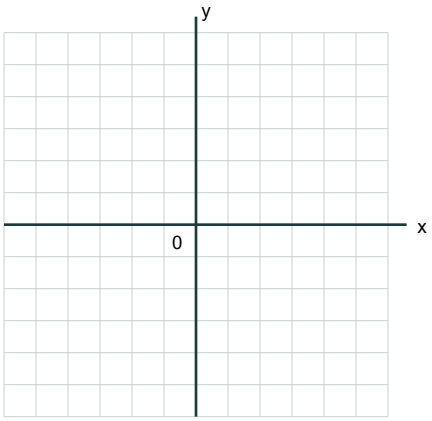
Plot A(-6, -5), B(3, -5), and C(-6, 5). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

09 / Coordinate plotting and numeric entry

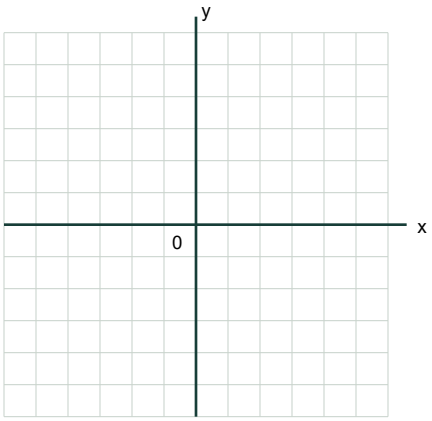
Plot A(-5, -6), B(3, -6), and C(-5, 6). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

10 / Constructed response

Plot (-5, 2) and (7, 2). Find their distance. Explain why adding magnitudes works or does not work for these points.



Label the scale and units.

Day 044 | Unit 03 | Lesson

Integers and Rational Numbers

Solve Coordinate-Plane Problems

Standards: 6.NS.C.6b, 6.NS.C.6c, 6.NS.C.8

Learning target: I can solve Coordinate-Plane Problems, and explain and check my reasoning.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

01 / Learn

Start at the origin. Move horizontally by x, then vertically by y. Quadrant signs are I (+,+), II (-,+), III (-,-), IV (+,-). A point on an axis belongs to no quadrant. Across the x-axis, keep x and reverse the sign of y. Across the y-axis, reverse x and keep y. Across both axes, reverse both signs. The reflected point stays the same distance from the axis. Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes.

Worked example 1

Plot P(7, -2) on a coordinate plane and name its quadrant.

1. Start at the origin.

2. Move horizontally by x, then vertically by y.

3. Move 7 right, then 2 down.

4. State the result in the form requested: Quadrant IV.

Answer: Quadrant IV

Worked example 2

Plot A(-4, -3), B(5, -3), and C(-4, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.

1. Plot both points.

2. AB is horizontal: 4 + 5 = 9.

3. AC is vertical: 3 + 3 = 6.

4. Each segment crosses an axis.

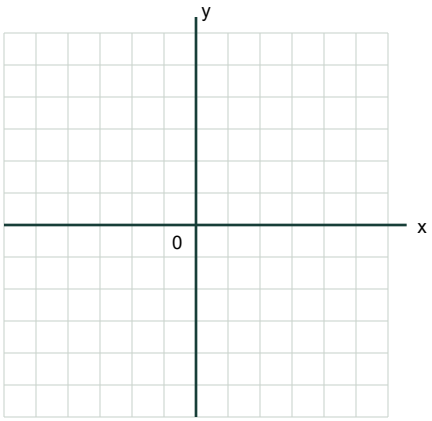
Answer: AB = 9 units; AC = 6 units

Day 044 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

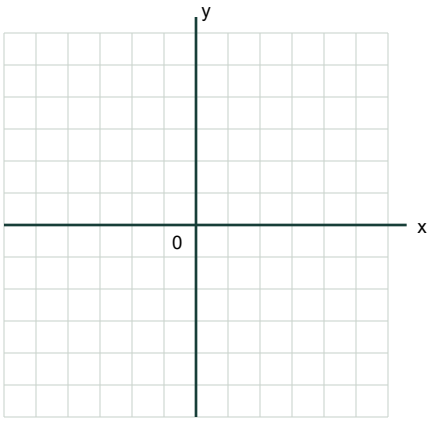
Plot P(-6, -3) on a coordinate plane and name its quadrant.



Label the scale and units.

02 / Constructed response

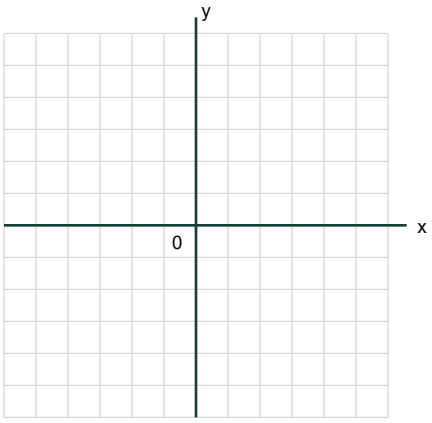
Plot (-5, -3), reflect it across y-axis, and give the reflected coordinates.



Label the scale and units.

03 / Constructed response

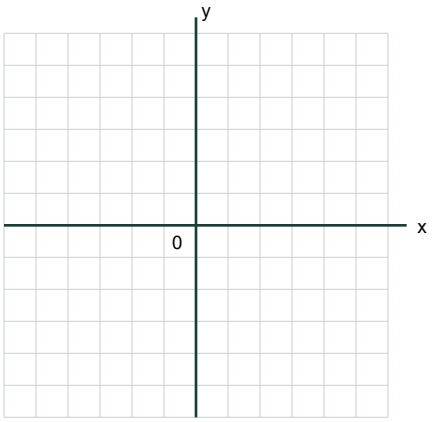
Plot (5, -3) and (5, -8). Find their distance.



Label the scale and units.

04 / Constructed response

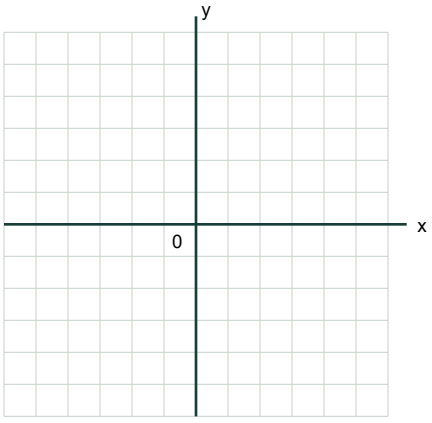
Plot P(8, -5) on a coordinate plane and name its quadrant.



Label the scale and units.

05 / Constructed response

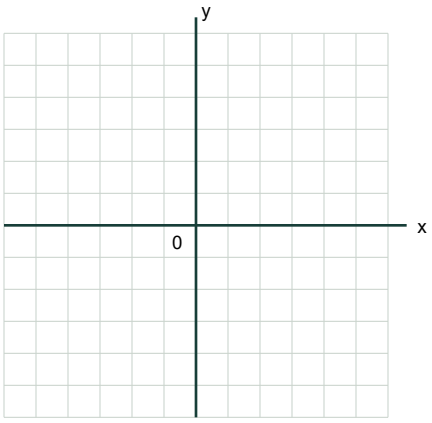
Plot (-4, 2), reflect it across x-axis, and give the reflected coordinates.



Label the scale and units.

06 / Constructed response

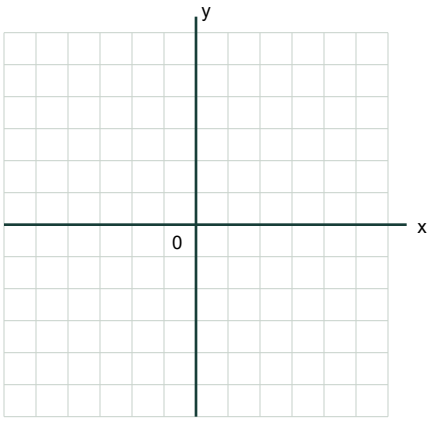
Plot (-8, 3) and (2, 3). Find their distance.



Label the scale and units.

07 / Constructed response

Plot P(-5/2, -2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



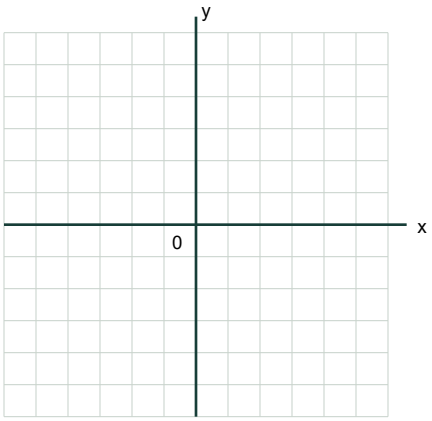
Label the scale and units.

08 / Multiple select

Point P is (-5, 4). Select all correct reflections.
A. Across the y-axis: (5, 4).
B. Across the x-axis: (-5, -4).
C. Across the x-axis: (5, 4).
D. Across both axes: (-5, 4).

09 / Coordinate plotting and numeric entry

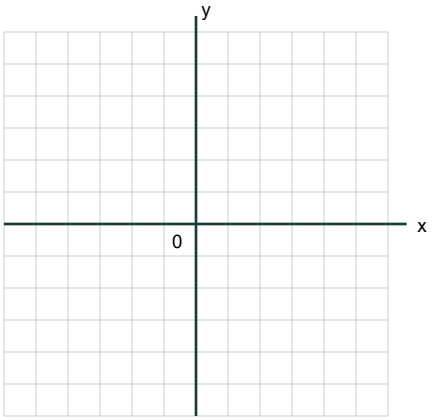
Plot A(-6, -3), B(5, -3), and C(-6, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

10 / Constructed response

Plot P(3, -7/2) on a coordinate plane and name its quadrant. Then plot Q on the y-axis at the same height and name its coordinates.



Label the scale and units.

Day 045 | Unit 03 | Unit Test

Integers and Rational Numbers

Unit 3 Test

Standards: 6.NS.A.1, 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.NS.C.7a, 6.NS.C.7b, 6.NS.C.7c, 6.NS.C.7d, 6.NS.C.6b, 6.NS.C.8

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 2: Integers and Rational Numbers | Spectrum companion Chapter 4

Directions

The first 4 questions (20%) review Unit 2. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Calculate 3/4 × 1/2. Simplify.

02 / Previous unit review | Constructed response

Calculate 3 1/4 × 1 1/3. Simplify.

03 / Previous unit review | Constructed response

Draw a bar partitioned into 12 equal parts. Shade 11 parts. Circle groups of 7 parts. Use the model to find 11/12 ÷ 7/12.

04 / Previous unit review | Constructed response

Draw a bar partitioned into 6 equal parts. Shade 6 parts. Circle groups of 3 parts. Use the model to find 1 ÷ 1/2.

05 / Constructed response

A marker is 8 m above sea level. Write its signed elevation and explain what elevation 0 means.

06 / Constructed response

Find the opposite of -8. Show both values on a number line.

07 / Constructed response

Draw a horizontal or vertical number line. Plot and label -3, 0, and 5.

08 / Constructed response

Draw a horizontal or vertical number line. Plot and label -8/3, 0, and 4/3.

09 / Constructed response

Draw a horizontal or vertical number line. Plot and label -1, 0, and 3/2.

10 / Constructed response

Compare -4/5 ___ -1 using <, >, or =. Sketch a number line to justify.

11 / Constructed response

Compare -4/5 ___ -7/10 using <, >, or =. Sketch a number line to justify.

12 / Multiple select

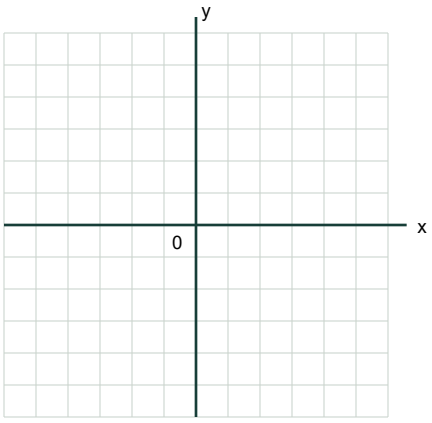
Consider -8 and 6. Select all true statements.
A. -8 > 6
B. |-8| = -8
C. |-8| = 8
D. -8 is farther from zero than 6.

13 / Multiple select

Consider -5 and 3. Select all true statements.
A. |-5| = -5
B. |-5| = 5
C. -5 is farther from zero than 3.
D. -5 > 3

14 / Coordinate plotting and explanation

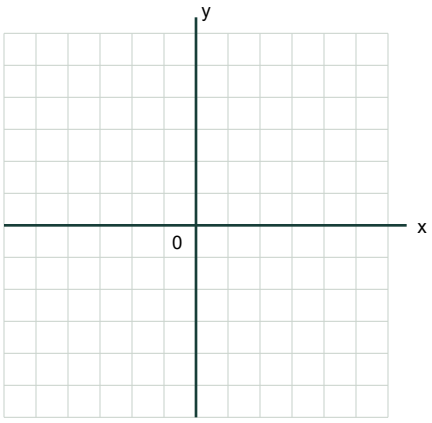
Plot P(-2, 6), Q(2, -6), and R(0, -6). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

15 / Coordinate plotting and explanation

Plot P(-3, 5), Q(3, -5), and R(0, -5). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



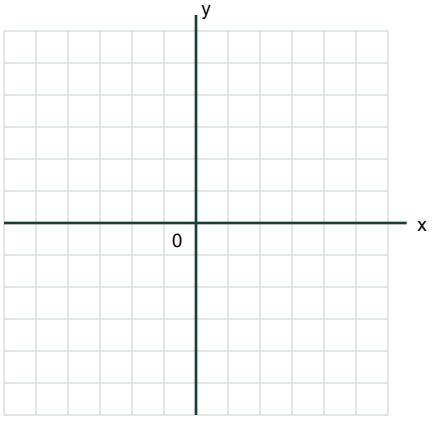
Label the scale and units.

16 / Multiple select

Point P is (-6, 3). Select all correct reflections.
A. Across the y-axis: (6, 3).
B. Across the x-axis: (6, 3).
C. Across both axes: (-6, 3).
D. Across the x-axis: (-6, -3).

17 / Coordinate plotting and numeric entry

Plot A(-5, -6), B(2, -6), and C(-5, 6). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

18 / Multiple select

Point P is (-5, 5). Select all correct reflections.
A. Across the x-axis: (-5, -5).
B. Across the x-axis: (5, 5).
C. Across both axes: (-5, 5).
D. Across the y-axis: (5, 5).

19 / Multiple select

A sensor is 7 meters below a reference point. Select all correct interpretations.
A. Its signed position is -7 m.
B. Its signed position is +7 m.
C. The opposite position is 7 m below the reference.
D. A reading of 0 m means at the reference point.

20 / Multiple select

Select all true statements about -7/2 and its opposite. Explain the role of zero.
A. The opposite of the opposite of -7/2 is -7/2.
B. The opposite of 0 is 1.
C. -7/2 and 7/2 are on the same side of zero.
D. The opposite of -7/2 is 7/2.

Day 046 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Write and Evaluate Powers

Standards: 6.EE.A.1

Learning target: I can write repeated multiplication using a whole-number exponent.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

The base is the repeated factor and the exponent tells how many copies to multiply. For a positive exponent, 4^3 = 4 × 4 × 4, not 4 × 3. Parentheses show when a whole product is the base.

Worked example 1

Write 3^2 as repeated multiplication and evaluate.

1. The base is the repeated factor and the exponent tells how many copies to multiply.

2. For a positive exponent, 4^3 = 4 × 4 × 4, not 4 × 3.

3. The exponent counts 2 factors of 3

4. multiply all factors.

5. State the result in the form requested: 3 × 3 = 9.

Answer: 3 × 3 = 9

Worked example 2

Select all expressions equal to 2^3.
A. 9
B. 2 × 3
C. 8
D. 2 × 2 × 2

1. The base is the repeated factor and the exponent tells how many copies to multiply.

2. The exponent 3 means three factors of 2

3. their product is 8.

4. It does not mean 2 times 3.

Answer: C, D - 8; 2 × 2 × 2

Day 046 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 3^3 as repeated multiplication and evaluate.

02 / Constructed response

Write 2^4 as repeated multiplication and evaluate.

03 / Constructed response

Write 4^3 as repeated multiplication and evaluate.

04 / Constructed response

Write 6^4 as repeated multiplication and evaluate.

05 / Constructed response

Write 7^2 as repeated multiplication and evaluate.

06 / Constructed response

Write 2^2 as repeated multiplication and evaluate.

07 / Constructed response

Write 4^3 as repeated multiplication and evaluate. Explain why it differs from 4 × 3.

08 / Multiple select

Select all expressions equal to 3^3.
A. 3 × 3
B. 27
C. 28
D. 3 × 3 × 3

09 / Multiple select

Select all expressions equal to 6^3.
A. 6 × 3
B. 6 × 6 × 6
C. 217
D. 216

10 / Constructed response

Write 3^3 as repeated multiplication and evaluate. Explain why it differs from 3 × 3.

Day 047 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Use the Order of Operations

Standards: 6.EE.A.1, 6.EE.A.2c

Learning target: I can evaluate expressions using the order of operations.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Work inside parentheses, then evaluate exponents. Multiply and divide from left to right; then add and subtract from left to right. Multiplication does not automatically come before division.

Worked example 1

Evaluate 7 + 3 × 5. Show the intermediate steps.

1. Work inside parentheses, then evaluate exponents.

2. Multiply and divide from left to right; then add and subtract from left to right.

3. Use parentheses and exponents first, then × and ÷ left to right, then + and -.

4. Result: 22.

5. State the result in the form requested: 22.

Answer: 22

Worked example 2

Evaluate $5^2 + 3(8 - 3)$. Part A: enter the value. Part B: a student begins by adding 5^2 and 3; explain why that changes the expression.

1. Work inside parentheses, then evaluate exponents.

2. Parentheses: 8 - 3 = 5.

3. Exponent: $5^2 = 25$.

4. Multiply: 3 × 5 = 15.

5. Add: 25 + 15 = 40.

6. The product must be completed before the final sum.

Answer: 40

Day 047 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Evaluate 3 + 2 × 6. Show the intermediate steps.

02 / Constructed response

Evaluate 3 + 5 × 6. Show the intermediate steps.

03 / Constructed response

Evaluate 5 + 7 × 5. Show the intermediate steps.

04 / Constructed response

Evaluate (6 + 7) × 6^2. Show the intermediate steps.

05 / Constructed response

Evaluate (6 + 7) × 3^2. Show the intermediate steps.

06 / Constructed response

Evaluate (3 + 3) × 4^2. Show the intermediate steps.

07 / Constructed response

Evaluate 6 ÷ 3 × 6 + 3^2. Show the intermediate steps.

08 / Numeric entry and error analysis

Evaluate $2^2 + 4(7 - 3)$. Part A: enter the value. Part B: a student begins by adding 2^2 and 4; explain why that changes the expression.

09 / Numeric entry and error analysis

Evaluate $6^2 + 5(5 - 3)$. Part A: enter the value. Part B: a student begins by adding 6^2 and 5; explain why that changes the expression.

10 / Constructed response

Evaluate 30 ÷ 6 × 6 + 6^2. Show the intermediate steps.

Day 048 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Find Greatest Common Factors

Standards: 6.NS.B.4a

Learning target: I can find the greatest common factor and interpret equal groups.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

01 / Learn

List factor pairs or use prime factorizations. The greatest factor shared by both numbers is the GCF. For identical kits with no leftovers, the GCF gives the greatest number of kits.

Worked example 1

Find the GCF of 6 and 9. Show factor pairs or prime factors.

1. List factor pairs or use prime factorizations.

2. The greatest factor shared by both numbers is the GCF.

3. Common factors: 1, 3

4. greatest is 3.

5. State the result in the form requested: 3.

Answer: 3

Worked example 2

Select all true statements about 36 and 54. Explain why the greatest factor fits the kit situation.
A. 90 divides both counts.
B. 19 is their greatest common factor.
C. 18 divides both 36 and 54.
D. 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

1. List factor pairs or use prime factorizations.

2. GCF(36, 54) = 18.

3. Divide each count by 18: 2 pencils and 3 erasers per kit.

4. A larger shared divisor is impossible.

Answer: C, D - 18 divides both 36 and 54.; 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

Day 048 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the GCF of 8 and 12. Show factor pairs or prime factors.

02 / Constructed response

Find the GCF of 10 and 15. Show factor pairs or prime factors.

03 / Constructed response

Find the GCF of 12 and 16. Show factor pairs or prime factors.

04 / Constructed response

Find the GCF of 45 and 60. Show factor pairs or prime factors.

05 / Constructed response

Find the GCF of 70 and 100. Show factor pairs or prime factors.

06 / Constructed response

Find the GCF of 28 and 42. Show factor pairs or prime factors.

07 / Constructed response

Make the greatest possible number of identical kits from 24 pencils and 36 erasers with no leftovers. Find the number of kits and each kit’s contents.

08 / Multiple select

Select all true statements about 18 and 27. Explain why the greatest factor fits the kit situation.
A. 9 is the largest number of identical no-leftover kits from 18 pencils and 27 erasers.
B. 10 is their greatest common factor.
C. 9 divides both 18 and 27.
D. 45 divides both counts.

09 / Multiple select

Select all true statements about 30 and 45. Explain why the greatest factor fits the kit situation.
A. 15 is the largest number of identical no-leftover kits from 30 pencils and 45 erasers.
B. 15 divides both 30 and 45.
C. 16 is their greatest common factor.
D. 75 divides both counts.

10 / Constructed response

Make the greatest possible number of identical kits from 18 pencils and 30 erasers with no leftovers. Find the number of kits and each kit’s contents.

Day 049 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Find Least Common Multiples

Standards: 6.NS.B.4b

Learning target: I can find the least common multiple of two numbers no greater than 12.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

01 / Learn

List positive multiples of both numbers. The smallest shared multiple is the LCM. Repeating events that begin together next occur together after this many time units.

Worked example 1

Find the LCM of 2 and 3 by listing multiples.

1. List positive multiples of both numbers.

2. The smallest shared multiple is the LCM.

3. The first positive multiple shared by 2 and 3 is 6.

4. State the result in the form requested: 6.

Answer: 6

Worked example 2

Two signals flash together now and then every 9 and 2 seconds. Select all true statements.
A. The next shared time is 19 seconds.
B. They next flash together after 18 seconds.
C. 36 seconds is also a time they flash together.
D. They next flash together after 1 seconds.

1. List positive multiples of both numbers.

2. List multiples or use prime factors.

3. The least shared positive multiple is 18

4. multiples of 18 are later shared times.

Answer: B, C - They next flash together after 18 seconds.; 36 seconds is also a time they flash together.

Day 049 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the LCM of 2 and 4 by listing multiples.

02 / Constructed response

Find the LCM of 4 and 6 by listing multiples.

03 / Constructed response

Find the LCM of 2 and 6 by listing multiples.

04 / Constructed response

Find the LCM of 6 and 2 by listing multiples.

05 / Constructed response

Find the LCM of 9 and 11 by listing multiples.

06 / Constructed response

Find the LCM of 7 and 9 by listing multiples.

07 / Constructed response

Two lights flash together now, then every 8 and 2 seconds. When do they next flash together? Explain.

08 / Multiple select

Two signals flash together now and then every 6 and 8 seconds. Select all true statements.
A. The next shared time is 25 seconds.
B. They next flash together after 24 seconds.
C. They next flash together after 2 seconds.
D. 48 seconds is also a time they flash together.

09 / Multiple select

Two signals flash together now and then every 10 and 9 seconds. Select all true statements.
A. They next flash together after 90 seconds.
B. They next flash together after 1 seconds.
C. 180 seconds is also a time they flash together.
D. The next shared time is 91 seconds.

10 / Constructed response

Two lights flash together now, then every 12 and 5 seconds. When do they next flash together? Explain.

Day 050 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Use Factors and Multiples to Explain Structure

Standards: 6.NS.B.4a, 6.NS.B.4b

Learning target: I can use Factors and Multiples to Explain Structure, and explain and check my reasoning.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

01 / Learn

List factor pairs or use prime factorizations. The greatest factor shared by both numbers is the GCF. For identical kits with no leftovers, the GCF gives the greatest number of kits. List positive multiples of both numbers. The smallest shared multiple is the LCM. Repeating events that begin together next occur together after this many time units.

Worked example 1

Find the GCF of 8 and 12. Show factor pairs or prime factors.

1. List factor pairs or use prime factorizations.

2. The greatest factor shared by both numbers is the GCF.

3. Common factors: 1, 2, 4

4. greatest is 4.

5. State the result in the form requested: 4.

Answer: 4

Worked example 2

Two signals flash together now and then every 2 and 9 seconds. Select all true statements.
A. They next flash together after 18 seconds.
B. They next flash together after 1 seconds.
C. 36 seconds is also a time they flash together.
D. The next shared time is 19 seconds.

1. List positive multiples of both numbers.

2. List multiples or use prime factors.

3. The least shared positive multiple is 18

4. multiples of 18 are later shared times.

Answer: A, C - They next flash together after 18 seconds.; 36 seconds is also a time they flash together.

Day 050 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find the GCF of 12 and 16. Show factor pairs or prime factors.

02 / Constructed response

Find the LCM of 5 and 4 by listing multiples.

03 / Constructed response

Find the GCF of 6 and 9. Show factor pairs or prime factors.

04 / Constructed response

Find the LCM of 12 and 6 by listing multiples.

05 / Constructed response

Find the GCF of 60 and 84. Show factor pairs or prime factors.

06 / Constructed response

Find the LCM of 3 and 8 by listing multiples.

07 / Constructed response

Make the greatest possible number of identical kits from 70 pencils and 100 erasers with no leftovers. Find the number of kits and each kit’s contents.

08 / Multiple select

Two signals flash together now and then every 3 and 11 seconds. Select all true statements.
A. 66 seconds is also a time they flash together.
B. They next flash together after 1 seconds.
C. The next shared time is 34 seconds.
D. They next flash together after 33 seconds.

09 / Multiple select

Select all true statements about 36 and 54. Explain why the greatest factor fits the kit situation.
A. 90 divides both counts.
B. 19 is their greatest common factor.
C. 18 divides both 36 and 54.
D. 18 is the largest number of identical no-leftover kits from 36 pencils and 54 erasers.

10 / Constructed response

Two lights flash together now, then every 8 and 3 seconds. When do they next flash together? Explain.

Day 051 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Factor Sums with the Distributive Property

Standards: 6.NS.B.4c

Learning target: I can factor a sum using the greatest common factor.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 1

01 / Learn

Find the GCF of the two addends. Divide each addend by that factor and place the quotients inside parentheses. The inside numbers should share no common factor greater than 1.

Worked example 1

Factor 8 + 12 using its GCF. Make the inside numbers relatively prime.

1. Find the GCF of the two addends.

2. Divide each addend by that factor and place the quotients inside parentheses.

3. GCF = 4

4. 8 = 4 × 2 and 12 = 4 × 3.

5. State the result in the form requested: 4(2 + 3).

Answer: 4(2 + 3)

Worked example 2

Select all expressions equal to 36 + 18.
A. 6(6 + 3)
B. 55
C. 6 × 10
D. 36 + 18

1. Find the GCF of the two addends.

2. Distribute 6 to both terms: 6 × 6 + 6 × 3 = 54.

3. The other choices are 6 or 1 too large.

Answer: A, D - 6(6 + 3); 36 + 18

Day 051 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Factor 6 + 9 using its GCF. Make the inside numbers relatively prime.

02 / Constructed response

Factor 10 + 15 using its GCF. Make the inside numbers relatively prime.

03 / Constructed response

Factor 12 + 16 using its GCF. Make the inside numbers relatively prime.

04 / Constructed response

Factor 28 + 42 using its GCF. Make the inside numbers relatively prime.

05 / Constructed response

Factor 48 + 72 using its GCF. Make the inside numbers relatively prime.

06 / Constructed response

Factor 12 + 18 using its GCF. Make the inside numbers relatively prime.

07 / Constructed response

Factor 18 + 30 using its GCF. Make the inside numbers relatively prime.

08 / Multiple select

Select all expressions equal to 6 + 6.
A. 13
B. 2(3 + 3)
C. 2 × 7
D. 6 + 6

09 / Multiple select

Select all expressions equal to 25 + 15.
A. 5 × 9
B. 5(5 + 3)
C. 41
D. 25 + 15

10 / Constructed response

Factor 70 + 100 using its GCF. Make the inside numbers relatively prime.

Day 052 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Identify Parts of Expressions

Standards: 6.EE.A.2b

Learning target: I can name terms, coefficients, factors, and grouped parts.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Addition and subtraction separate terms. A coefficient multiplies a variable; a constant has no variable. A parenthesized sum can be one factor of a larger product, so describe both the inside and the whole expression.

Worked example 1

For 4x + 7y + 3, list terms, coefficients, and the constant.

1. Addition and subtraction separate terms.

2. A coefficient multiplies a variable; a constant has no variable.

3. Terms are separated by addition

4. coefficients multiply variables.

5. State the result in the form requested: Terms 4x, 7y, 3; coefficients 4, 7; constant 3.

Answer: Terms 4x, 7y, 3; coefficients 4, 7; constant 3

Worked example 2

For the expression 6(x + 2) + 4, select all true statements.
A. 6 and (x + 2) are factors of the first term.
B. 2 is the coefficient of x in the fully expanded expression.
C. 4 is a constant term.
D. The entire expression is a product with 4.

1. Addition and subtraction separate terms.

2. The outer sum has two terms.

3. The first term is a product with two factors.

4. Expanding gives 6x + 12 + 4, so the coefficient is 6.

Answer: A, C - 6 and (x + 2) are factors of the first term.; 4 is a constant term.

Day 052 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

For 4x + 4y + 6, list terms, coefficients, and the constant.

02 / Constructed response

For 3x + 7y + 5, list terms, coefficients, and the constant.

03 / Constructed response

For (6x + 7) ÷ 3, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

04 / Constructed response

For 2x + 5y + 2, list terms, coefficients, and the constant.

05 / Constructed response

For (4x + 4) ÷ 4, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

06 / Constructed response

For 6x + 5y + 5, list terms, coefficients, and the constant.

07 / Constructed response

For (2x + 3) ÷ 5, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

08 / Multiple select

For the expression 7(x + 4) + 2, select all true statements.
A. The entire expression is a product with 2.
B. 2 is a constant term.
C. 4 is the coefficient of x in the fully expanded expression.
D. 7 and (x + 4) are factors of the first term.

09 / Multiple select

For the expression 3(x + 4) + 3, select all true statements.
A. 4 is the coefficient of x in the fully expanded expression.
B. 3 is a constant term.
C. 3 and (x + 4) are factors of the first term.
D. The entire expression is a product with 3.

10 / Constructed response

In 7(x + 4), identify the outer operation, both factors, and the operation inside the grouped factor.

Day 053 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Write Expressions from Words

Standards: 6.EE.A.2a, 6.EE.B.6

Learning target: I can translate words and situations into expressions.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Define the variable first. Use parentheses for a sum or difference that must stay together. Read subtraction carefully: five less than n means n - 5, while five minus n means 5 - n.

Worked example 1

Write an expression for 2 less than n.

1. Define the variable first.

2. Use parentheses for a sum or difference that must stay together.

3. Start with n, then subtract the named amount.

4. State the result in the form requested: n - 2.

Answer: n - 2

Worked example 2

A student buys x markers at $4 each and y folders at $5 each, then uses one $3 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

1. Define the variable first.

2. Marker cost is 4x

3. folder cost is 5y.

4. Add these subtotals, then subtract the one fixed coupon of 3.

5. An expression does not need an equals sign.

Answer: 4x + 5y - 3

Day 053 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write an expression for 7 less than n.

02 / Constructed response

Write an expression for 5 less than n.

03 / Constructed response

Write an expression for 8 less than n.

04 / Constructed response

Write an expression for 6 times the sum of x and 4.

05 / Constructed response

Write an expression for 7 times the sum of x and 6.

06 / Constructed response

Write an expression for 3 times the sum of x and 7.

07 / Constructed response

A ticket costs $4. A group buys n tickets and pays a single $7 booking fee. Define n and write the total-cost expression. Evaluate it for n = 4.

08 / Expression entry and reasoning

A student buys x markers at $6 each and y folders at $4 each, then uses one $3 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

09 / Expression entry and reasoning

A student buys x markers at $4 each and y folders at $2 each, then uses one $3 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

10 / Constructed response

A ticket costs $7. A group buys n tickets and pays a single $3 booking fee. Define n and write the total-cost expression. Evaluate it for n = 3.

Day 054 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Evaluate Expressions and Formulas

Standards: 6.EE.A.2c

Learning target: I can substitute values and evaluate expressions or formulas.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units.

Worked example 1

Evaluate 4x + 4 for x = 6.

1. Replace every variable with its assigned value, using parentheses.

2. Apply the order of operations.

3. Substitute first: 4 × 6 + 4 = 28.

4. State the result in the form requested: 28.

Answer: 28

Worked example 2

Let x = 5/2 and y = 5. Evaluate 4x + y^2. Part A: enter the value. Part B: show substitution and the order of operations.

1. Replace every variable with its assigned value, using parentheses.

2. Substitute: 4 × (5/2) + 5^2.

3. The product is 10 and the square is 25.

4. Add to get 35.

Answer: 35

Day 054 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Evaluate 8x + 5 for x = 2.

02 / Constructed response

Evaluate 8x + 6 for x = 2.

03 / Constructed response

Evaluate 4x + 6 for x = 3.

04 / Constructed response

Evaluate x^2 + 5y for x = 6 and y = 4.

05 / Constructed response

Evaluate x^2 + 3y for x = 3 and y = 3.

06 / Constructed response

Evaluate x^2 + 2y for x = 4 and y = 3.

07 / Constructed response

A rectangle has length x + 6 and width 4. Use A = lw to write its area expression and find A when x = 6.

08 / Numeric entry and reasoning

Let x = 3/2 and y = 5. Evaluate 3x + y^2. Part A: enter the value. Part B: show substitution and the order of operations.

09 / Numeric entry and reasoning

Let x = 1 and y = 5. Evaluate 7x + y^2. Part A: enter the value. Part B: show substitution and the order of operations.

10 / Constructed response

A rectangle has length x + 6 and width 6. Use A = lw to write its area expression and find A when x = 2.

Day 055 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Connect Distribution and Expression Structure

Standards: 6.EE.A.3, 6.EE.A.2c

Learning target: I can connect Distribution and Expression Structure, and explain and check my reasoning.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Multiply the factor outside parentheses by every term inside. The two expressions have the same value for every allowed input. An area model can show how one rectangle splits into smaller rectangles. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units.

Worked example 1

Expand 6(x + 4).

1. Multiply the factor outside parentheses by every term inside.

2. The two expressions have the same value for every allowed input.

3. Multiply the outside factor by every inside term, including the constant.

4. State the result in the form requested: 6x + 24.

Answer: 6x + 24

Worked example 2

Let x = 2 and y = 5. Evaluate 4x + y². Part A: enter the value. Part B: show substitution and the order of operations.

1. Replace every variable with its assigned value, using parentheses.

2. Substitute: 4 × (2) + 5².

3. The product is 8 and the square is 25.

4. Add to get 33.

Answer: 33

Day 055 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Expand 2(x + 6).

02 / Constructed response

Evaluate 8x + 4 for x = 2.

03 / Constructed response

Expand 5(x + 2).

04 / Constructed response

Evaluate x^2 + 8y for x = 4 and y = 5.

05 / Constructed response

Expand 7(6x + 3).

06 / Constructed response

Evaluate x^2 + 2y for x = 4 and y = 2.

07 / Constructed response

Expand 7(7x + 6). Verify your result at x = 2.

08 / Numeric entry and reasoning

Let x = 1 and y = 4. Evaluate 3x + y². Part A: enter the value. Part B: show substitution and the order of operations.

09 / Multiple select

Select all expressions equivalent to 3(x + 2).
A. 6 + 3x
B. 5x
C. 3x + 6
D. 3x + 2

10 / Constructed response

A rectangle has length x + 3 and width 6. Use A = lw to write its area expression and find A when x = 5.

Day 056 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Generate Equivalent Expressions

Standards: 6.EE.A.3

Learning target: I can use distribution to generate an equivalent expression.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Multiply the factor outside parentheses by every term inside. The two expressions have the same value for every allowed input. An area model can show how one rectangle splits into smaller rectangles.

Worked example 1

Expand 5(x + 3).

1. Multiply the factor outside parentheses by every term inside.

2. The two expressions have the same value for every allowed input.

3. Multiply the outside factor by every inside term, including the constant.

4. State the result in the form requested: 5x + 15.

Answer: 5x + 15

Worked example 2

Select all expressions equivalent to 6(x + 6).
A. 6x + 6
B. 12x
C. 6x + 36
D. 36 + 6x

1. Multiply the factor outside parentheses by every term inside.

2. Distribute 6 to both terms to get 6x + 36.

3. Reversing the addends gives the same value.

4. The other expressions fail, for example at x = 0.

Answer: C, D - 6x + 36; 36 + 6x

Day 056 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Expand 2(x + 6).

02 / Constructed response

Expand 3(x + 6).

03 / Constructed response

Expand 3(x + 7).

04 / Constructed response

Expand 2(4x + 4).

05 / Constructed response

Expand 3(4x + 2).

06 / Constructed response

Expand 7(7x + 6).

07 / Constructed response

Expand 2(6x + 5). Verify your result at x = 2.

08 / Multiple select

Select all expressions equivalent to 3(x + 4).
A. 12 + 3x
B. 7x
C. 3x + 4
D. 3x + 12

09 / Multiple select

Select all expressions equivalent to 4(x + 3).
A. 12 + 4x
B. 7x
C. 4x + 3
D. 4x + 12

10 / Constructed response

Expand 5(7x + 2). Verify your result at x = 2.

Day 057 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Combine Like Terms

Standards: 6.EE.A.3

Learning target: I can combine terms with the same variable part.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Terms with the same variable part count the same kind of object. Add their coefficients. Keep constants separate from variable terms and distribute before combining when parentheses are present.

Worked example 1

Simplify 8x + 4x. Explain which terms combine.

1. Terms with the same variable part count the same kind of object.

2. Add their coefficients.

3. Add coefficients of x terms

4. add only constants to constants.

5. Distribute first if needed.

6. State the result in the form requested: 12x.

Answer: 12x

Worked example 2

Select all expressions equivalent to 6x + 6x + 3.
A. 12x + 3
B. 12x + 4
C. 15x
D. (6 + 6)x + 3

1. Terms with the same variable part count the same kind of object.

2. Combine coefficients of like x terms: 6 + 6 = 12.

3. The constant 3 stays separate.

Answer: A, D - 12x + 3; (6 + 6)x + 3

Day 057 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Simplify 6x + 3x. Explain which terms combine.

02 / Constructed response

Simplify 4x + 3x. Explain which terms combine.

03 / Constructed response

Simplify 3x + 4x. Explain which terms combine.

04 / Constructed response

Simplify 4x + 3 + 4x + 4. Explain which terms combine.

05 / Constructed response

Simplify 2x + 6 + 7x + 2. Explain which terms combine.

06 / Constructed response

Simplify 4x + 3 + 3x + 4. Explain which terms combine.

07 / Constructed response

Simplify 2(x + 5) + 3x. Explain which terms combine.

08 / Multiple select

Select all expressions equivalent to 7x + 5x + 4.
A. (7 + 5)x + 4
B. 12x + 5
C. 16x
D. 12x + 4

09 / Multiple select

Select all expressions equivalent to 4x + 4x + 4.
A. 12x
B. 8x + 4
C. (4 + 4)x + 4
D. 8x + 5

10 / Constructed response

Simplify 7(x + 2) + 7x. Explain which terms combine.

Day 058 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Identify Equivalent Expressions

Standards: 6.EE.A.4

Learning target: I can justify whether expressions are equivalent.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

01 / Learn

Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent.

Worked example 1

Are 6(x + 7) and 6x + 42 equivalent for every x? Use a property or a counterexample.

1. Use operation properties to rewrite an expression.

2. Matching values at one input do not prove equivalence for all inputs.

3. 6(x + 7) = 6x + 42.

4. This matches.

5. State the result in the form requested: Yes; distribution gives 6x + 42.

Answer: Yes; distribution gives 6x + 42

Worked example 2

Select all expressions equivalent to 6(x + 3).
A. 9x
B. 6x + 18
C. 6x + 3
D. 18 + 6x

1. Use operation properties to rewrite an expression.

2. Distribute 6 to both terms to get 6x + 18.

3. Reversing the addends gives the same value.

4. The other expressions fail, for example at x = 0.

Answer: B, D - 6x + 18; 18 + 6x

Day 058 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Are 7(x + 2) and 7x + 2 equivalent for every x? Use a property or a counterexample.

02 / Constructed response

Are 2(x + 5) and 2x + 10 equivalent for every x? Use a property or a counterexample.

03 / Constructed response

Are 8(x + 7) and 8x + 7 equivalent for every x? Use a property or a counterexample.

04 / Constructed response

Are 8(x + 2) and 8x + 16 equivalent for every x? Use a property or a counterexample.

05 / Constructed response

Are 3(x + 2) and 3x + 2 equivalent for every x? Use a property or a counterexample.

06 / Constructed response

Are 2(x + 7) and 2x + 14 equivalent for every x? Use a property or a counterexample.

07 / Constructed response

Are 6(x + 6) and 6x + 6 equivalent for every x? Use a property or a counterexample.

08 / Multiple select

Select all expressions equivalent to 7(x + 4).
A. 11x
B. 7x + 28
C. 7x + 4
D. 28 + 7x

09 / Multiple select

Select all expressions equivalent to 4(x + 6).
A. 10x
B. 4x + 6
C. 4x + 24
D. 24 + 4x

10 / Constructed response

Are 3(x + 6) and 3x + 18 equivalent for every x? Use a property or a counterexample.

Day 059 | Unit 04 | Lesson

Numeric and Algebraic Expressions

Apply Expression Structure

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4c, 6.EE.A.4

Learning target: I can apply Expression Structure, and explain and check my reasoning.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapters 5, 1

01 / Learn

Work inside parentheses, then evaluate exponents. Multiply and divide from left to right; then add and subtract from left to right. Multiplication does not automatically come before division. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units. Find the GCF of the two addends. Divide each addend by that factor and place the quotients inside parentheses. The inside numbers should share no common factor greater than 1. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent.

Worked example 1

Evaluate 8 + 3 × 6. Show the intermediate steps.

1. Work inside parentheses, then evaluate exponents.

2. Multiply and divide from left to right; then add and subtract from left to right.

3. Use parentheses and exponents first, then × and ÷ left to right, then + and -.

4. Result: 26.

5. State the result in the form requested: 26.

Answer: 26

Worked example 2

Select all expressions equivalent to 7(x + 4).
A. 11x
B. 7x + 28
C. 7x + 4
D. 28 + 7x

1. Use operation properties to rewrite an expression.

2. Distribute 7 to both terms to get 7x + 28.

3. Reversing the addends gives the same value.

4. The other expressions fail, for example at x = 0.

Answer: B, D - 7x + 28; 28 + 7x

Day 059 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Evaluate 2 + 5 × 5. Show the intermediate steps.

02 / Constructed response

Evaluate 4x + 3 for x = 2.

03 / Constructed response

Factor 8 + 12 using its GCF. Make the inside numbers relatively prime.

04 / Constructed response

Are 7(x + 3) and 7x + 21 equivalent for every x? Use a property or a counterexample.

05 / Constructed response

Evaluate (8 + 6) × 3^2. Show the intermediate steps.

06 / Constructed response

Evaluate x^2 + 8y for x = 5 and y = 7.

07 / Constructed response

Factor 32 + 48 using its GCF. Make the inside numbers relatively prime.

08 / Multiple select

Select all expressions equivalent to 3(x + 2).
A. 6 + 3x
B. 5x
C. 3x + 6
D. 3x + 2

09 / Numeric entry and error analysis

Evaluate $4^2 + 2(8 - 3)$. Part A: enter the value. Part B: a student begins by adding 4^2 and 2; explain why that changes the expression.

10 / Constructed response

A rectangle has length x + 4 and width 7. Use A = lw to write its area expression and find A when x = 4.

Day 060 | Unit 04 | Unit Test

Numeric and Algebraic Expressions

Unit 4 Test

Standards: 6.NS.C.5, 6.NS.C.6a, 6.NS.C.6c, 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.NS.B.4c, 6.EE.A.2b, 6.EE.A.2a, 6.EE.B.6, 6.EE.A.3, 6.EE.A.4

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 3: Numeric and Algebraic Expressions | Spectrum companion Chapter 5

Directions

The first 4 questions (20%) review Unit 3. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

A marker is 28 m above sea level. Write its signed elevation and explain what elevation 0 means.

02 / Previous unit review | Constructed response

Find the opposite of -8. Show both values on a number line.

03 / Previous unit review | Constructed response

Draw a horizontal or vertical number line. Plot and label -8, 0, and 7.

04 / Previous unit review | Constructed response

Draw a horizontal or vertical number line. Plot and label -5, 0, and 3.

05 / Constructed response

Write 4^2 as repeated multiplication and evaluate.

06 / Constructed response

Evaluate 5 + 4 × 4. Show the intermediate steps.

07 / Constructed response

Find the GCF of 10 and 15. Show factor pairs or prime factors.

08 / Constructed response

Find the LCM of 8 and 9 by listing multiples.

09 / Constructed response

Find the GCF of 24 and 36. Show factor pairs or prime factors.

10 / Constructed response

Factor 28 + 42 using its GCF. Make the inside numbers relatively prime.

11 / Constructed response

For (5x + 7) ÷ 2, name the outer operation, the dividend as a single entity, and the divisor. Identify the sum inside the parentheses.

12 / Expression entry and reasoning

A student buys x markers at $4 each and y folders at $3 each, then uses one $2 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

13 / Numeric entry and reasoning

Let x = 1 and y = 3. Evaluate $4x + y^2$. Part A: enter the value. Part B: show substitution and the order of operations.

14 / Multiple select

Select all expressions equivalent to 7(x + 6).
A. 42 + 7x
B. 13x
C. 7x + 6
D. 7x + 42

15 / Multiple select

Select all expressions equivalent to 6(x + 5).
A. 11x
B. 6x + 5
C. 30 + 6x
D. 6x + 30

16 / Multiple select

Select all expressions equivalent to 2x + 5x + 4.
A. 7x + 4
B. (2 + 5)x + 4
C. 7x + 5
D. 11x

17 / Multiple select

Select all expressions equivalent to 4(x + 6).
A. 10x
B. 4x + 6
C. 4x + 24
D. 24 + 4x

18 / Numeric entry and reasoning

Let x = 3/2 and y = 2. Evaluate 5x + y^2. Part A: enter the value. Part B: show substitution and the order of operations.

19 / Multiple select

Select all expressions equal to 6^3.
A. 6 × 3
B. 6 × 6 × 6
C. 217
D. 216

20 / Numeric entry and error analysis

Evaluate 3^2 + 3(6 - 3). Part A: enter the value. Part B: a student begins by adding 3^2 and 3; explain why that changes the expression.

Day 061 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Test Solutions by Substitution

Standards: 6.EE.B.5

Learning target: I can test a proposed solution by substitution.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Replace the variable with the proposed value, compute each side, and compare. A number is a solution only when it makes the original equation true.

Worked example 1

Does x = 4 solve 3x + 6 = 18? Show substitution.

1. Replace the variable with the proposed value, compute each side, and compare.

2. A number is a solution only when it makes the original equation true.

3. 3 × 4 + 6 = 18

4. compare with 18.

5. State the result in the form requested: Yes; left side = 18.

Answer: Yes; left side = 18

Worked example 2

Select all values of x from the list that make x + 6 > 8 true. Explain why each rejected value fails.
A. 3
B. 1
C. 2
D. 4

1. Replace the variable with the proposed value, compute each side, and compare.

2. Subtract 6 conceptually: x must exceed 2.

3. Equality at 2 fails because the inequality is strict.

Answer: A, D - 3; 4

Day 061 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Does x = 6 solve 4x + 5 = 25? Show substitution.

02 / Constructed response

Does x = 5 solve 3x + 2 = 17? Show substitution.

03 / Constructed response

Does x = 4 solve 3x + 6 = 15? Show substitution.

04 / Constructed response

Does x = 4 solve 8x + 5 = 37? Show substitution.

05 / Constructed response

Does x = 5 solve 8x + 2 = 34? Show substitution.

06 / Constructed response

Does x = 3 solve 5x + 6 = 21? Show substitution.

07 / Constructed response

Does x = 4 solve 5x + 5 = 20? Show substitution.

08 / Multiple select

Select all values of x from the list that make x + 3 > 5 true. Explain why each rejected value fails.
A. 3
B. 4
C. 1
D. 2

09 / Multiple select

Select all values of x from the list that make x + 6 > 10 true. Explain why each rejected value fails.
A. 4
B. 6
C. 3
D. 5

10 / Constructed response

Does x = 2 solve 7x + 7 = 21? Show substitution.

Day 062 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Solve Addition Equations

Standards: 6.EE.B.7

Learning target: I can solve x + p = q and check.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Subtract p from both sides to preserve equality. Then substitute the answer into the original equation. The value of the left side must equal the right side.

Worked example 1

Solve x + 6 = 12. Check in the original equation.

1. Subtract p from both sides to preserve equality.

2. Then substitute the answer into the original equation.

3. Subtract 6 on both sides.

4. Then x = 6

5. substitution gives 12 on each side.

6. State the result in the form requested: x = 6.

Answer: x = 6

Worked example 2

Solve x + 1 = 11/2. Select every representation of the solution.
A. 18/4
B. 13/2
C. 9/2
D. 11/2

1. Subtract p from both sides to preserve equality.

2. Use the inverse operation on both sides.

3. The solution is 9/2

4. the doubled numerator and denominator name the same value.

5. Substitution verifies the solution.

Answer: A, C - 18/4; 9/2

Day 062 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Solve x + 2 = 4. Check in the original equation.

02 / Constructed response

Solve x + 5 = 9. Check in the original equation.

03 / Constructed response

Solve x + 5 = 8. Check in the original equation.

04 / Constructed response

Solve x + 8 = 19/2. Check in the original equation.

05 / Constructed response

Solve x + 3 = 4. Check in the original equation.

06 / Constructed response

Solve x + 8 = 21/2. Check in the original equation.

07 / Constructed response

Solve x + 2 = 7/2. Check in the original equation.

08 / Multiple select

Solve x + 2 = 15/2. Select every representation of the solution.
A. 15/2
B. 11/2
C. 22/4
D. 13/2

09 / Multiple select

Solve x + 2 = 17/2. Select every representation of the solution.
A. 26/4
B. 15/2
C. 13/2
D. 17/2

10 / Constructed response

Solve x + 4 = 13/2. Check in the original equation.

Day 063 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Solve Subtraction Equations

Standards: 6.EE.B.7

Learning target: I can solve x - p = q and check.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Add p to both sides to undo subtraction. Check by subtracting p from your answer. Values in this lesson are nonnegative rational numbers.

Worked example 1

Solve x - 4 = 3. Check in the original equation.

1. Add p to both sides to undo subtraction.

2. Check by subtracting p from your answer.

3. Add 4 on both sides.

4. Then x = 7

5. substitution gives 3 on each side.

6. State the result in the form requested: x = 7.

Answer: x = 7

Worked example 2

Solve x - 2 = 3/2. Select every representation of the solution.
A. 14/4
B. 7/2
C. 11/2
D. 9/2

1. Add p to both sides to undo subtraction.

2. Use the inverse operation on both sides.

3. The solution is 7/2

4. the doubled numerator and denominator name the same value.

5. Substitution verifies the solution.

Answer: A, B - 14/4; 7/2

Day 063 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Solve x - 4 = 1. Check in the original equation.

02 / Constructed response

Solve x - 4 = 2. Check in the original equation.

03 / Constructed response

Solve x - 5 = 1. Check in the original equation.

04 / Constructed response

Solve x - 3 = 1/2. Check in the original equation.

05 / Constructed response

Solve x - 3 = 5/2. Check in the original equation.

06 / Constructed response

Solve x - 6 = 3. Check in the original equation.

07 / Constructed response

Solve x - 3/2 = 1/2. Check in the original equation.

08 / Multiple select

Solve x - 5/2 = 0. Select every representation of the solution.
A. 7/2
B. 9/2
C. 5/2
D. 10/4

09 / Multiple select

Solve x - 3/2 = 5. Select every representation of the solution.
A. 17/2
B. 26/4
C. 15/2
D. 13/2

10 / Constructed response

Solve x - 1 = 2. Check in the original equation.

Day 064 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Solve Multiplication Equations

Standards: 6.EE.B.7

Learning target: I can solve px = q using division.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Divide both sides by the nonzero coefficient p. Keep the equation balanced. Check by multiplying the proposed solution by p.

Worked example 1

Solve 8 × x = 16. Check in the original equation.

1. Divide both sides by the nonzero coefficient p.

2. Keep the equation balanced.

3. Divide by 8 on both sides.

4. Then x = 2

5. substitution gives 16 on each side.

6. State the result in the form requested: x = 2.

Answer: x = 2

Worked example 2

Solve 3/2 × x = 27/4. Select every representation of the solution.
A. 9/2
B. 11/2
C. 13/2
D. 18/4

1. Divide both sides by the nonzero coefficient p.

2. Use the inverse operation on both sides.

3. The solution is 9/2

4. the doubled numerator and denominator name the same value.

5. Substitution verifies the solution.

Answer: A, D - 9/2; 18/4

Day 064 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Solve 5 × x = 20. Check in the original equation.

02 / Constructed response

Solve 2 × x = 6. Check in the original equation.

03 / Constructed response

Solve 2 × x = 4. Check in the original equation.

04 / Constructed response

Solve 4 × x = 14. Check in the original equation.

05 / Constructed response

Solve 4 × x = 10. Check in the original equation.

06 / Constructed response

Solve 6 × x = 6. Check in the original equation.

07 / Constructed response

Solve 1 × x = 3. Check in the original equation.

08 / Multiple select

Solve 2 × x = 7. Select every representation of the solution.
A. 14/4
B. 9/2
C. 11/2
D. 7/2

09 / Multiple select

Solve 3/2 × x = 15/4. Select every representation of the solution.
A. 7/2
B. 5/2
C. 10/4
D. 9/2

10 / Constructed response

Solve 2 × x = 7. Check in the original equation.

Day 065 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Explain Balance and Inverse Operations

Standards: 6.EE.B.7

Learning target: I can explain Balance and Inverse Operations, and explain and check my reasoning.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Subtract p from both sides to preserve equality. Then substitute the answer into the original equation. The value of the left side must equal the right side. Add p to both sides to undo subtraction. Check by subtracting p from your answer. Values in this lesson are nonnegative rational numbers. Divide both sides by the nonzero coefficient p. Keep the equation balanced. Check by multiplying the proposed solution by p.

Worked example 1

Solve x + 7 = 14. Check in the original equation.

1. Subtract p from both sides to preserve equality.

2. Then substitute the answer into the original equation.

3. Subtract 7 on both sides.

4. Then x = 7

5. substitution gives 14 on each side.

6. State the result in the form requested: x = 7.

Answer: x = 7

Worked example 2

Solve 3/2 × x = 15/4. Select every representation of the solution.
A. 7/2
B. 5/2
C. 10/4
D. 9/2

1. Divide both sides by the nonzero coefficient p.

2. Use the inverse operation on both sides.

3. The solution is 5/2

4. the doubled numerator and denominator name the same value.

5. Substitution verifies the solution.

Answer: B, C - 5/2; 10/4

Day 065 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Solve x + 5 = 7. Check in the original equation.

02 / Constructed response

Solve x - 3 = 2. Check in the original equation.

03 / Constructed response

Solve 6 × x = 18. Check in the original equation.

04 / Constructed response

Solve x + 8 = 19/2. Check in the original equation.

05 / Constructed response

Solve x - 6 = 3/2. Check in the original equation.

06 / Constructed response

Solve 8 × x = 8. Check in the original equation.

07 / Constructed response

Solve x + 4 = 11/2. Check in the original equation.

08 / Multiple select

Solve x - 5/2 = 1. Select every representation of the solution.
A. 9/2
B. 7/2
C. 11/2
D. 14/4

09 / Multiple select

Solve 2 × x = 9. Select every representation of the solution.
A. 9/2
B. 13/2
C. 11/2
D. 18/4

10 / Constructed response

Solve x + 1 = 9/2. Check in the original equation.

Day 066 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Solve Division Equations

Standards: 6.EE.B.7

Learning target: I can solve x/p = q using multiplication.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Multiply both sides by the nonzero divisor p. Check by dividing the solution by p. Name the unknown quantity and attach its unit in a story problem.

Worked example 1

Solve x ÷ 7 = 1. Check in the original equation.

1. Multiply both sides by the nonzero divisor p.

2. Check by dividing the solution by p.

3. Multiply by 7 on both sides.

4. Then x = 7

5. substitution gives 1 on each side.

6. State the result in the form requested: x = 7.

Answer: x = 7

Worked example 2

Solve x ÷ 5/2 = 9/5. Select every representation of the solution.
A. 11/2
B. 18/4
C. 13/2
D. 9/2

1. Multiply both sides by the nonzero divisor p.

2. Use the inverse operation on both sides.

3. The solution is 9/2

4. the doubled numerator and denominator name the same value.

5. Substitution verifies the solution.

Answer: B, D - 18/4; 9/2

Day 066 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Solve x ÷ 5 = 3/5. Check in the original equation.

02 / Constructed response

Solve x ÷ 4 = 7/4. Check in the original equation.

03 / Constructed response

Solve x ÷ 8 = 7/8. Check in the original equation.

04 / Constructed response

Solve x ÷ 7 = 5/14. Check in the original equation.

05 / Constructed response

Solve x ÷ 6 = 5/12. Check in the original equation.

06 / Constructed response

Solve x ÷ 5 = 3/10. Check in the original equation.

07 / Constructed response

Solve x ÷ 1 = 7/2. Check in the original equation.

08 / Multiple select

Solve x ÷ 5/2 = 1. Select every representation of the solution.
A. 10/4
B. 7/2
C. 5/2
D. 9/2

09 / Multiple select

Solve x ÷ 1 = 13/2. Select every representation of the solution.
A. 17/2
B. 13/2
C. 26/4
D. 15/2

10 / Constructed response

Solve x ÷ 5/2 = 3/5. Check in the original equation.

Day 067 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Write One-Step Equations from Context

Standards: 6.EE.B.6, 6.EE.B.7

Learning target: I can model a situation with a one-step equation and solve it.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Define the unknown quantity. Translate the relationship into an equation before calculating. Use the inverse operation and check whether the answer makes sense in the situation.

Worked example 1

A collection grows by 6 cards to 42 cards. Let x be the starting count. Write and solve an equation.

1. Define the unknown quantity.

2. Translate the relationship into an equation before calculating.

3. Subtract 6 from both sides

4. 36 + 6 = 42.

5. State the result in the form requested: x + 6 = 42; x = 36 cards.

Answer: x + 6 = 42; x = 36 cards

Worked example 2

5 identical ropes have total length 15/2 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

1. Define the unknown quantity.

2. Total = number of ropes × one length.

3. Divide both sides by 5

4. 15/2 ÷ 5 = 3/2.

5. Check 5 × 3/2 = 15/2.

Answer: 5x = 15/2; x = 3/2 meters

Day 067 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A collection grows by 7 cards to 35 cards. Let x be the starting count. Write and solve an equation.

02 / Constructed response

A collection grows by 2 cards to 6 cards. Let x be the starting count. Write and solve an equation.

03 / Constructed response

A collection grows by 7 cards to 49 cards. Let x be the starting count. Write and solve an equation.

04 / Constructed response

7 identical lengths of rope total 14 m. Let x be the length of one piece. Write and solve an equation, then check.

05 / Constructed response

2 identical lengths of rope total 16 m. Let x be the length of one piece. Write and solve an equation, then check.

06 / Constructed response

5 identical lengths of rope total 35 m. Let x be the length of one piece. Write and solve an equation, then check.

07 / Constructed response

7 identical lengths of rope total 28 m. Let x be the length of one piece. Write and solve an equation, then check.

08 / Equation entry and two-part response

2 identical ropes have total length 3 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

09 / Equation entry and two-part response

3 identical ropes have total length 21/2 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

10 / Constructed response

6 identical lengths of rope total 18 m. Let x be the length of one piece. Write and solve an equation, then check.

Day 068 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Write and Graph Inequalities

Standards: 6.EE.B.8

Learning target: I can write and graph a constraint using an inequality.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions.

Worked example 1

Let x be any real number. Write and graph: x is more than 48. Label the endpoint and shade the solution ray.

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. Graph a strict bound with an open circle; use a closed circle for an included endpoint.

3. “more than” means >

4. exclude the endpoint.

5. State the result in the form requested: x > 48; open circle at 48, shade right.

Answer: x > 48; open circle at 48, shade right

Worked example 2

A club needs at least 35 members. Select all true statements for its membership count n.
A. 35 is an allowed membership count.
B. 34 is an allowed membership count.
C. n > 35
D. n ≥ 35

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. At least includes the boundary, so n ≥ 35.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: A, D - 35 is an allowed membership count.; n ≥ 35

Day 068 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Let x be any real number. Write and graph: x is at most 49. Label the endpoint and shade the solution ray.

02 / Constructed response

Let x be any real number. Write and graph: x is at least 42. Label the endpoint and shade the solution ray.

03 / Constructed response

Let x be any real number. Write and graph: x is at least 35. Label the endpoint and shade the solution ray.

04 / Constructed response

Let x be any real number. Write and graph: x is at most 16. Label the endpoint and shade the solution ray.

05 / Constructed response

Let x be any real number. Write and graph: x is more than 8. Label the endpoint and shade the solution ray.

06 / Constructed response

Let x be any real number. Write and graph: x is more than 25. Label the endpoint and shade the solution ray.

07 / Constructed response

Let x be any real number. Write and graph: x is at least 24. Give one included value and explain whether the endpoint is included.

08 / Multiple select

A club needs at least 10 members. Select all true statements for its membership count n.
A. n > 10
B. 10 is an allowed membership count.
C. 9 is an allowed membership count.
D. n ≥ 10

09 / Multiple select

A club needs at least 8 members. Select all true statements for its membership count n.
A. 7 is an allowed membership count.
B. n ≥ 8
C. 8 is an allowed membership count.
D. n > 8

10 / Constructed response

A museum group must have at least 25 participants. Let p be the participant count. Write an inequality and graph its real-number solution set. State which solutions make sense as participant counts.

Day 069 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Test Inequality Solutions

Standards: 6.EE.B.5, 6.EE.B.8

Learning target: I can test values and explain the many solutions of an inequality.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

Worked example 1

Which values in {5, 6, 7} satisfy x ≤ 6? Test each by substitution.

1. Substitute each value and decide whether the comparison is true.

2. A continuous number-line ray represents infinitely many real-number solutions.

3. Values below 6 work

4. the endpoint works too.

5. State the result in the form requested: 5, 6.

Answer: 5, 6

Worked example 2

A club needs at least 18 members. Select all true statements for its membership count n.
A. n ≥ 18
B. n > 18
C. 17 is an allowed membership count.
D. 18 is an allowed membership count.

1. Substitute each value and decide whether the comparison is true.

2. At least includes the boundary, so n ≥ 18.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: A, D - n ≥ 18; 18 is an allowed membership count.

Day 069 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Which values in {6, 7, 8} satisfy x < 7? Test each by substitution.

02 / Constructed response

Which values in {11, 12, 13} satisfy x ≤ 12? Test each by substitution.

03 / Constructed response

Which values in {14, 15, 16} satisfy x < 15? Test each by substitution.

04 / Constructed response

Which values in {10, 11, 12} satisfy x ≤ 11? Test each by substitution.

05 / Constructed response

Which values in {5, 6, 7} satisfy x < 6? Test each by substitution.

06 / Constructed response

Which values in {8, 9, 10} satisfy x ≤ 9? Test each by substitution.

07 / Constructed response

Which values in {8, 9, 10} satisfy x < 9? Does the real-number inequality have only these solutions? Explain.

08 / Multiple select

A club needs at least 6 members. Select all true statements for its membership count n.
A. n > 6
B. 5 is an allowed membership count.
C. n ≥ 6
D. 6 is an allowed membership count.

09 / Multiple select

A club needs at least 10 members. Select all true statements for its membership count n.
A. n > 10
B. 10 is an allowed membership count.
C. 9 is an allowed membership count.
D. n ≥ 10

10 / Constructed response

Which values in {10, 11, 12} satisfy x ≤ 11? Does the real-number inequality have only these solutions? Explain.

Day 070 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Connect Constraints and Solution Sets

Standards: 6.EE.B.8, 6.EE.B.5

Learning target: I can connect Constraints and Solution Sets, and explain and check my reasoning.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

Worked example 1

Let x be any real number. Write and graph: x is less than 36. Label the endpoint and shade the solution ray.

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. Graph a strict bound with an open circle; use a closed circle for an included endpoint.

3. “less than” means <

4. exclude the endpoint.

5. State the result in the form requested: x < 36; open circle at 36, shade left.

Answer: x < 36; open circle at 36, shade left

Worked example 2

A club needs at least 18 members. Select all true statements for its membership count n.
A. n ≥ 18
B. n > 18
C. 17 is an allowed membership count.
D. 18 is an allowed membership count.

1. Substitute each value and decide whether the comparison is true.

2. At least includes the boundary, so n ≥ 18.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: A, D - n ≥ 18; 18 is an allowed membership count.

Day 070 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Let x be any real number. Write and graph: x is at least 21. Label the endpoint and shade the solution ray.

02 / Constructed response

Which values in {9, 10, 11} satisfy x ≤ 10? Test each by substitution.

03 / Constructed response

Let x be any real number. Write and graph: x is more than 4. Label the endpoint and shade the solution ray.

04 / Constructed response

Which values in {11, 12, 13} satisfy x ≤ 12? Test each by substitution.

05 / Constructed response

Let x be any real number. Write and graph: x is more than 28. Label the endpoint and shade the solution ray.

06 / Constructed response

Which values in {7, 8, 9} satisfy x ≤ 8? Test each by substitution.

07 / Constructed response

Let x be any real number. Write and graph: x is at most 30. Give one included value and explain whether the endpoint is included.

08 / Multiple select

A club needs at least 28 members. Select all true statements for its membership count n.
A. 28 is an allowed membership count.
B. n ≥ 28
C. n > 28
D. 27 is an allowed membership count.

09 / Multiple select

A club needs at least 16 members. Select all true statements for its membership count n.
A. n ≥ 16
B. 15 is an allowed membership count.
C. n > 16
D. 16 is an allowed membership count.

10 / Constructed response

Which values in {4, 5, 6} satisfy x ≤ 5? Does the real-number inequality have only these solutions? Explain.

Day 071 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Interpret Endpoints and Infinitely Many Solutions

Standards: 6.EE.B.8, 6.EE.B.5

Learning target: I can interpret Endpoints and Infinitely Many Solutions, and explain and check my reasoning.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Use > or < for a strict bound and ≥ or ≤ when the endpoint is included. Graph a strict bound with an open circle; use a closed circle for an included endpoint. Shade the direction of all solutions. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers.

Worked example 1

Let x be any real number. Write and graph: x is at most 12. Label the endpoint and shade the solution ray.

1. Use > or < for a strict bound and ≥ or ≤ when the endpoint is included.

2. Graph a strict bound with an open circle; use a closed circle for an included endpoint.

3. “at most” means ≤

4. include the endpoint.

5. State the result in the form requested: x ≤ 12; closed circle at 12, shade left.

Answer: x ≤ 12; closed circle at 12, shade left

Worked example 2

A club needs at least 6 members. Select all true statements for its membership count n.
A. n > 6
B. 5 is an allowed membership count.
C. n ≥ 6
D. 6 is an allowed membership count.

1. Substitute each value and decide whether the comparison is true.

2. At least includes the boundary, so n ≥ 6.

3. On a real number line use a closed point and shade right

4. actual member counts are whole numbers.

Answer: C, D - n ≥ 6; 6 is an allowed membership count.

Day 071 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Let x be any real number. Write and graph: x is more than 24. Label the endpoint and shade the solution ray.

02 / Constructed response

Which values in {7, 8, 9} satisfy x ≤ 8? Test each by substitution.

03 / Constructed response

Let x be any real number. Write and graph: x is at least 8. Label the endpoint and shade the solution ray.

04 / Constructed response

Which values in {4, 5, 6} satisfy x ≤ 5? Test each by substitution.

05 / Constructed response

Let x be any real number. Write and graph: x is more than 14. Label the endpoint and shade the solution ray.

06 / Constructed response

Which values in {11, 12, 13} satisfy x ≤ 12? Test each by substitution.

07 / Constructed response

Let x be any real number. Write and graph: x is more than 24. Give one included value and explain whether the endpoint is included.

08 / Multiple select

A club needs at least 18 members. Select all true statements for its membership count n.
A. n ≥ 18
B. n > 18
C. 17 is an allowed membership count.
D. 18 is an allowed membership count.

09 / Multiple select

A club needs at least 30 members. Select all true statements for its membership count n.
A. 29 is an allowed membership count.
B. n > 30
C. n ≥ 30
D. 30 is an allowed membership count.

10 / Constructed response

Which values in {10, 11, 12} satisfy x ≤ 11? Does the real-number inequality have only these solutions? Explain.

Day 072 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Identify Independent and Dependent Variables

Standards: 6.EE.C.9

Learning target: I can identify quantities that change together and write a rule.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

The independent variable is the input you choose; the dependent variable changes with it. Define both and write the output in terms of the input. A table and graph must show the same rule and units.

Worked example 1

A shop charges $7 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

1. The independent variable is the input you choose; the dependent variable changes with it.

2. Define both and write the output in terms of the input.

3. The chosen notebook count is the input

4. multiply it by price and add any fixed fee to get the cost.

5. State the result in the form requested: Independent n = notebooks; dependent C = cost; C = 7n.

Answer: Independent n = notebooks; dependent C = cost; C = 7n

Worked example 2

A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (3, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.

1. The independent variable is the input you choose; the dependent variable changes with it.

2. Hours are the input.

3. Multiply each hour value by 4 to obtain liters.

4. Graph hours horizontally and liters vertically.

Answer: (0,0), (1,4), (3,12); independent h; L = 4h

Day 072 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A shop charges $5 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

02 / Constructed response

A shop charges $4 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

03 / Constructed response

A shop charges $3 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

04 / Constructed response

A shop charges $2 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule.

05 / Constructed response

A shop charges $2 for each of n notebooks plus a single $4 delivery fee. Name the independent and dependent variables and write a cost rule.

06 / Constructed response

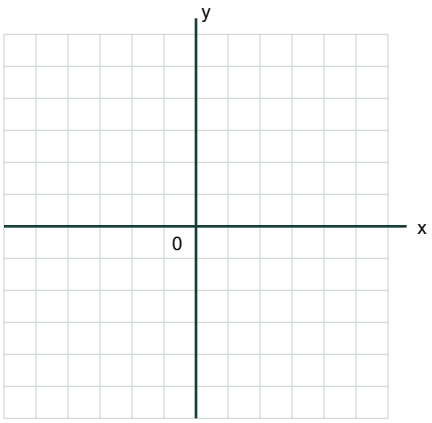
A shop charges $8 for each of n notebooks plus a single $2 delivery fee. Name the independent and dependent variables and write a cost rule.

07 / Constructed response

A shop charges $4 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule. What does the cost at n = 0 mean in this model?

08 / Table completion, equation, and graph

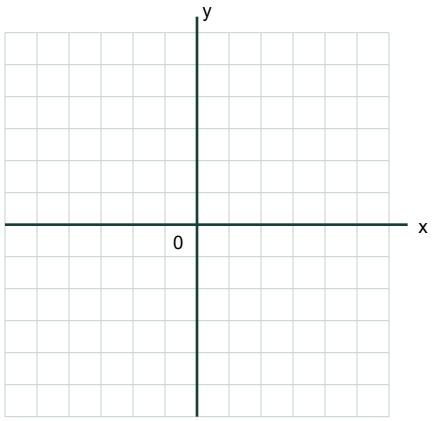
A machine uses 6 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

09 / Table completion, equation, and graph

A machine uses 3 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

10 / Constructed response

A shop charges $5 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule. What does the cost at n = 0 mean in this model?

Day 073 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Connect Rules, Tables, and Graphs

Standards: 6.EE.C.9

Learning target: I can connect an equation, a table, and a graph.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

Choose input values, substitute to find outputs, and write ordered pairs. Label horizontal input and vertical output axes. Plot each pair and interpret a point in the situation.

Worked example 1

For y = 7x, make a table for x = 0, 1, 2, 3. Plot the ordered pairs and label both axes.

1. Choose input values, substitute to find outputs, and write ordered pairs.

2. Label horizontal input and vertical output axes.

3. Substitute each input into y = 7x

4. x is horizontal and y vertical.

5. State the result in the form requested: (0, 0); (1, 7); (2, 14); (3, 21).

Answer: (0, 0); (1, 7); (2, 14); (3, 21)

Worked example 2

A machine uses 6 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.

1. Choose input values, substitute to find outputs, and write ordered pairs.

2. Hours are the input.

3. Multiply each hour value by 6 to obtain liters.

4. Graph hours horizontally and liters vertically.

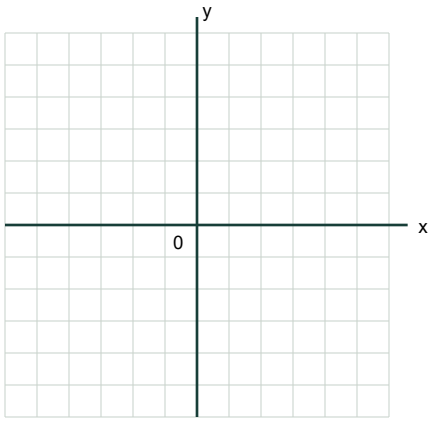
Answer: (0,0), (1,6), (2,12); independent h; L = 6h

Day 073 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

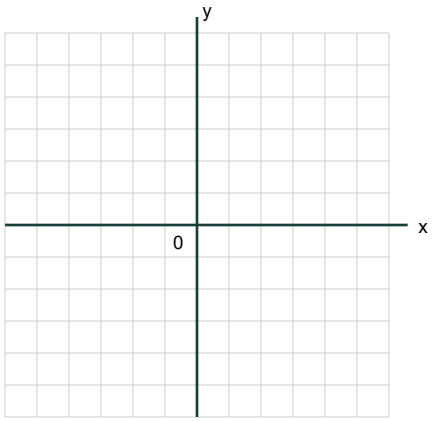
For y = 2x, make a table for x = 0, 1, 2, 3. Plot the ordered pairs and label both axes.



Label the scale and units.

02 / Constructed response

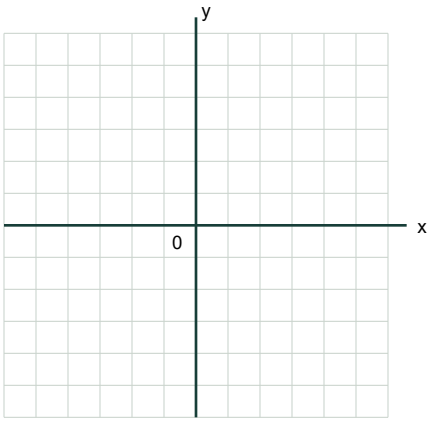
For y = 2x, make a table for x = 0, 1, 2, 6. Plot the ordered pairs and label both axes.



Label the scale and units.

03 / Constructed response

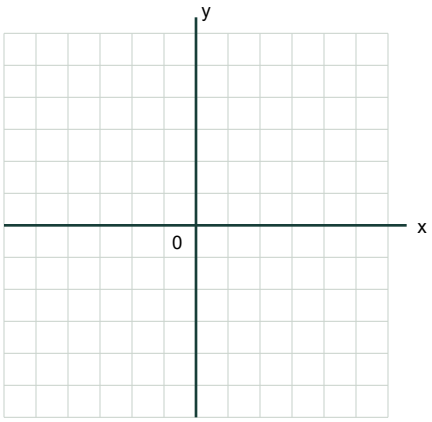
For y = 6x, make a table for x = 0, 1, 2, 6. Plot the ordered pairs and label both axes.



Label the scale and units.

04 / Constructed response

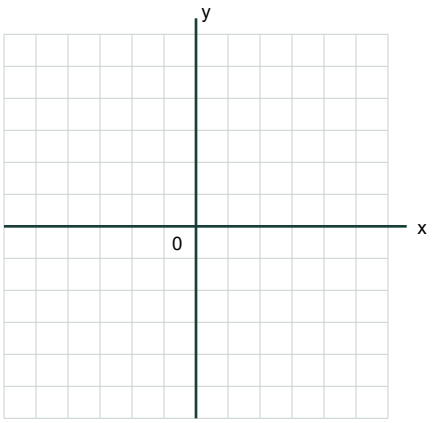
For y = 6x + 3, make a table for x = 0, 1, 2, 4. Plot the ordered pairs and label both axes.



Label the scale and units.

05 / Constructed response

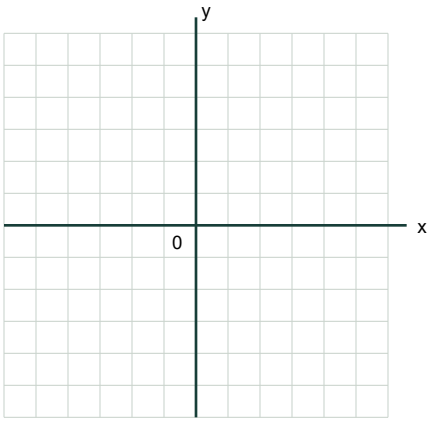
For y = 2x + 5, make a table for x = 0, 1, 2, 5. Plot the ordered pairs and label both axes.



Label the scale and units.

06 / Constructed response

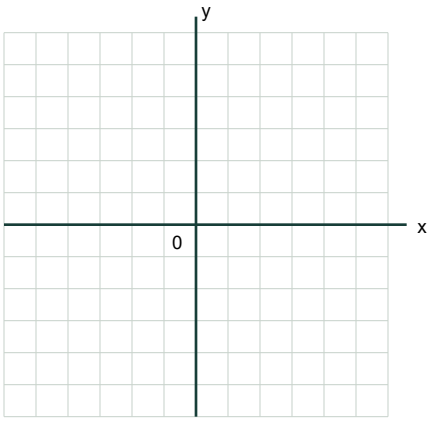
For y = 8x + 7, make a table for x = 0, 1, 2, 4. Plot the ordered pairs and label both axes.



Label the scale and units.

07 / Constructed response

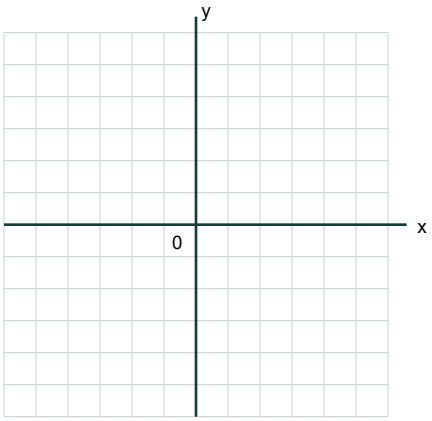
For y = 5x + 4, make a table for x = 0, 1, 2, 2. Plot the ordered pairs and label both axes. Explain what happens to y each time x increases by 1.



Label the scale and units.

08 / Table completion, equation, and graph

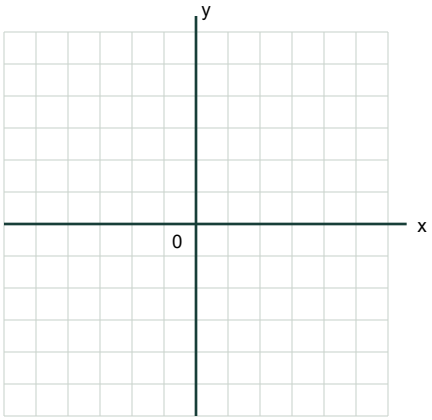
A machine uses 2 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

09 / Table completion, equation, and graph

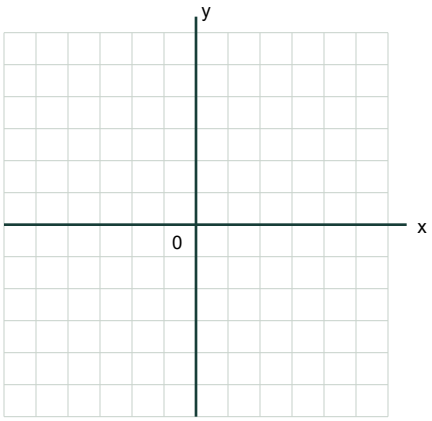
A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

10 / Constructed response

For y = 8x + 2, make a table for x = 0, 1, 2, 6. Plot the ordered pairs and label both axes. Explain what happens to y each time x increases by 1.



Label the scale and units.

Day 074 | Unit 05 | Lesson

Represent and Solve Equations and Inequalities

Interpret Relationships in Context

Standards: 6.EE.C.9

Learning target: I can interpret Relationships in Context, and explain and check my reasoning.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

01 / Learn

The independent variable is the input you choose; the dependent variable changes with it. Define both and write the output in terms of the input. A table and graph must show the same rule and units. Choose input values, substitute to find outputs, and write ordered pairs. Label horizontal input and vertical output axes. Plot each pair and interpret a point in the situation.

Worked example 1

A shop charges $5 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

1. The independent variable is the input you choose; the dependent variable changes with it.

2. Define both and write the output in terms of the input.

3. The chosen notebook count is the input

4. multiply it by price and add any fixed fee to get the cost.

5. State the result in the form requested: Independent n = notebooks; dependent C = cost; C = 5n.

Answer: Independent n = notebooks; dependent C = cost; C = 5n

Worked example 2

A machine uses 7 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.

1. Choose input values, substitute to find outputs, and write ordered pairs.

2. Hours are the input.

3. Multiply each hour value by 7 to obtain liters.

4. Graph hours horizontally and liters vertically.

Answer: (0,0), (1,7), (2,14); independent h; L = 7h

Day 074 | Your turn

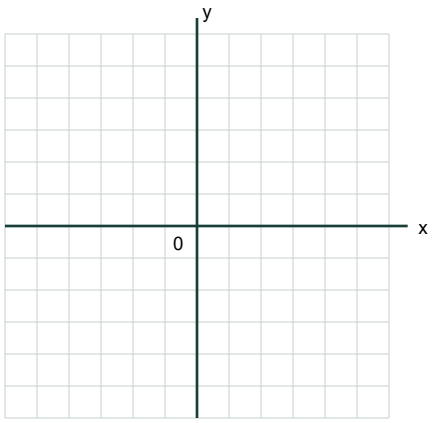
Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A shop charges $6 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

02 / Constructed response

For y = 2x, make a table for x = 0, 1, 2, 3. Plot the ordered pairs and label both axes.



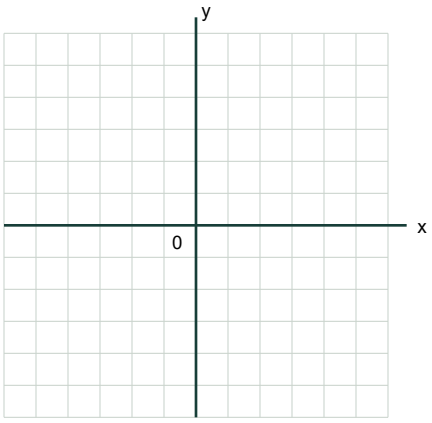
Label the scale and units.

03 / Constructed response

A shop charges $8 for each of n notebooks. Name the independent and dependent variables and write a cost rule.

04 / Constructed response

For y = 8x + 7, make a table for x = 0, 1, 2, 5. Plot the ordered pairs and label both axes.



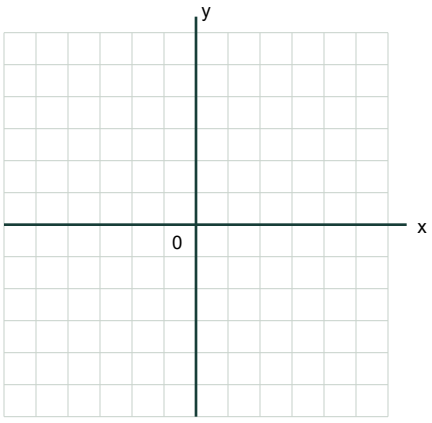
Label the scale and units.

05 / Constructed response

A shop charges $4 for each of n notebooks plus a single $7 delivery fee. Name the independent and dependent variables and write a cost rule.

06 / Constructed response

For y = 6x + 3, make a table for x = 0, 1, 2, 5. Plot the ordered pairs and label both axes.



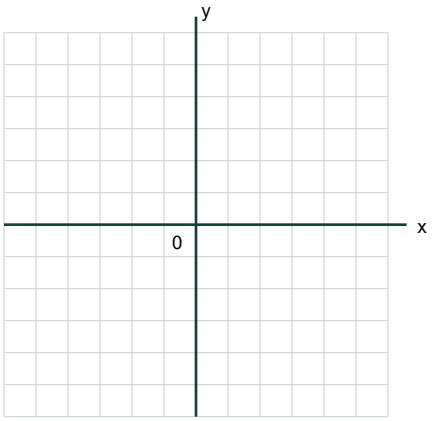
Label the scale and units.

07 / Constructed response

A shop charges $8 for each of n notebooks plus a single $3 delivery fee. Name the independent and dependent variables and write a cost rule. What does the cost at n = 0 mean in this model?

08 / Table completion, equation, and graph

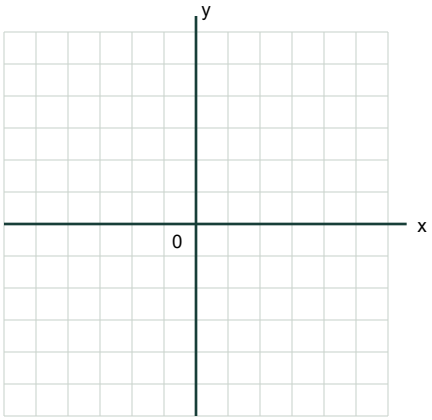
A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (3, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

09 / Table completion, equation, and graph

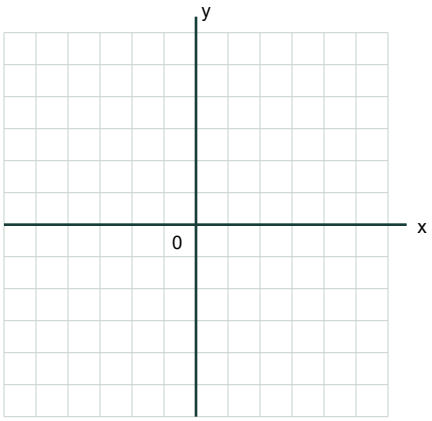
A machine uses 2 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (5, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

10 / Constructed response

For y = 7x + 6, make a table for x = 0, 1, 2, 3. Plot the ordered pairs and label both axes. Explain what happens to y each time x increases by 1.



Label the scale and units.

Day 075 | Unit 05 | Unit Test

Represent and Solve Equations and Inequalities

Unit 5 Test

Standards: 6.EE.A.1, 6.EE.A.2c, 6.NS.B.4a, 6.NS.B.4b, 6.EE.B.5, 6.EE.B.7, 6.EE.B.6, 6.EE.B.8, 6.EE.C.9

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

Directions

The first 4 questions (20%) review Unit 4. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Write 3^2 as repeated multiplication and evaluate.

02 / Previous unit review | Constructed response

Evaluate 8 + 5 × 2. Show the intermediate steps.

03 / Previous unit review | Constructed response

Find the GCF of 10 and 15. Show factor pairs or prime factors.

04 / Previous unit review | Constructed response

Find the LCM of 2 and 6 by listing multiples.

05 / Constructed response

Does x = 7 solve 3x + 7 = 25? Show substitution.

06 / Constructed response

Solve x + 3 = 5. Check in the original equation.

07 / Constructed response

Solve x - 7 = 5/2. Check in the original equation.

08 / Constructed response

Solve 7 × x = 21/2. Check in the original equation.

09 / Constructed response

Solve x + 4 = 15/2. Check in the original equation.

10 / Constructed response

Solve x ÷ 8 = 7/16. Check in the original equation.

11 / Constructed response

7 identical lengths of rope total 28 m. Let x be the length of one piece. Write and solve an equation, then check.

12 / Multiple select

A club needs at least 12 members. Select all true statements for its membership count n.
A. 11 is an allowed membership count.
B. n > 12
C. 12 is an allowed membership count.
D. n ≥ 12

13 / Multiple select

A club needs at least 15 members. Select all true statements for its membership count n.
A. n > 15
B. 14 is an allowed membership count.
C. n ≥ 15
D. 15 is an allowed membership count.

14 / Multiple select

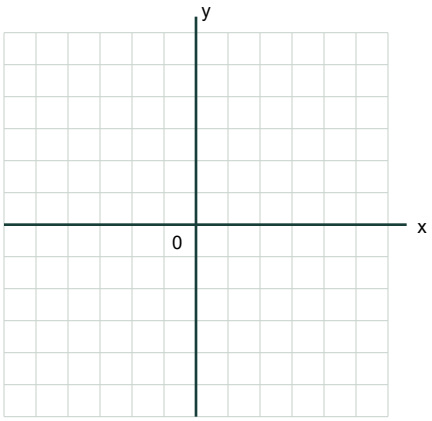
A club needs at least 42 members. Select all true statements for its membership count n.
A. n ≥ 42
B. 41 is an allowed membership count.
C. 42 is an allowed membership count.
D. n > 42

15 / Multiple select

A club needs at least 25 members. Select all true statements for its membership count n.
A. n > 25
B. 24 is an allowed membership count.
C. 25 is an allowed membership count.
D. n ≥ 25

16 / Table completion, equation, and graph

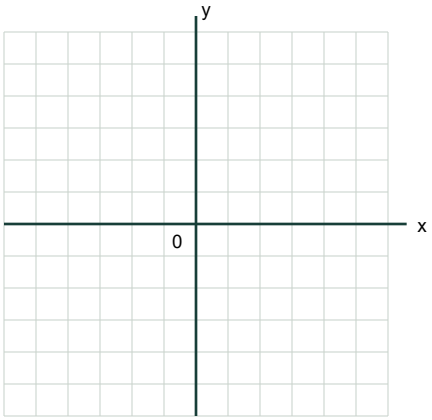
A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

17 / Table completion, equation, and graph

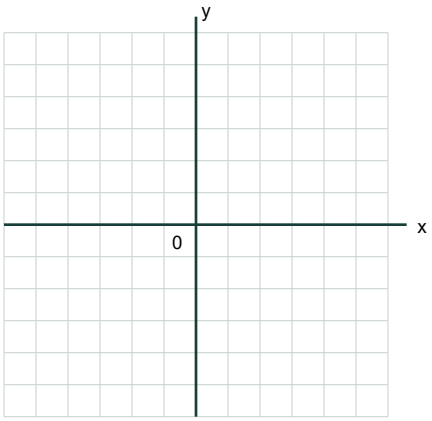
A machine uses 3 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (6, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

18 / Table completion, equation, and graph

A machine uses 6 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

19 / Multiple select

Select all values of x from the list that make x + 7 > 12 true. Explain why each rejected value fails.

A. 5
B. 4
C. 6
D. 7

20 / Multiple select

Solve x + 5/2 = 7. Select every representation of the solution.

A. 11/2
B. 18/4
C. 13/2
D. 9/2

Day 076 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Describe Ratios with Ordered Labels

Standards: 6.RP.A.1

Learning target: I can describe Ratios with Ordered Labels, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

A ratio compares two quantities multiplicatively. In a:b, a names the first quantity and b names the second. Divide both terms by a common factor to simplify, while keeping the labels and order. For a:b, the total is a+b equal parts. The first part is a/(a+b) of the total; a/b compares it to the second part. Write labels before choosing the denominator.

Worked example 1

A tray has 3 red tiles and 5 blue tiles. Write the red-to-blue ratio in simplest form.

1. A ratio compares two quantities multiplicatively.

2. In a:b, a names the first quantity and b names the second.

3. Keep red first

4. divide 3:5 by 1.

5. State the result in the form requested: 3:5.

Answer: 3:5

Worked example 2

A bag contains 25 red tiles and 30 blue tiles. Select all true statements.
A. The fraction of all tiles that are red is 5/11.
B. The red-to-blue ratio is 5:11.
C. The red-to-blue ratio is 5:6.
D. There are 56 tiles in all.

1. For a:b, the total is a+b equal parts.

2. Divide both color counts by 5.

3. For the fraction of the whole, use total 55 as denominator

4. red-to-blue compares just the two colors.

Answer: A, C - The fraction of all tiles that are red is 5/11.; The red-to-blue ratio is 5:6.

Day 076 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A tray has 3 red tiles and 2 blue tiles. Write the red-to-blue ratio in simplest form.

02 / Constructed response

A collection has 10 green and 15 yellow counters. Write the fraction that is green.

03 / Constructed response

A tray has 5 red tiles and 4 blue tiles. Write the red-to-blue ratio in simplest form.

04 / Constructed response

A collection has 25 green and 20 yellow counters. Write the fraction that is green.

05 / Constructed response

A tray has 25 red tiles and 25 blue tiles. Write the red-to-blue ratio in simplest form.

06 / Constructed response

A collection has 40 green and 20 yellow counters. Write the fraction that is green.

07 / Constructed response

A tray has 4 red tiles and 12 blue tiles. Write the red-to-blue ratio in simplest form and complete: there are ___ times as many red as blue.

08 / Multiple select

A bag contains 10 red tiles and 15 blue tiles. Select all true statements.
A. The red-to-blue ratio is 2:3.
B. There are 26 tiles in all.
C. The red-to-blue ratio is 2:5.
D. The fraction of all tiles that are red is 2/5.

09 / Multiple select

A bag contains 10 red tiles and 30 blue tiles. Select all true statements.
A. The red-to-blue ratio is 2:8.
B. The red-to-blue ratio is 2:6.
C. The fraction of all tiles that are red is 1/4.
D. There are 41 tiles in all.

10 / Constructed response

A collection has 18 green and 36 yellow counters. Write the fraction that is green and the green-to-yellow ratio in simplest form. Explain the different denominators.

Day 077 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Generate Equivalent Ratios

Standards: 6.RP.A.3a

Learning target: I can complete an equivalent-ratio table.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order.

Worked example 1

A mix uses 2 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (2, 6), (4, ___), (8, ___).

1. Multiply both values in a ratio by the same scale factor.

2. Organize pairs in a table and check that every row compares the same quantities in the same order.

3. Multiply both parts by 2 and by 4.

4. State the result in the form requested: 12; 24.

Answer: 12; 24

Worked example 2

A recipe uses 2 cups oats for 3 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (4, 6)
B. (9, 12)
C. (6, 9)
D. (4, 7)

1. Multiply both values in a ratio by the same scale factor.

2. Equivalent ratios multiply both quantities by the same factor.

3. A uses 2

4. B uses 3.

5. C and D change only one scaled quantity.

Answer: A, C - (4, 6); (6, 9)

Day 077 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A mix uses 8 cups of oats for 7 cups of seeds. Complete the table (oats, seeds): (8, 7), (16, ___), (16, ___).

02 / Constructed response

A mix uses 3 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (3, 3), (6, ___), (18, ___).

03 / Constructed response

A mix uses 5 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (5, 3), (10, ___), (10, ___).

04 / Constructed response

A mix uses 3 cups of oats for 4 cups of seeds. Complete the table (oats, seeds): (3, 4), (6, ___), (15, ___).

05 / Constructed response

A mix uses 8 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (8, 2), (16, ___), (16, ___).

06 / Constructed response

A mix uses 4 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (4, 3), (8, ___), (24, ___).

07 / Constructed response

A mix uses 7 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (7, 3), (14, ___), (28, ___) and explain how you know the last two rows are equivalent.

08 / Multiple select

A recipe uses 5 cups oats for 3 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (21, 12)
B. (15, 9)
C. (10, 7)
D. (10, 6)

09 / Multiple select

A recipe uses 2 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (4, 5)
B. (6, 6)
C. (4, 4)
D. (11, 10)

10 / Constructed response

A mix uses 6 cups of oats for 5 cups of seeds. Complete the table (oats, seeds): (6, 5), (12, ___), (42, ___) and explain how you know the last two rows are equivalent.

Day 078 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Compare Ratios Fairly

Standards: 6.RP.A.3a

Learning target: I can compare ratios using a common quantity.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Scale two ratios until one quantity matches, then compare the other quantity. The larger count alone is not enough: compare the amount per equal-sized batch or per one item.

Worked example 1

Mix A has 2 cups juice for 6 cups water. Mix B has 7 cups juice for 18 cups water. Which is stronger in juice? Show equivalent ratios.

1. Scale two ratios until one quantity matches, then compare the other quantity.

2. The larger count alone is not enough: compare the amount per equal-sized batch or per one item.

3. For 18 cups water, A uses 6 cups juice

4. B uses 7.

5. B has more juice per equal water amount.

6. State the result in the form requested: Mix B.

Answer: Mix B

Worked example 2

Mix A uses 2 cups concentrate per 5 cups water. Mix B uses 9 per 20. Select all true statements.
A. The mixtures have equivalent ratios.
B. Mix B is stronger.
C. For 20 cups water, A uses 8 cups concentrate.
D. The larger water amount alone proves A is stronger.

1. Scale two ratios until one quantity matches, then compare the other quantity.

2. Scale A by 4 to the same water amount.

3. B has 1 more cup of concentrate with equal water, so B is stronger.

Answer: B, C - Mix B is stronger.; For 20 cups water, A uses 8 cups concentrate.

Day 078 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Mix A has 5 cups juice for 7 cups water. Mix B has 26 cups juice for 35 cups water. Which is stronger in juice? Show equivalent ratios.

02 / Constructed response

Mix A has 5 cups juice for 2 cups water. Mix B has 33 cups juice for 12 cups water. Which is stronger in juice? Show equivalent ratios.

03 / Constructed response

Mix A has 7 cups juice for 4 cups water. Mix B has 45 cups juice for 24 cups water. Which is stronger in juice? Show equivalent ratios.

04 / Constructed response

Mix A has 6 cups juice for 6 cups water. Mix B has 39 cups juice for 36 cups water. Which is stronger in juice? Show equivalent ratios.

05 / Constructed response

Mix A has 7 cups juice for 6 cups water. Mix B has 38 cups juice for 30 cups water. Which is stronger in juice? Show equivalent ratios.

06 / Constructed response

Mix A has 6 cups juice for 2 cups water. Mix B has 20 cups juice for 6 cups water. Which is stronger in juice? Show equivalent ratios.

07 / Constructed response

Mix A has 7 cups juice for 5 cups water. Mix B has 44 cups juice for 30 cups water. Which is stronger in juice? Show equivalent ratios.

08 / Multiple select

Mix A uses 5 cups concentrate per 5 cups water. Mix B uses 26 per 25. Select all true statements.
A. For 25 cups water, A uses 25 cups concentrate.
B. The larger water amount alone proves A is stronger.
C. Mix B is stronger.
D. The mixtures have equivalent ratios.

09 / Multiple select

Mix A uses 5 cups concentrate per 6 cups water. Mix B uses 26 per 30. Select all true statements.
A. For 30 cups water, A uses 25 cups concentrate.
B. The larger water amount alone proves A is stronger.
C. The mixtures have equivalent ratios.
D. Mix B is stronger.

10 / Constructed response

Mix A has 7 cups juice for 5 cups water. Mix B has 39 cups juice for 25 cups water. Which is stronger in juice? Show equivalent ratios.

Day 079 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Represent Ratios with Tape Diagrams

Standards: 6.RP.A.3a

Learning target: I can use equal-size tape sections to solve a ratio problem.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

For a ratio a:b, draw a sections for one quantity and b sections for the other. A total covers a+b sections. Divide the total by a+b to find one section, then multiply by each part count.

Worked example 1

The ratio of oak trees to pine trees is 6:5. There are 66 trees altogether. Draw a tape diagram and find each count.

1. For a ratio a:b, draw a sections for one quantity and b sections for the other.

2. A total covers a+b sections.

3. 11 equal sections total

4. each section is 66 ÷ 11 = 6.

5. State the result in the form requested: 36 oaks and 30 pines.

Answer: 36 oaks and 30 pines

Worked example 2

The ratio of paper flags to fabric flags is 3:2. There are 20 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 13 paper flags. Use your model to explain the error.

1. For a ratio a:b, draw a sections for one quantity and b sections for the other.

2. There are 5 equal sections.

3. Each is 20 ÷ 5 = 4.

4. Multiply the section counts by 4.

Answer: 12 paper; 8 fabric; the claim is 1 paper flag too many.

Day 079 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

The ratio of oak trees to pine trees is 2:6. There are 16 trees altogether. Draw a tape diagram and find each count.

02 / Constructed response

The ratio of oak trees to pine trees is 6:3. There are 54 trees altogether. Draw a tape diagram and find each count.

03 / Constructed response

The ratio of oak trees to pine trees is 2:5. There are 35 trees altogether. Draw a tape diagram and find each count.

04 / Constructed response

The ratio of oak trees to pine trees is 8:6. There are 28 trees altogether. Draw a tape diagram and find each count.

05 / Constructed response

The ratio of oak trees to pine trees is 5:3. There are 48 trees altogether. Draw a tape diagram and find each count.

06 / Constructed response

The ratio of oak trees to pine trees is 5:7. There are 24 trees altogether. Draw a tape diagram and find each count.

07 / Constructed response

The ratio of oak trees to pine trees is 6:7. There are 52 trees altogether. Draw a tape diagram and find each count and the difference between the counts.

08 / Model and error analysis

The ratio of paper flags to fabric flags is 4:4. There are 16 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 9 paper flags. Use your model to explain the error.

09 / Model and error analysis

The ratio of paper flags to fabric flags is 7:5. There are 36 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 22 paper flags. Use your model to explain the error.

10 / Constructed response

The ratio of oak trees to pine trees is 5:3. There are 32 trees altogether. Draw a tape diagram and find each count and the difference between the counts.

Day 080 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Explain Equivalent Ratios in Two Models

Standards: 6.RP.A.3a

Learning target: I can explain Equivalent Ratios in Two Models, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order. For a ratio a:b, draw a sections for one quantity and b sections for the other. A total covers a+b sections. Divide the total by a+b to find one section, then multiply by each part count.

Worked example 1

A mix uses 6 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (6, 6), (12, ___), (18, ___).

1. Multiply both values in a ratio by the same scale factor.

2. Organize pairs in a table and check that every row compares the same quantities in the same order.

3. Multiply both parts by 2 and by 3.

4. State the result in the form requested: 12; 18.

Answer: 12; 18

Worked example 2

The ratio of paper flags to fabric flags is 3:3. There are 12 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 7 paper flags. Use your model to explain the error.

1. For a ratio a:b, draw a sections for one quantity and b sections for the other.

2. There are 6 equal sections.

3. Each is 12 ÷ 6 = 2.

4. Multiply the section counts by 2.

Answer: 6 paper; 6 fabric; the claim is 1 paper flag too many.

Day 080 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A mix uses 6 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (6, 3), (12, ___), (18, ___).

02 / Constructed response

The ratio of oak trees to pine trees is 7:6. There are 52 trees altogether. Draw a tape diagram and find each count.

03 / Constructed response

A mix uses 6 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (6, 6), (12, ___), (12, ___).

04 / Constructed response

The ratio of oak trees to pine trees is 8:3. There are 22 trees altogether. Draw a tape diagram and find each count.

05 / Constructed response

A mix uses 3 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (3, 2), (6, ___), (18, ___).

06 / Constructed response

The ratio of oak trees to pine trees is 4:7. There are 44 trees altogether. Draw a tape diagram and find each count.

07 / Constructed response

A mix uses 5 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (5, 2), (10, ___), (15, ___) and explain how you know the last two rows are equivalent.

08 / Model and error analysis

The ratio of paper flags to fabric flags is 5:2. There are 21 flags. Part A: draw a tape diagram and enter each count. Part B: a student says there are 16 paper flags. Use your model to explain the error.

09 / Multiple select

A recipe uses 6 cups oats for 4 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (25, 16)
B. (12, 8)
C. (12, 9)
D. (18, 12)

10 / Constructed response

The ratio of oak trees to pine trees is 3:6. There are 36 trees altogether. Draw a tape diagram and find each count and the difference between the counts.

Day 081 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Use Double Number Lines

Standards: 6.RP.A.3a

Learning target: I can represent a ratio on a double number line.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Place zero on both parallel number lines. Align a known pair above and below one tick. Add equal jumps on each line so aligned values remain in the same ratio.

Worked example 1

Draw aligned number lines for 3 liters of paint covering 5 square meters. Mark 0, one batch, and 6 batches. How much area do 18 liters cover?

1. Place zero on both parallel number lines.

2. Align a known pair above and below one tick.

3. 18 ÷ 3 = 6 batches

4. 6 × 5 = 30.

5. State the result in the form requested: 30 square meters; aligned pairs (0,0), (3,5), (18,30).

Answer: 30 square meters; aligned pairs (0,0), (3,5), (18,30)

Worked example 2

Draw a double number line for 5 liters of paint covering 4 m². Part A: align 0, 1, 2, and 4 batches. Part B: enter the area covered by 20 liters and explain why the scales can be different.

1. Place zero on both parallel number lines.

2. Each aligned jump adds 5 liters and 4 m².

3. Both quantities scale by 4

4. the number lines measure different units.

Answer: Aligned pairs (0,0), (5,4), (10,8), (20,16); 16 m²

Day 081 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw aligned number lines for 3 liters of paint covering 7 square meters. Mark 0, one batch, and 5 batches. How much area do 15 liters cover?

02 / Constructed response

Draw aligned number lines for 8 liters of paint covering 2 square meters. Mark 0, one batch, and 3 batches. How much area do 24 liters cover?

03 / Constructed response

Draw aligned number lines for 2 liters of paint covering 7 square meters. Mark 0, one batch, and 6 batches. How much area do 12 liters cover?

04 / Constructed response

Draw aligned number lines for 4 liters of paint covering 3 square meters. Mark 0, one batch, and 5 batches. How much area do 20 liters cover?

05 / Constructed response

Draw aligned number lines for 6 liters of paint covering 5 square meters. Mark 0, one batch, and 5 batches. How much area do 30 liters cover?

06 / Constructed response

Draw aligned number lines for 4 liters of paint covering 3 square meters. Mark 0, one batch, and 2 batches. How much area do 8 liters cover?

07 / Constructed response

Draw aligned number lines for 6 liters of paint covering 2 square meters. Mark 0, one batch, and 4 batches. How much area do 24 liters cover?

08 / Double number line and explanation

Draw a double number line for 6 liters of paint covering 4 m². Part A: align 0, 1, 2, and 7 batches. Part B: enter the area covered by 42 liters and explain why the scales can be different.

09 / Double number line and explanation

Draw a double number line for 7 liters of paint covering 6 m². Part A: align 0, 1, 2, and 6 batches. Part B: enter the area covered by 42 liters and explain why the scales can be different.

10 / Constructed response

Draw aligned number lines for 6 liters of paint covering 7 square meters. Mark 0, one batch, and 4 batches. How much area do 24 liters cover?

Day 082 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Represent and Graph Ratios

Standards: 6.RP.A.3a

Learning target: I can plot equivalent-ratio pairs and explain a point.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Put the first quantity on the horizontal axis and the second on the vertical axis. Build a table, label units, and plot ordered pairs. A point (x,y) describes both quantities together.

Worked example 1

For every 6 tickets sold, a club earns $2. Make a table for 0, 1, 2, and 3 batches. Plot (tickets, dollars), label axes, and interpret (18, 6).

1. Put the first quantity on the horizontal axis and the second on the vertical axis.

2. Build a table, label units, and plot ordered pairs.

3. Scale both quantities by each batch count

4. plot horizontal tickets and vertical dollars.

5. State the result in the form requested: (0,0), (6,2), (12,4), (18,6); 18 tickets earns $6.

Answer: (0,0), (6,2), (12,4), (18,6); 18 tickets earns $6

Worked example 2

A club earns $5 for every 5 tickets. Part A: plot ordered pairs for 0, 1, 2, and 6 batches (tickets, dollars). Part B: explain the meaning of (30, 30).

1. Put the first quantity on the horizontal axis and the second on the vertical axis.

2. For each batch count, multiply both 5 and 5 by that count.

3. Keep tickets horizontal and dollars vertical.

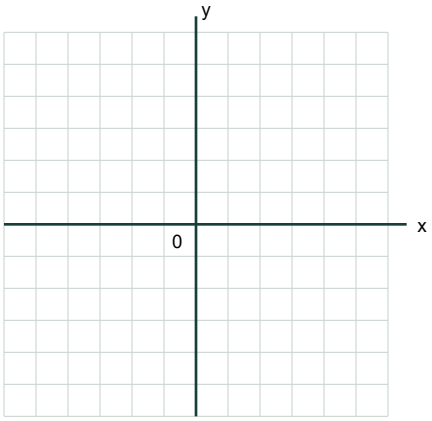
Answer: (0,0), (5,5), (10,10), (30,30); 30 tickets earn $30.

Day 082 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

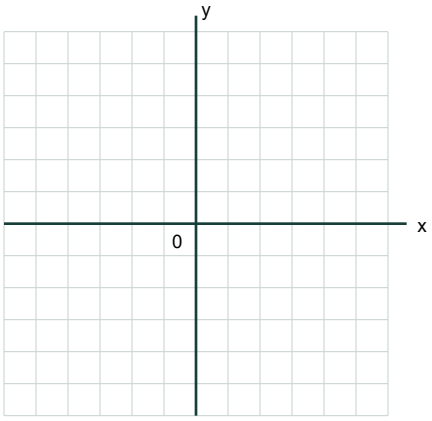
For every 6 tickets sold, a club earns $5. Make a table for 0, 1, 2, and 6 batches. Plot (tickets, dollars), label axes, and interpret (36, 30).



Label the scale and units.

02 / Constructed response

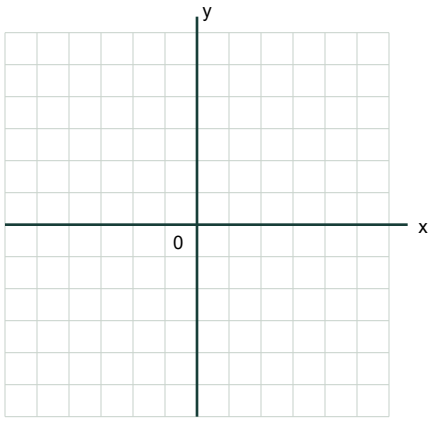
For every 2 tickets sold, a club earns $7. Make a table for 0, 1, 2, and 5 batches. Plot (tickets, dollars), label axes, and interpret (10, 35).



Label the scale and units.

03 / Constructed response

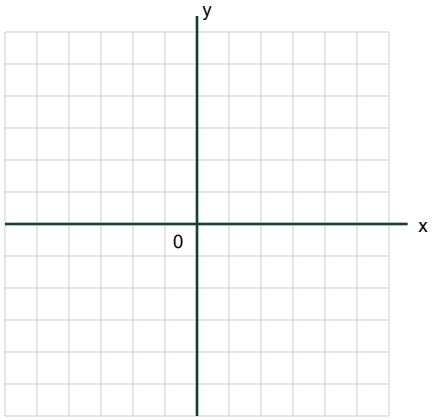
For every 7 tickets sold, a club earns $6. Make a table for 0, 1, 2, and 2 batches. Plot (tickets, dollars), label axes, and interpret (14, 12).



Label the scale and units.

04 / Constructed response

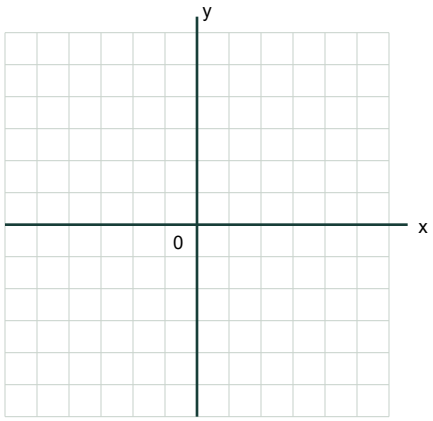
For every 2 tickets sold, a club earns $6. Make a table for 0, 1, 2, and 4 batches. Plot (tickets, dollars), label axes, and interpret (8, 24).



Label the scale and units.

05 / Constructed response

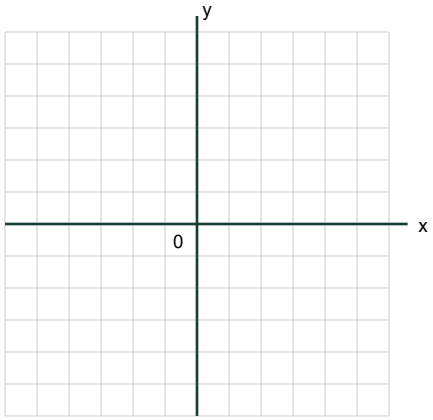
For every 2 tickets sold, a club earns $3. Make a table for 0, 1, 2, and 3 batches. Plot (tickets, dollars), label axes, and interpret (6, 9).



Label the scale and units.

06 / Constructed response

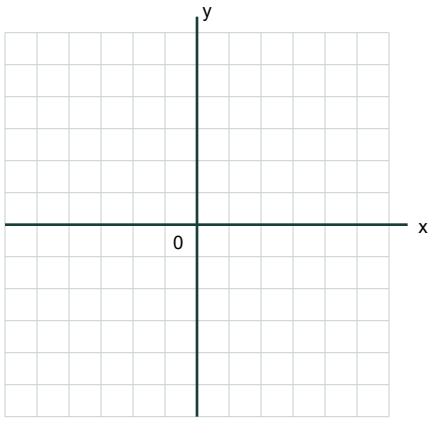
For every 5 tickets sold, a club earns $6. Make a table for 0, 1, 2, and 2 batches. Plot (tickets, dollars), label axes, and interpret (10, 12).



Label the scale and units.

07 / Constructed response

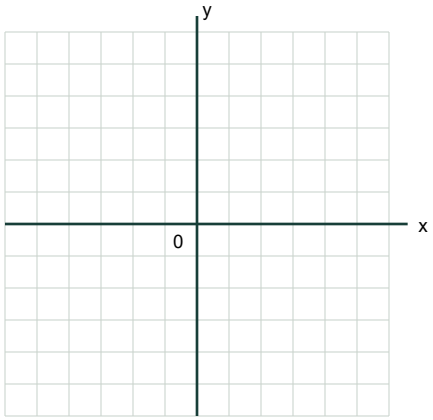
For every 2 tickets sold, a club earns $2. Make a table for 0, 1, 2, and 3 batches. Plot (tickets, dollars), label axes, and interpret (6, 6).



Label the scale and units.

08 / Table, graph, and interpretation

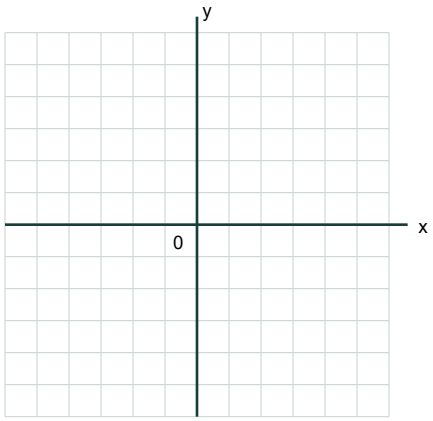
A club earns $2 for every 7 tickets. Part A: plot ordered pairs for 0, 1, 2, and 5 batches (tickets, dollars). Part B: explain the meaning of (35, 10).



Label the scale and units.

09 / Table, graph, and interpretation

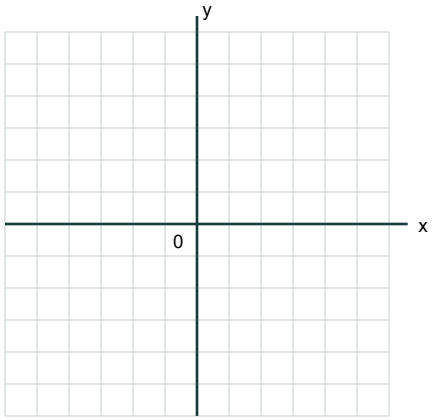
A club earns $6 for every 4 tickets. Part A: plot ordered pairs for 0, 1, 2, and 4 batches (tickets, dollars). Part B: explain the meaning of (16, 24).



Label the scale and units.

10 / Constructed response

For every 5 tickets sold, a club earns $4. Make a table for 0, 1, 2, and 6 batches. Plot (tickets, dollars), label axes, and interpret (30, 24).



Label the scale and units.

Day 083 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Understand Rates and Unit Rates

Standards: 6.RP.A.2, 6.RP.A.3b

Learning target: I can find a unit rate and explain its units.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

A unit rate compares an amount with one unit of another quantity. Divide the first quantity by the second. Write the units, such as dollars per notebook, so you know what the quotient means.

Worked example 1

5 identical notebooks cost $10. Find the cost per notebook.

1. A unit rate compares an amount with one unit of another quantity.

2. Divide the first quantity by the second.

3. Unit rate = 10 ÷ 5 = 2 dollars per notebook.

4. State the result in the form requested: $2 per notebook.

Answer: $2 per notebook

Worked example 2

6 notebooks cost $42. Select all statements at the same price per notebook.
A. 5 notebooks cost $36.
B. 5 notebooks cost $35.
C. Each notebook costs $252.
D. Each notebook costs $7.

1. A unit rate compares an amount with one unit of another quantity.

2. Cost per notebook = 42 ÷ 6 = 7.

3. Multiply $7 by the chosen notebook count.

Answer: B, D - 5 notebooks cost $35.; Each notebook costs $7.

Day 083 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

6 identical notebooks cost $18. Find the cost per notebook.

02 / Constructed response

4 identical notebooks cost $12. Find the cost per notebook.

03 / Constructed response

7 identical notebooks cost $42. Find the cost per notebook.

04 / Constructed response

7 identical notebooks cost $21. Find the cost per notebook.

05 / Constructed response

2 identical notebooks cost $12. Find the cost per notebook.

06 / Constructed response

6 identical notebooks cost $42. Find the cost per notebook.

07 / Constructed response

7 identical notebooks cost $14. Find the cost per notebook and the cost of 10 notebooks at this rate.

08 / Multiple select

6 notebooks cost $12. Select all statements at the same price per notebook.
A. Each notebook costs $2.
B. 4 notebooks cost $8.
C. 4 notebooks cost $9.
D. Each notebook costs $72.

09 / Multiple select

5 notebooks cost $10. Select all statements at the same price per notebook.
A. 5 notebooks cost $10.
B. Each notebook costs $50.
C. Each notebook costs $2.
D. 5 notebooks cost $11.

10 / Constructed response

6 identical notebooks cost $15. Find the cost per notebook and the cost of 9 notebooks at this rate.

Day 084 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Compare Unit Rates and Prices

Standards: 6.RP.A.3b

Learning target: I can compare unit prices to find the better buy.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price.

Worked example 1

Pack A has 2 markers for $4; Pack B has 4 markers for $10. Find each unit price and the better buy.

1. Divide total cost by the number of items or units.

2. Compare costs for exactly one of the same unit.

3. Divide each price by its marker count

4. compare 2 with 2.5.

5. State the result in the form requested: A $2/marker; B $2.5/marker; A is cheaper.

Answer: A $2/marker; B $2.5/marker; A is cheaper

Worked example 2

Store A sells 4 notebooks for $20. Store B sells 5 notebooks for $30. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

1. Divide total cost by the number of items or units.

2. Divide each cost by its own notebook count: 20/4 = 5

3. 30/5 = 6.

4. Compare the same one-notebook unit.

Answer: A: $5 each; B: $6 each; A is the better buy.

Day 084 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Pack A has 4 markers for $16; Pack B has 7 markers for $31.5. Find each unit price and the better buy.

02 / Constructed response

Pack A has 3 markers for $24; Pack B has 5 markers for $42.5. Find each unit price and the better buy.

03 / Constructed response

Pack A has 5 markers for $20; Pack B has 9 markers for $40.5. Find each unit price and the better buy.

04 / Constructed response

Pack A has 2 markers for $3; Pack B has 5 markers for $10. Find each unit price and the better buy.

05 / Constructed response

Pack A has 5 markers for $5; Pack B has 7 markers for $10.5. Find each unit price and the better buy.

06 / Constructed response

Pack A has 2 markers for $8; Pack B has 6 markers for $27. Find each unit price and the better buy.

07 / Constructed response

Pack A has 4 markers for $4; Pack B has 9 markers for $13.5. Find each unit price and the better buy. How much would 36 markers cost using each pack size?

08 / Two-part numeric entry and reasoning

Store A sells 3 notebooks for $9. Store B sells 4 notebooks for $16. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

09 / Two-part numeric entry and reasoning

Store A sells 2 notebooks for $4. Store B sells 3 notebooks for $9. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

10 / Constructed response

Pack A has 5 markers for $15; Pack B has 11 markers for $38.5. Find each unit price and the better buy. How much would 55 markers cost using each pack size?

Day 085 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Compare Rate Models and Units

Standards: 6.RP.A.2, 6.RP.A.3b

Learning target: I can compare Rate Models and Units, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

A unit rate compares an amount with one unit of another quantity. Divide the first quantity by the second. Write the units, such as dollars per notebook, so you know what the quotient means. Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating.

Worked example 1

7 identical notebooks cost $49. Find the cost per notebook.

1. A unit rate compares an amount with one unit of another quantity.

2. Divide the first quantity by the second.

3. Unit rate = 49 ÷ 7 = 7 dollars per notebook.

4. State the result in the form requested: $7 per notebook.

Answer: $7 per notebook

Worked example 2

A bicycle travels 140 km in 4 hours at constant speed. Select all true statements.
A. In 1 hour it travels 36 km.
B. The speed is 35 km per hour.
C. The speed is 144 km per hour.
D. In 3 hours it travels 105 km.

1. At a constant speed, distance = speed × time.

2. Divide distance by time: 140 ÷ 4 = 35 km/h.

3. Multiply this unit rate by 3 for the later distance.

Answer: B, D - The speed is 35 km per hour.; In 3 hours it travels 105 km.

Day 085 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

4 identical notebooks cost $16. Find the cost per notebook.

02 / Constructed response

Pack A has 3 markers for $18; Pack B has 6 markers for $39. Find each unit price and the better buy.

03 / Constructed response

A cyclist travels 80 km in 2 hours at constant speed. Find the speed.

04 / Constructed response

4 identical notebooks cost $28. Find the cost per notebook.

05 / Constructed response

Pack A has 4 markers for $16; Pack B has 9 markers for $40.5. Find each unit price and the better buy.

06 / Constructed response

A bus travels at 10 km/h. How long does 30 km take?

07 / Constructed response

2 identical notebooks cost $6. Find the cost per notebook and the cost of 5 notebooks at this rate.

08 / Two-part numeric entry and reasoning

Store A sells 6 notebooks for $12. Store B sells 7 notebooks for $21. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

09 / Multiple select

A bicycle travels 70 km in 2 hours at constant speed. Select all true statements.
A. The speed is 35 km per hour.
B. In 4 hours it travels 140 km.
C. In 1 hour it travels 36 km.
D. The speed is 72 km per hour.

10 / Constructed response

4 identical notebooks cost $14. Find the cost per notebook and the cost of 7 notebooks at this rate.

Day 086 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Solve Constant-Speed Problems

Standards: 6.RP.A.3b

Learning target: I can solve constant-speed problems using a rate per hour.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating.

Worked example 1

A bus travels at 40 km/h. How long does 280 km take?

1. At a constant speed, distance = speed × time.

2. Divide distance by time to find speed, or distance by speed to find time.

3. Time = distance ÷ speed = 280 ÷ 40 = 7.

4. State the result in the form requested: 7 hours.

Answer: 7 hours

Worked example 2

A bicycle travels 180 km in 6 hours at constant speed. Select all true statements.
A. The speed is 186 km per hour.
B. In 5 hours it travels 150 km.
C. The speed is 30 km per hour.
D. In 1 hour it travels 31 km.

1. At a constant speed, distance = speed × time.

2. Divide distance by time: 180 ÷ 6 = 30 km/h.

3. Multiply this unit rate by 5 for the later distance.

Answer: B, C - In 5 hours it travels 150 km.; The speed is 30 km per hour.

Day 086 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A cyclist travels 60 km in 6 hours at constant speed. Find the speed.

02 / Constructed response

A bus travels at 20 km/h. How long does 140 km take?

03 / Constructed response

A cyclist travels 210 km in 7 hours at constant speed. Find the speed.

04 / Constructed response

A bus travels at 10 km/h. How long does 50 km take?

05 / Constructed response

A cyclist travels 200 km in 5 hours at constant speed. Find the speed.

06 / Constructed response

A bus travels at 40 km/h. How long does 200 km take?

07 / Constructed response

A cyclist travels 45 km in 3/2 hours at constant speed. Find the speed and the distance in 5 hours.

08 / Multiple select

A bicycle travels 50 km in 2 hours at constant speed. Select all true statements.
A. The speed is 52 km per hour.
B. In 5 hours it travels 125 km.
C. The speed is 25 km per hour.
D. In 1 hour it travels 26 km.

09 / Multiple select

A bicycle travels 40 km in 4 hours at constant speed. Select all true statements.
A. The speed is 44 km per hour.
B. In 5 hours it travels 50 km.
C. In 1 hour it travels 11 km.
D. The speed is 10 km per hour.

10 / Constructed response

A bus travels at 25 km/h. How long does 25 km take?

Day 087 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Convert Customary Units with Ratios

Standards: 6.RP.A.3d

Learning target: I can convert measurements using equivalent ratios.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

Worked example 1

Convert 4 yards to feet. Show the conversion ratio.

1. Write an equality such as 1 foot = 12 inches.

2. Multiply to change larger units into smaller units; divide for the reverse.

3. 4 × 3 = 12.

4. Smaller units require more units for the same amount.

5. State the result in the form requested: 12 feet.

Answer: 12 feet

Worked example 2

Use 1 feet = 12 inches. Select all statements equivalent to 9/2 feet.
A. 54 inches
B. 3/8 inches
C. 55 inches
D. 9/2 × 12 inches

1. Write an equality such as 1 foot = 12 inches.

2. There are 12 smaller units in each larger one.

3. Multiply 9/2 by 12, giving 54 inches.

4. The physical length does not change.

Answer: A, D - 54 inches; 9/2 × 12 inches

Day 087 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Convert 6 yards to feet. Show the conversion ratio.

02 / Constructed response

Convert 6 feet to inches. Show the conversion ratio.

03 / Constructed response

Convert 5 yards to feet. Show the conversion ratio.

04 / Constructed response

Convert 84 inches to feet. Show the conversion ratio.

05 / Constructed response

Convert 3 feet to inches. Show the conversion ratio.

06 / Constructed response

Convert 36 inches to feet. Show the conversion ratio.

07 / Constructed response

Convert 3/2 feet to inches. Show the conversion ratio and explain why the numerical value grows.

08 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 7/2 feet.
A. 42 inches
B. 43 inches
C. 7/2 × 12 inches
D. 7/24 inches

09 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 13/2 feet.
A. 78 inches
B. 79 inches
C. 13/2 × 12 inches
D. 13/24 inches

10 / Constructed response

Convert 6 feet to yards. Show the conversion ratio.

Day 088 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Convert Metric Units with Ratios

Standards: 6.RP.A.3d

Learning target: I can convert measurements using equivalent ratios.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

Worked example 1

Convert 5 liters to milliliters. Show the conversion ratio.

1. Write an equality such as 1 foot = 12 inches.

2. Multiply to change larger units into smaller units; divide for the reverse.

3. 5 × 1000 = 5000.

4. Smaller units require more units for the same amount.

5. State the result in the form requested: 5000 milliliters.

Answer: 5000 milliliters

Worked example 2

Use 1 meters = 100 centimeters. Select all statements equivalent to 11/2 meters.
A. 551 centimeters
B. 550 centimeters
C. 11/2 × 100 centimeters
D. 11/200 centimeters

1. Write an equality such as 1 foot = 12 inches.

2. There are 100 smaller units in each larger one.

3. Multiply 11/2 by 100, giving 550 centimeters.

4. The physical length does not change.

Answer: B, C - 550 centimeters; 11/2 × 100 centimeters

Day 088 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Convert 7 liters to milliliters. Show the conversion ratio.

02 / Constructed response

Convert 3 liters to milliliters. Show the conversion ratio.

03 / Constructed response

Convert 5 meters to centimeters. Show the conversion ratio.

04 / Constructed response

Convert 8000 milliliters to liters. Show the conversion ratio.

05 / Constructed response

Convert 7 meters to centimeters. Show the conversion ratio.

06 / Constructed response

Convert 600 centimeters to meters. Show the conversion ratio.

07 / Constructed response

Convert 3 meters to centimeters. Show the conversion ratio and explain why the numerical value grows.

08 / Multiple select

Use 1 meters = 100 centimeters. Select all statements equivalent to 13/2 meters.
A. 651 centimeters
B. 650 centimeters
C. 13/200 centimeters
D. 13/2 × 100 centimeters

09 / Multiple select

Use 1 meters = 100 centimeters. Select all statements equivalent to 7/2 meters.
A. 351 centimeters
B. 7/200 centimeters
C. 7/2 × 100 centimeters
D. 350 centimeters

10 / Constructed response

Convert 2000 milliliters to liters. Show the conversion ratio.

Day 089 | Unit 06 | Lesson

Understand and Use Ratios and Rates

Apply Ratio and Rate Reasoning

Standards: 6.RP.A.3a, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can apply Ratio and Rate Reasoning, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

01 / Learn

Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating. Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price. Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

Worked example 1

A mix uses 7 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (7, 2), (14, ___), (35, ___).

1. Multiply both values in a ratio by the same scale factor.

2. Organize pairs in a table and check that every row compares the same quantities in the same order.

3. Multiply both parts by 2 and by 5.

4. State the result in the form requested: 4; 10.

Answer: 4; 10

Worked example 2

Use 1 feet = 12 inches. Select all statements equivalent to 5/2 feet.
A. 31 inches
B. 5/24 inches
C. 30 inches
D. 5/2 × 12 inches

1. Write an equality such as 1 foot = 12 inches.

2. There are 12 smaller units in each larger one.

3. Multiply 5/2 by 12, giving 30 inches.

4. The physical length does not change.

Answer: C, D - 30 inches; 5/2 × 12 inches

Day 089 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A mix uses 6 cups of oats for 5 cups of seeds. Complete the table (oats, seeds): (6, 5), (12, ___), (18, ___).

02 / Constructed response

A bus travels at 25 km/h. How long does 175 km take?

03 / Constructed response

Pack A has 2 markers for $12; Pack B has 7 markers for $45.5. Find each unit price and the better buy.

04 / Constructed response

Convert 120 minutes to hours. Show the conversion ratio.

05 / Constructed response

A mix uses 7 cups of oats for 5 cups of seeds. Complete the table (oats, seeds): (7, 5), (14, ___), (21, ___).

06 / Constructed response

A bus travels at 40 km/h. How long does 160 km take?

07 / Constructed response

Pack A has 2 markers for $7; Pack B has 7 markers for $28. Find each unit price and the better buy. How much would 14 markers cost using each pack size?

08 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 9/2 feet.
A. 54 inches
B. 3/8 inches
C. 55 inches
D. 9/2 × 12 inches

09 / Multiple select

A recipe uses 7 cups oats for 4 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (14, 9)
B. (14, 8)
C. (36, 20)
D. (21, 12)

10 / Constructed response

A bus travels at 20 km/h. How long does 60 km take?

Day 090 | Unit 06 | Unit Test

Understand and Use Ratios and Rates

Unit 6 Test

Standards: 6.EE.B.5, 6.EE.B.7, 6.RP.A.1, 6.RP.A.3a, 6.RP.A.2, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

Directions

The first 4 questions (20%) review Unit 5. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Does x = 3 solve 2x + 7 = 11? Show substitution.

02 / Previous unit review | Constructed response

Solve x + 4 = 6. Check in the original equation.

03 / Previous unit review | Constructed response

Solve x - 8 = 5/2. Check in the original equation.

04 / Previous unit review | Constructed response

Solve 2 × x = 12. Check in the original equation.

05 / Constructed response

A tray has 7 red tiles and 2 blue tiles. Write the red-to-blue ratio in simplest form.

06 / Constructed response

A mix uses 6 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (6, 3), (12, ___), (36, ___).

07 / Constructed response

Mix A has 2 cups juice for 4 cups water. Mix B has 12 cups juice for 16 cups water. Which is stronger in juice? Show equivalent ratios.

08 / Constructed response

The ratio of oak trees to pine trees is 6:3. There are 45 trees altogether. Draw a tape diagram and find each count.

09 / Constructed response

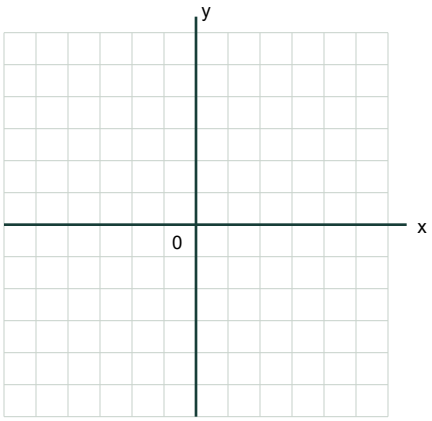
A mix uses 7 cups of oats for 3 cups of seeds. Complete the table (oats, seeds): (7, 3), (14, ___), (14, ___).

10 / Constructed response

Draw aligned number lines for 3 liters of paint covering 6 square meters. Mark 0, one batch, and 5 batches. How much area do 15 liters cover?

11 / Constructed response

For every 6 tickets sold, a club earns $3. Make a table for 0, 1, 2, and 5 batches. Plot (tickets, dollars), label axes, and interpret (30, 15).



Label the scale and units.

12 / Multiple select

3 notebooks cost $18. Select all statements at the same price per notebook.
A. 4 notebooks cost $24.
B. 4 notebooks cost $25.
C. Each notebook costs $54.
D. Each notebook costs $6.

13 / Two-part numeric entry and reasoning

Store A sells 5 notebooks for $20. Store B sells 6 notebooks for $30. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

14 / Multiple select

3 notebooks cost $6. Select all statements at the same price per notebook.
A. 2 notebooks cost $5.
B. 2 notebooks cost $4.
C. Each notebook costs $2.
D. Each notebook costs $18.

15 / Multiple select

A bicycle travels 120 km in 4 hours at constant speed. Select all true statements.
A. The speed is 124 km per hour.
B. The speed is 30 km per hour.
C. In 1 hour it travels 31 km.
D. In 3 hours it travels 90 km.

16 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 5/2 feet.
A. 31 inches
B. 5/24 inches
C. 30 inches
D. 5/2 × 12 inches

17 / Multiple select

Use 1 meters = 100 centimeters. Select all statements equivalent to 5/2 meters.
A. 1/40 centimeters
B. 5/2 × 100 centimeters
C. 250 centimeters
D. 251 centimeters

18 / Multiple select

A bicycle travels 75 km in 3 hours at constant speed. Select all true statements.
A. In 5 hours it travels 125 km.
B. The speed is 25 km per hour.
C. In 1 hour it travels 26 km.
D. The speed is 78 km per hour.

19 / Multiple select

A bag contains 4 red tiles and 10 blue tiles. Select all true statements.
A. There are 15 tiles in all.
B. The fraction of all tiles that are red is 2/7.
C. The red-to-blue ratio is 2:7.
D. The red-to-blue ratio is 2:5.

20 / Multiple select

A recipe uses 5 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (15, 6)
B. (10, 5)
C. (10, 4)
D. (21, 8)

Day 091 | Unit 07 | Lesson

Understand and Use Percent

Understand Percent as a Rate per 100

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

Worked example 1

Write 25% as a simplified fraction and decimal.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 25% = 25/100 = 1/4

4. divide by 100 for 0.25.

5. State the result in the form requested: 1/4; 0.25.

Answer: 1/4; 0.25

Worked example 2

Select all representations of 15%.
A. 15/10
B. 0.15
C. 15
D. 15/100

1. Percent means per 100.

2. Percent means per 100.

3. Divide 15 by 100 to get 0.15

4. simplify 15/100 to 3/20.

Answer: B, D - 0.15; 15/100

Day 091 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 10% as a simplified fraction and decimal.

02 / Constructed response

Write 20% as a simplified fraction and decimal.

03 / Constructed response

Write 50% as a simplified fraction and decimal.

04 / Constructed response

Write 40% as a simplified fraction and decimal.

05 / Constructed response

Write 30% as a simplified fraction and decimal.

06 / Constructed response

Write 15% as a simplified fraction and decimal.

07 / Constructed response

Write 75% as a simplified fraction and decimal.

08 / Multiple select

Select all representations of 75%.
A. 75/100
B. 0.75
C. 75
D. 75/10

09 / Multiple select

Select all representations of 325%.
A. 325/100
B. 325/10
C. 325
D. 3.25

10 / Constructed response

Write 60% as a simplified fraction and decimal.

Day 092 | Unit 07 | Lesson

Understand and Use Percent

Relate Fractions, Decimals, and Percents

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

Worked example 1

Write 50% as a simplified fraction and decimal.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 50% = 50/100 = 1/2

4. divide by 100 for 0.5.

5. State the result in the form requested: 1/2; 0.5.

Answer: 1/2; 0.5

Worked example 2

Select all representations of 60%.
A. 60
B. 0.6
C. 60/10
D. 60/100

1. Percent means per 100.

2. Percent means per 100.

3. Divide 60 by 100 to get 0.6

4. simplify 60/100 to 3/5.

Answer: B, D - 0.6; 60/100

Day 092 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 25% as a simplified fraction and decimal.

02 / Constructed response

Write 20% as a simplified fraction and decimal.

03 / Constructed response

Write 10% as a simplified fraction and decimal.

04 / Constructed response

Write 5% as a simplified fraction and decimal.

05 / Constructed response

Write 60% as a simplified fraction and decimal.

06 / Constructed response

Write 75% as a simplified fraction and decimal.

07 / Constructed response

Write 30% as a simplified fraction and decimal.

08 / Multiple select

Select all representations of 40%.
A. 0.4
B. 40/100
C. 40
D. 40/10

09 / Multiple select

Select all representations of 350%.
A. 350/10
B. 350
C. 3.5
D. 350/100

10 / Constructed response

Write 15% as a simplified fraction and decimal.

Day 093 | Unit 07 | Lesson

Understand and Use Percent

Represent Percents Above 100

Standards: 6.RP.A.3c

Learning target: I can connect a percent with a fraction out of 100 and a decimal.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100.

Worked example 1

Write 170% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 170% = 170/100 = 17/10.

4. Dividing by 100 gives 1.7, which exceeds 1 because 170 exceeds 100.

5. State the result in the form requested: 17/10; 1.7.

Answer: 17/10; 1.7

Worked example 2

Select all representations of 375%.
A. 375
B. 375/100
C. 375/10
D. 3.75

1. Percent means per 100.

2. Percent means per 100.

3. Divide 375 by 100 to get 3.75

4. simplify 375/100 to 15/4.

Answer: B, D - 375/100; 3.75

Day 093 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 150% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

02 / Constructed response

Write 210% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

03 / Constructed response

Write 155% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

04 / Constructed response

Write 110% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

05 / Constructed response

Write 290% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

06 / Constructed response

Write 235% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

07 / Constructed response

Write 160% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

08 / Multiple select

Select all representations of 275%.
A. 275
B. 275/100
C. 2.75
D. 275/10

09 / Multiple select

Select all representations of 325%.
A. 325/100
B. 325/10
C. 325
D. 3.25

10 / Constructed response

Write 115% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

Day 094 | Unit 07 | Lesson

Understand and Use Percent

Use Benchmark Percents

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

Find 25% of 40.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 25/100 × 40 = 10.

4. State the result in the form requested: 10.

Answer: 10

Worked example 2

A school club has 40 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 15% as 3/20.

3. Multiply by 40 to obtain 6.

4. Compare the rate with 1.

Answer: 6 participants; n = (15/100) × 40; less than the whole because the percent is below 100

Day 094 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 10% of 200.

02 / Constructed response

Find 50% of 180.

03 / Constructed response

Find 20% of 200.

04 / Constructed response

Find 125% of 100.

05 / Constructed response

Find 12% of 140.

06 / Constructed response

Find 5% of 220.

07 / Constructed response

Find 40% of 100. Then explain a mental-math or fraction strategy that checks your answer.

08 / Numeric entry and explanation

A poster is enlarged to 125% of its original 100 cm width. Part A: enter the new width. Part B: explain why it exceeds the original.

09 / Numeric entry and equation

A school club has 80 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

10 / Constructed response

Find 125% of 100. Then explain a mental-math or fraction strategy that checks your answer.

Day 095 | Unit 07 | Lesson

Understand and Use Percent

Connect Percent Models and Benchmarks

Standards: 6.RP.A.3c

Learning target: I can connect Percent Models and Benchmarks, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Percent means per 100. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal. To turn a decimal into a percent, multiply by 100. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

Write 10% as a simplified fraction and decimal.

1. Percent means per 100.

2. Write p% as p/100, simplify the fraction, and divide by 100 to write a decimal.

3. 10% = 10/100 = 1/10

4. divide by 100 for 0.1.

5. State the result in the form requested: 1/10; 0.1.

Answer: 1/10; 0.1

Worked example 2

A school club has 60 members; 30% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 30% as 3/10.

3. Multiply by 60 to obtain 18.

4. Compare the rate with 1.

Answer: 18 participants; n = (30/100) × 60; less than the whole because the percent is below 100

Day 095 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Write 50% as a simplified fraction and decimal.

02 / Constructed response

Find 10% of 240.

03 / Constructed response

Write 20% as a simplified fraction and decimal.

04 / Constructed response

Find 125% of 140.

05 / Constructed response

Write 25% as a simplified fraction and decimal.

06 / Constructed response

Find 20% of 200.

07 / Constructed response

Write 125% as a simplified fraction and decimal. Explain why the decimal can be greater than 1.

08 / Numeric entry and equation

A school club has 40 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Multiple select

Select all representations of 375%.
A. 375
B. 375/100
C. 375/10
D. 3.75

10 / Constructed response

Find 15% of 120. Then explain a mental-math or fraction strategy that checks your answer.

Day 096 | Unit 07 | Lesson

Understand and Use Percent

Find the Percent of a Number

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

Find 50% of 200.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 50/100 × 200 = 100.

4. State the result in the form requested: 100.

Answer: 100

Worked example 2

A school club has 100 members; 60% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 60% as 3/5.

3. Multiply by 100 to obtain 60.

4. Compare the rate with 1.

Answer: 60 participants; n = (60/100) × 100; less than the whole because the percent is below 100

Day 096 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 50% of 220.

02 / Constructed response

Find 20% of 200.

03 / Constructed response

Find 25% of 120.

04 / Constructed response

Find 60% of 180.

05 / Constructed response

Find 15% of 200.

06 / Constructed response

Find 75% of 200.

07 / Constructed response

Find 125% of 40. Then explain a mental-math or fraction strategy that checks your answer.

08 / Numeric entry and equation

A school club has 100 members; 20% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Numeric entry and equation

A school club has 80 members; 40% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

10 / Constructed response

Find 15% of 240. Then explain a mental-math or fraction strategy that checks your answer.

Day 097 | Unit 07 | Lesson

Understand and Use Percent

Use a Percent Model in Context

Standards: 6.RP.A.3c

Learning target: I can find the part when I know the percent and whole.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

Find 25% of 180.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 25/100 × 180 = 45.

4. State the result in the form requested: 45.

Answer: 45

Worked example 2

A school club has 60 members; 60% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 60% as 3/5.

3. Multiply by 60 to obtain 36.

4. Compare the rate with 1.

Answer: 36 participants; n = (60/100) × 60; less than the whole because the percent is below 100

Day 097 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 50% of 220.

02 / Constructed response

Find 10% of 160.

03 / Constructed response

Find 20% of 60.

04 / Constructed response

Find 15% of 200.

05 / Constructed response

Find 5% of 220.

06 / Constructed response

Find 30% of 100.

07 / Constructed response

Find 75% of 160. Then explain a mental-math or fraction strategy that checks your answer.

08 / Numeric entry and equation

A school club has 120 members; 25% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Numeric entry and equation

A school club has 40 members; 25% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

10 / Constructed response

Find 75% of 100. Then explain a mental-math or fraction strategy that checks your answer.

Day 098 | Unit 07 | Lesson

Understand and Use Percent

Find the Whole from Part and Percent

Standards: 6.RP.A.3c

Learning target: I can find the whole when a part and percent are known.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

Worked example 1

20% of a number is 16. Find the number and check.

1. Write part = decimal rate × whole.

2. Divide the part by the decimal rate to find the whole.

3. Whole = 16 ÷ 1/5 = 80.

4. Check 20% × 80 = 16.

5. State the result in the form requested: 80.

Answer: 80

Worked example 2

72 is 60% of an unknown quantity w. Select all equations that find w.
A. (60/100)w = 72
B. w = 72 + 60
C. w = 72 ÷ (60/100)
D. w = 72 × (60/100)

1. Write part = decimal rate × whole.

2. The part is percent rate times whole.

3. Divide 72 by 3/5

4. w = 120.

5. Check by multiplication.

Answer: A, C - (60/100)w = 72; w = 72 ÷ (60/100)

Day 098 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

50% of a number is 70. Find the number and check.

02 / Constructed response

10% of a number is 22. Find the number and check.

03 / Constructed response

25% of a number is 40. Find the number and check.

04 / Constructed response

25% of a number is 15. Find the number and check.

05 / Constructed response

15% of a number is 27. Find the number and check.

06 / Constructed response

75% of a number is 75. Find the number and check.

07 / Constructed response

33 students are 15% of a school’s enrollment. Find the enrollment.

08 / Multiple select

12 is 30% of an unknown quantity w. Select all equations that find w.
A. (30/100)w = 12
B. w = 12 ÷ (30/100)
C. w = 12 × (30/100)
D. w = 12 + 30

09 / Multiple select

60 is 60% of an unknown quantity w. Select all equations that find w.
A. w = 60 ÷ (60/100)
B. (60/100)w = 60
C. w = 60 × (60/100)
D. w = 60 + 60

10 / Constructed response

56 students are 40% of a school’s enrollment. Find the enrollment.

Day 099 | Unit 07 | Lesson

Understand and Use Percent

Check the Reference Whole

Standards: 6.RP.A.3c

Learning target: I can find the whole when a part and percent are known.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

Worked example 1

50% of a number is 60. Find the number and check.

1. Write part = decimal rate × whole.

2. Divide the part by the decimal rate to find the whole.

3. Whole = 60 ÷ 1/2 = 120.

4. Check 50% × 120 = 60.

5. State the result in the form requested: 120.

Answer: 120

Worked example 2

18 is 15% of an unknown quantity w. Select all equations that find w.
A. (15/100)w = 18
B. w = 18 × (15/100)
C. w = 18 + 15
D. w = 18 ÷ (15/100)

1. Write part = decimal rate × whole.

2. The part is percent rate times whole.

3. Divide 18 by 3/20

4. w = 120.

5. Check by multiplication.

Answer: A, D - (15/100)w = 18; w = 18 ÷ (15/100)

Day 099 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

20% of a number is 36. Find the number and check.

02 / Constructed response

50% of a number is 30. Find the number and check.

03 / Constructed response

50% of a number is 50. Find the number and check.

04 / Constructed response

12% of a number is 144/5. Find the number and check.

05 / Constructed response

25% of a number is 30. Find the number and check.

06 / Constructed response

125% of a number is 200. Find the number and check.

07 / Constructed response

48 students are 30% of a school’s enrollment. Find the enrollment.

08 / Multiple select

24 is 60% of an unknown quantity w. Select all equations that find w.
A. w = 24 × (60/100)
B. w = 24 + 60
C. w = 24 ÷ (60/100)
D. (60/100)w = 24

09 / Multiple select

12 is 30% of an unknown quantity w. Select all equations that find w.
A. (30/100)w = 12
B. w = 12 ÷ (30/100)
C. w = 12 × (30/100)
D. w = 12 + 30

10 / Constructed response

80 students are 40% of a school’s enrollment. Find the enrollment.

Day 100 | Unit 07 | Lesson

Understand and Use Percent

Explain the Reference Whole in Percent Models

Standards: 6.RP.A.3c

Learning target: I can explain the Reference Whole in Percent Models, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole.

Worked example 1

50% of a number is 60. Find the number and check.

1. Write part = decimal rate × whole.

2. Divide the part by the decimal rate to find the whole.

3. Whole = 60 ÷ 1/2 = 120.

4. Check 50% × 120 = 60.

5. State the result in the form requested: 120.

Answer: 120

Worked example 2

A school club has 80 members; 30% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

1. Use part = (percent/100) × whole.

2. Rename 30% as 3/10.

3. Multiply by 80 to obtain 24.

4. Compare the rate with 1.

Answer: 24 participants; n = (30/100) × 80; less than the whole because the percent is below 100

Day 100 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

25% of a number is 45. Find the number and check.

02 / Constructed response

Find 25% of 220.

03 / Constructed response

25% of a number is 35. Find the number and check.

04 / Constructed response

Find 15% of 120.

05 / Constructed response

125% of a number is 50. Find the number and check.

06 / Constructed response

Find 40% of 220.

07 / Constructed response

11 students are 5% of a school’s enrollment. Find the enrollment.

08 / Numeric entry and equation

A school club has 100 members; 25% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

09 / Multiple select

24 is 20% of an unknown quantity w. Select all equations that find w.
A. w = 24 × (20/100)
B. w = 24 + 20
C. (20/100)w = 24
D. w = 24 ÷ (20/100)

10 / Constructed response

Find 12% of 180. Then explain a mental-math or fraction strategy that checks your answer.

Day 101 | Unit 07 | Lesson

Understand and Use Percent

Find the Unknown Percent

Standards: 6.RP.A.3c

Learning target: I can find the percent that one quantity is of another.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

70 is what percent of 140?

1. Divide the part by the whole, then multiply by 100.

2. Keep the same units in part and whole.

3. 70 ÷ 140 × 100 = 50%

4. the reference whole is 140.

5. State the result in the form requested: 50%.

Answer: 50%

Worked example 2

A quantity changes from a reference whole of 60 to a compared amount of 45. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (45 ÷ 60) × 100 = 75%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 75%; denominator 60

Day 101 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

40 is what percent of 200?

02 / Constructed response

12 is what percent of 120?

03 / Constructed response

80 is what percent of 160?

04 / Constructed response

125 is what percent of 100?

05 / Constructed response

75 is what percent of 100?

06 / Constructed response

40 is what percent of 100?

07 / Constructed response

30 is what percent of 40? State which quantity is the reference whole.

08 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 15. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

09 / Numeric entry and explanation

A quantity changes from a reference whole of 60 to a compared amount of 36. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

24 is what percent of 60? State which quantity is the reference whole.

Day 102 | Unit 07 | Lesson

Understand and Use Percent

Solve Percent Problems with Any Unknown

Standards: 6.RP.A.3c

Learning target: I can solve Percent Problems with Any Unknown, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

Find 10% of 140.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 10/100 × 140 = 14.

4. State the result in the form requested: 14.

Answer: 14

Worked example 2

A quantity changes from a reference whole of 100 to a compared amount of 40. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (40 ÷ 100) × 100 = 40%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 40%; denominator 100

Day 102 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 25% of 200.

02 / Constructed response

10% of a number is 18. Find the number and check.

03 / Constructed response

10 is what percent of 40?

04 / Constructed response

Find 30% of 200.

05 / Constructed response

75% of a number is 150. Find the number and check.

06 / Constructed response

40 is what percent of 100?

07 / Constructed response

Find 20% of 60. Then explain a mental-math or fraction strategy that checks your answer.

08 / Multiple select

32 is 40% of an unknown quantity w. Select all equations that find w.
A. w = 32 + 40
B. w = 32 ÷ (40/100)
C. w = 32 × (40/100)
D. (40/100)w = 32

09 / Numeric entry and explanation

A quantity changes from a reference whole of 60 to a compared amount of 9. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

Find 125% of 180. Then explain a mental-math or fraction strategy that checks your answer.

Day 103 | Unit 07 | Lesson

Understand and Use Percent

Explain and Critique Percent Reasoning

Standards: 6.RP.A.3c

Learning target: I can explain and Critique Percent Reasoning, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer.

Worked example 1

40 is what percent of 200?

1. Divide the part by the whole, then multiply by 100.

2. Keep the same units in part and whole.

3. 40 ÷ 200 × 100 = 20%

4. the reference whole is 200.

5. State the result in the form requested: 20%.

Answer: 20%

Worked example 2

125 is 125% of an unknown quantity w. Select all equations that find w.
A. w = 125 ÷ (125/100)
B. w = 125 + 125
C. (125/100)w = 125
D. w = 125 × (125/100)

1. Write part = decimal rate × whole.

2. The part is percent rate times whole.

3. Divide 125 by 5/4

4. w = 100.

5. Check by multiplication.

Answer: A, C - w = 125 ÷ (125/100); (125/100)w = 125

Day 103 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

40 is what percent of 80?

02 / Constructed response

20% of a number is 24. Find the number and check.

03 / Constructed response

20 is what percent of 100?

04 / Constructed response

12% of a number is 24/5. Find the number and check.

05 / Constructed response

24 is what percent of 40?

06 / Constructed response

60% of a number is 108. Find the number and check.

07 / Constructed response

84 is what percent of 140? State which quantity is the reference whole.

08 / Multiple select

16 is 40% of an unknown quantity w. Select all equations that find w.
A. w = 16 ÷ (40/100)
B. w = 16 × (40/100)
C. (40/100)w = 16
D. w = 16 + 40

09 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 40. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

125% of a number is 150. Find the number and check.

Day 104 | Unit 07 | Lesson

Understand and Use Percent

Model a Percent Situation

Standards: 6.RP.A.3c

Learning target: I can model a Percent Situation, and explain and check my reasoning.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

01 / Learn

Use part = (percent/100) × whole. You may also find 10%, 5%, or 1% and combine equal pieces. Label the part in the same units as the whole. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

Find 20% of 160.

1. Use part = (percent/100) × whole.

2. You may also find 10%, 5%, or 1% and combine equal pieces.

3. 20/100 × 160 = 32.

4. State the result in the form requested: 32.

Answer: 32

Worked example 2

A quantity changes from a reference whole of 60 to a compared amount of 12. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (12 ÷ 60) × 100 = 20%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 20%; denominator 60

Day 104 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Find 50% of 80.

02 / Constructed response

20% of a number is 16. Find the number and check.

03 / Constructed response

70 is what percent of 140?

04 / Constructed response

Find 30% of 40.

05 / Constructed response

125% of a number is 225. Find the number and check.

06 / Constructed response

60 is what percent of 200?

07 / Constructed response

Find 30% of 200. Then explain a mental-math or fraction strategy that checks your answer.

08 / Multiple select

9 is 15% of an unknown quantity w. Select all equations that find w.
A. (15/100)w = 9
B. w = 9 ÷ (15/100)
C. w = 9 × (15/100)
D. w = 9 + 15

09 / Numeric entry and explanation

A quantity changes from a reference whole of 40 to a compared amount of 16. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

10 / Constructed response

Find 15% of 60. Then explain a mental-math or fraction strategy that checks your answer.

Day 105 | Unit 07 | Unit Test

Understand and Use Percent

Unit 7 Test

Standards: 6.RP.A.1, 6.RP.A.3a, 6.RP.A.3c

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 6: Understand and Use Percent | Spectrum companion Chapter 3

Directions

The first 4 questions (20%) review Unit 6. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

A tray has 6 red tiles and 6 blue tiles. Write the red-to-blue ratio in simplest form.

02 / Previous unit review | Constructed response

A mix uses 5 cups of oats for 6 cups of seeds. Complete the table (oats, seeds): (5, 6), (10, ___), (10, ___).

03 / Previous unit review | Constructed response

Mix A has 5 cups juice for 6 cups water. Mix B has 14 cups juice for 12 cups water. Which is stronger in juice? Show equivalent ratios.

04 / Previous unit review | Constructed response

The ratio of oak trees to pine trees is 2:6. There are 32 trees altogether. Draw a tape diagram and find each count.

05 / Constructed response

Write 50% as a simplified fraction and decimal.

06 / Constructed response

Write 10% as a simplified fraction and decimal.

07 / Constructed response

Write 300% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

08 / Constructed response

Find 40% of 100.

09 / Constructed response

Write 5% as a simplified fraction and decimal.

10 / Constructed response

Find 15% of 160.

11 / Constructed response

Find 40% of 60.

12 / Multiple select

100 is 125% of an unknown quantity w. Select all equations that find w.
A. w = 100 × (125/100)
B. w = 100 + 125
C. (125/100)w = 100
D. w = 100 ÷ (125/100)

13 / Multiple select

20 is 20% of an unknown quantity w. Select all equations that find w.
A. w = 20 × (20/100)
B. (20/100)w = 20
C. w = 20 + 20
D. w = 20 ÷ (20/100)

14 / Multiple select

15 is 25% of an unknown quantity w. Select all equations that find w.
A. w = 15 × (25/100)
B. w = 15 + 25
C. w = 15 ÷ (25/100)
D. (25/100)w = 15

15 / Numeric entry and explanation

A quantity changes from a reference whole of 60 to a compared amount of 75. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

16 / Multiple select

12 is 15% of an unknown quantity w. Select all equations that find w.
A. (15/100)w = 12
B. w = 12 + 15
C. w = 12 ÷ (15/100)
D. w = 12 × (15/100)

17 / Multiple select

24 is 40% of an unknown quantity w. Select all equations that find w.
A. (40/100)w = 24
B. w = 24 × (40/100)
C. w = 24 + 40
D. w = 24 ÷ (40/100)

18 / Multiple select

48 is 60% of an unknown quantity w. Select all equations that find w.
A. (60/100)w = 48
B. w = 48 ÷ (60/100)
C. w = 48 × (60/100)
D. w = 48 + 60

19 / Multiple select

Select all representations of 250%.
A. 250
B. 250/100
C. 250/10
D. 2.5

20 / Multiple select

Select all representations of 350%.

A. 350/10

B. 350

C. 3.5

D. 350/100

Day 106 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Find Triangle Area

Standards: 6.G.A.1

Learning target: I can find the area of any triangle using a perpendicular height.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2. A slanted side is not necessarily the height. Label area in square units.

Worked example 1

Sketch a triangle with base 4 cm and perpendicular height 5 cm. Find its area.

1. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2.

2. A slanted side is not necessarily the height.

3. A = bh/2 = 4 × 5 ÷ 2 = 10.

4. Two such triangles make a parallelogram.

5. State the result in the form requested: 10 cm².

Answer: 10 cm²

Worked example 2

A triangle has base 14 cm and perpendicular height 4 cm. Select all expressions that give its area in cm².
A. 7 × 4
B. (14 + 4) ÷ 2
C. 14 × 4
D. (14 × 4) ÷ 2

1. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2.

2. Triangle area is half the matching parallelogram: (14 × 4)/2 = 28 cm².

3. Multiplying half the base by the height gives the same result.

Answer: A, D - 7 × 4; (14 × 4) ÷ 2

Day 106 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 3 cm. Find its area.

02 / Constructed response

Sketch a triangle with base 16 cm and perpendicular height 5 cm. Find its area.

03 / Constructed response

Sketch a triangle with base 4 cm and perpendicular height 6 cm. Find its area.

04 / Constructed response

Sketch a triangle with base 14 cm and perpendicular height 5 cm. Find its area.

05 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 2 cm. Find its area.

06 / Constructed response

Sketch a triangle with base 12 cm and perpendicular height 5 cm. Find its area.

07 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 3/2 cm. Find its area. Explain why the slanted side is not needed.

08 / Multiple select

A triangle has base 8 cm and perpendicular height 6 cm. Select all expressions that give its area in cm².
A. 8 × 6
B. 4 × 6
C. (8 × 6) ÷ 2
D. (8 + 6) ÷ 2

09 / Multiple select

A triangle has base 10 cm and perpendicular height 5 cm. Select all expressions that give its area in cm².
A. (10 + 5) ÷ 2
B. 5 × 5
C. (10 × 5) ÷ 2
D. 10 × 5

10 / Constructed response

Sketch a triangle with base 10 cm and perpendicular height 7/2 cm. Find its area. Explain why the slanted side is not needed.

Day 107 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Find Parallelogram and Trapezoid Area

Standards: 6.G.A.1

Learning target: I can find parallelogram and trapezoid areas by rearranging shapes.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Cut and move a corner of a parallelogram to make a rectangle: A = bh. Two matching trapezoids form a parallelogram, giving A = (b1 + b2)h/2. Heights must be perpendicular.

Worked example 1

A parallelogram has base 2 m and perpendicular height 5 m. Sketch how it can be rearranged into a rectangle and find its area.

1. Cut and move a corner of a parallelogram to make a rectangle: A = bh.

2. Two matching trapezoids form a parallelogram, giving A = (b1 + b2)h/2.

3. Cut and shift a corner

4. A = bh = 2 × 5 = 10.

5. State the result in the form requested: 10 m².

Answer: 10 m²

Worked example 2

A trapezoid has parallel bases 14 m and 22 m, and perpendicular height 3 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

1. Cut and move a corner of a parallelogram to make a rectangle: A = bh.

2. Use average base × height: (14 + 22)/2 × 3 = 54.

3. A full rectangle at the longer base includes extra corner regions.

Answer: 54 m²

Day 107 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A parallelogram has base 8 m and perpendicular height 4 m. Sketch how it can be rearranged into a rectangle and find its area.

02 / Constructed response

A parallelogram has base 6 m and perpendicular height 4 m. Sketch how it can be rearranged into a rectangle and find its area.

03 / Constructed response

A parallelogram has base 8 m and perpendicular height 7 m. Sketch how it can be rearranged into a rectangle and find its area.

04 / Constructed response

A trapezoid has parallel bases 4 m and 14 m and perpendicular height 5 m. Sketch it and find its area.

05 / Constructed response

A trapezoid has parallel bases 8 m and 16 m and perpendicular height 7 m. Sketch it and find its area.

06 / Constructed response

A trapezoid has parallel bases 8 m and 12 m and perpendicular height 4 m. Sketch it and find its area.

07 / Constructed response

A trapezoid has parallel bases 6 m and 14 m and perpendicular height 5 m. Sketch it and find its area.

08 / Numeric entry and error analysis

A trapezoid has parallel bases 14 m and 22 m, and perpendicular height 4 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

09 / Numeric entry and error analysis

A trapezoid has parallel bases 4 m and 8 m, and perpendicular height 2 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

10 / Constructed response

A trapezoid has parallel bases 12 m and 24 m and perpendicular height 2 m. Sketch it and find its area.

Day 108 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Decompose Composite Polygons

Standards: 6.G.A.1

Learning target: I can decompose a polygon into familiar shapes to find area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure.

Worked example 1

An L-shaped garden is a 4 m by 4 m rectangle with an 2 m by 2 m corner rectangle removed. Sketch and find the garden area.

1. Sketch the figure and label every given length.

2. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle.

3. Outer area 16 minus cutout 4 = 12.

4. State the result in the form requested: 12 m².

Answer: 12 m²

Worked example 2

A 6 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

1. Sketch the figure and label every given length.

2. Outer area = 60.

3. Triangle = 4 × 6/2 = 12.

4. Subtract the removed region exactly once.

Answer: Removed 12 m²; remaining 48 m²

Day 108 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

An L-shaped garden is a 13 m by 11 m rectangle with an 8 m by 6 m corner rectangle removed. Sketch and find the garden area.

02 / Constructed response

An L-shaped garden is a 11 m by 9 m rectangle with an 5 m by 3 m corner rectangle removed. Sketch and find the garden area.

03 / Constructed response

An L-shaped garden is a 14 m by 11 m rectangle with an 8 m by 5 m corner rectangle removed. Sketch and find the garden area.

04 / Constructed response

An L-shaped garden is a 8 m by 9 m rectangle with an 3 m by 4 m corner rectangle removed. Sketch and find the garden area.

05 / Constructed response

An L-shaped garden is a 9 m by 9 m rectangle with an 3 m by 3 m corner rectangle removed. Sketch and find the garden area.

06 / Constructed response

An L-shaped garden is a 14 m by 8 m rectangle with an 8 m by 2 m corner rectangle removed. Sketch and find the garden area.

07 / Constructed response

A 4 m by 7 m rectangle has a right-triangle corner removed with perpendicular legs 2 m and 5 m. Sketch the remaining polygon and find its area.

08 / Two-part numeric entry

A 11 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

09 / Two-part numeric entry

A 8 m by 8 m rectangle has a right-triangle corner removed with perpendicular legs 5 m and 3 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

10 / Constructed response

A 9 m by 6 m rectangle has a right-triangle corner removed with perpendicular legs 3 m and 3 m. Sketch the remaining polygon and find its area.

Day 109 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Use Coordinates to Find Polygon Lengths and Area

Standards: 6.G.A.3, 6.G.A.1

Learning target: I can draw polygons from coordinates and use side lengths to find area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths.

Worked example 1

Plot and connect (0,0), (13,0), (13,8), (0,8). Find the side lengths and area.

1. Plot vertices in order and connect them, including the last to the first.

2. Equal x-values make a vertical side; equal y-values make a horizontal side.

3. Horizontal length 13 - (0) = 13

4. vertical length 8 - (0) = 8

5. area = 13 × 8.

6. State the result in the form requested: Sides 13 and 8; area 104 square units.

Answer: Sides 13 and 8; area 104 square units

Worked example 2

A rectangle has vertices (-3, -4), (5, -4), (5, 4), and (-3, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.

1. Plot vertices in order and connect them, including the last to the first.

2. Horizontal length = 3 + 5

3. vertical length = 4 + 4.

4. Multiply 8 × 8 = 64.

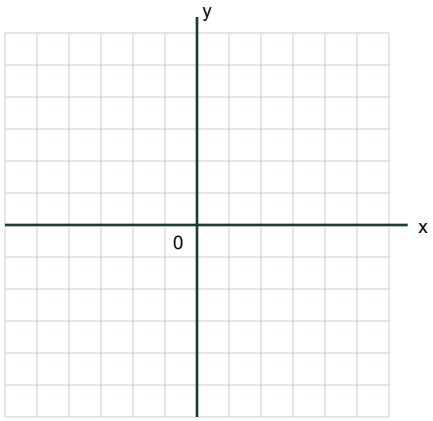
Answer: 8 by 8 units; area 64 square units

Day 109 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

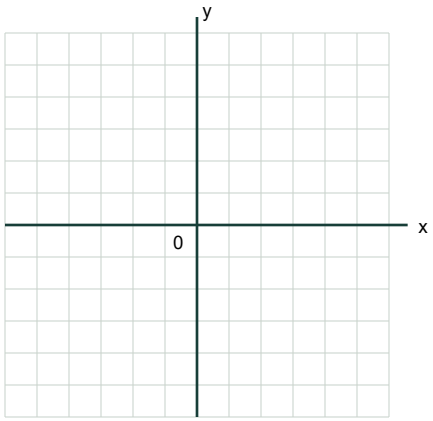
Plot and connect (0,0), (8,0), (8,7), (0,7). Find the side lengths and area.



Label the scale and units.

02 / Constructed response

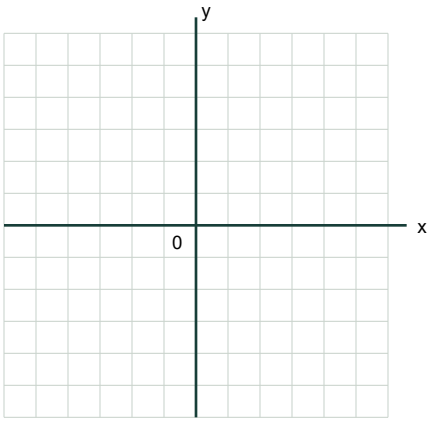
Plot and connect (0,0), (8,0), (8,5), (0,5). Find the side lengths and area.



Label the scale and units.

03 / Constructed response

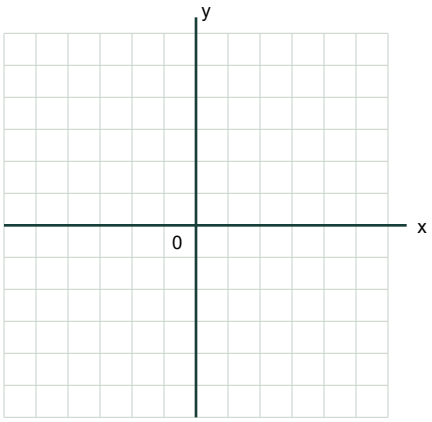
Plot and connect (0,0), (7,0), (7,12), (0,12). Find the side lengths and area.



Label the scale and units.

04 / Constructed response

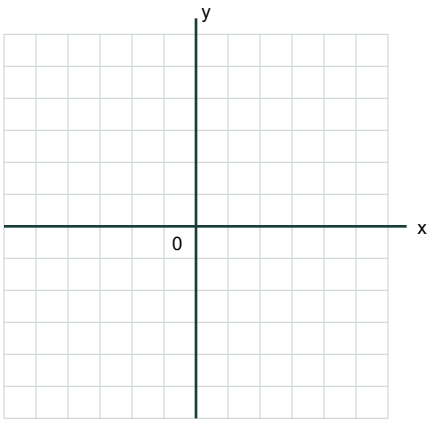
Plot and connect (-3,-5), (6,-5), (6,6), (-3,6). Find the side lengths and area.



Label the scale and units.

05 / Constructed response

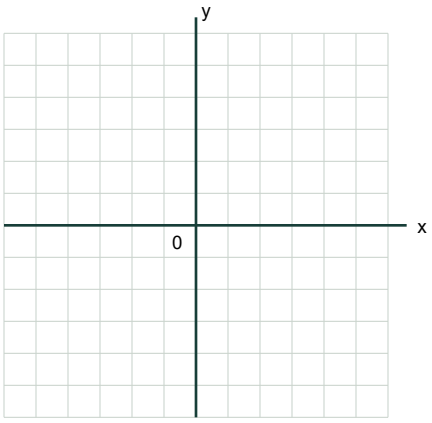
Plot and connect (-4,-2), (2,-2), (2,2), (-4,2). Find the side lengths and area.



Label the scale and units.

06 / Constructed response

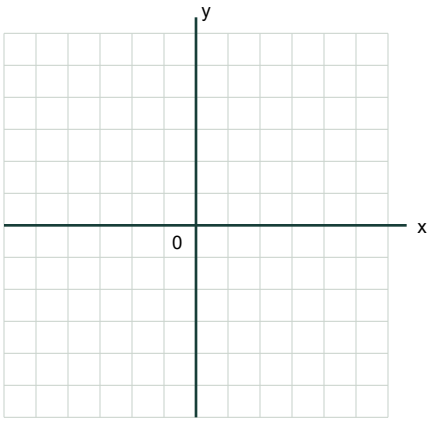
Plot and connect (-3,-4), (2,-4), (2,2), (-3,2). Find the side lengths and area.



Label the scale and units.

07 / Constructed response

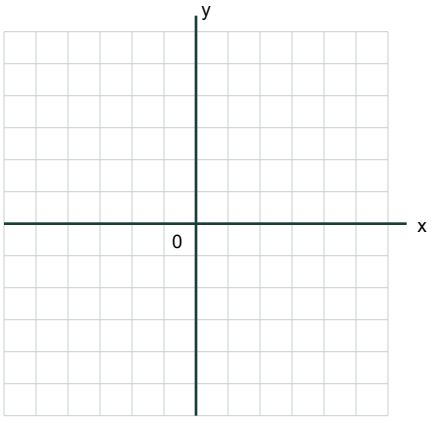
Plot and connect (-8,-3), (6,-3), (6,6), (-8,6). Find the side lengths and area. Find the perimeter too.



Label the scale and units.

08 / Coordinate plotting and two-part response

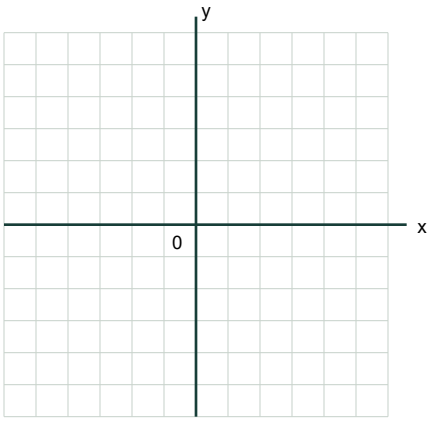
A rectangle has vertices (-6, -4), (4, -4), (4, 4), and (-6, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

09 / Coordinate plotting and two-part response

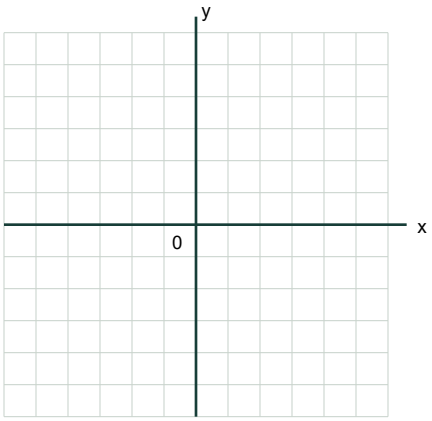
A rectangle has vertices (-5, -4), (5, -4), (5, 4), and (-5, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

10 / Constructed response

Plot and connect (-7,-3), (4,-3), (4,4), (-7,4). Find the side lengths and area. Find the perimeter too.



Label the scale and units.

Day 110 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Justify Polygon Area with Coordinates

Standards: 6.G.A.3, 6.G.A.1

Learning target: I can justify Polygon Area with Coordinates, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure.

Worked example 1

Plot and connect (0,0), (5,0), (5,9), (0,9). Find the side lengths and area.

1. Plot vertices in order and connect them, including the last to the first.

2. Equal x-values make a vertical side; equal y-values make a horizontal side.

3. Horizontal length 5 - (0) = 5

4. vertical length 9 - (0) = 9

5. area = 5 × 9.

6. State the result in the form requested: Sides 5 and 9; area 45 square units.

Answer: Sides 5 and 9; area 45 square units

Worked example 2

A 7 m by 9 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 5 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

1. Sketch the figure and label every given length.

2. Outer area = 63.

3. Triangle = 4 × 5/2 = 10.

4. Subtract the removed region exactly once.

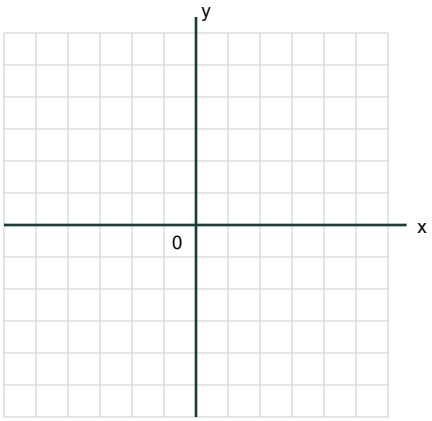
Answer: Removed 10 m²; remaining 53 m²

Day 110 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Plot and connect (0,0), (11,0), (11,5), (0,5). Find the side lengths and area.



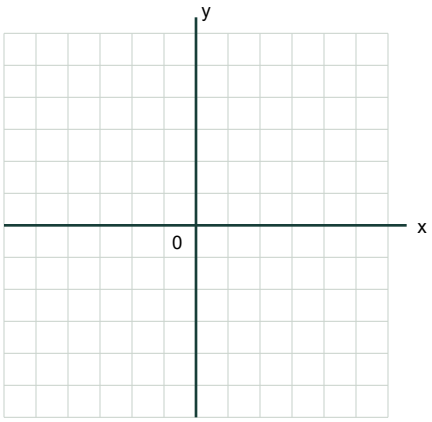
Label the scale and units.

02 / Constructed response

An L-shaped garden is a 13 m by 10 m rectangle with an 8 m by 5 m corner rectangle removed. Sketch and find the garden area.

03 / Constructed response

Plot and connect (0,0), (11,0), (11,10), (0,10). Find the side lengths and area.



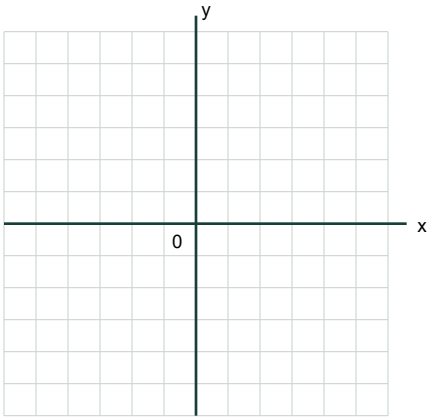
Label the scale and units.

04 / Constructed response

An L-shaped garden is a 6 m by 10 m rectangle with an 3 m by 7 m corner rectangle removed. Sketch and find the garden area.

05 / Constructed response

Plot and connect (-8,-4), (3,-4), (3,3), (-8,3). Find the side lengths and area.



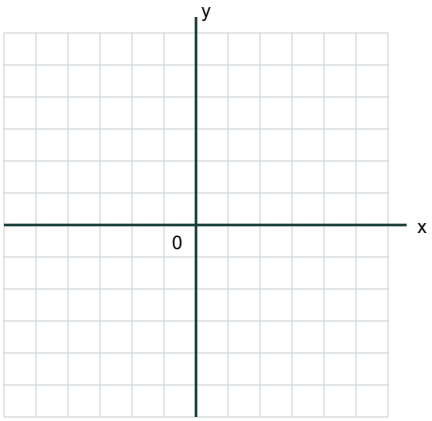
Label the scale and units.

06 / Constructed response

An L-shaped garden is a 6 m by 5 m rectangle with an 3 m by 2 m corner rectangle removed. Sketch and find the garden area.

07 / Constructed response

Plot and connect (-2,-4), (6,-4), (6,6), (-2,6). Find the side lengths and area. Find the perimeter too.



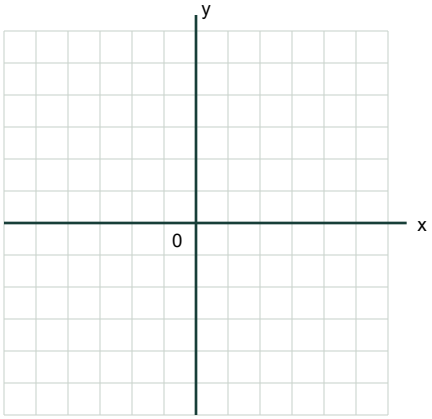
Label the scale and units.

08 / Two-part numeric entry

A 9 m by 9 m rectangle has a right-triangle corner removed with perpendicular legs 3 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

09 / Coordinate plotting and two-part response

A rectangle has vertices (-2, -2), (4, -2), (4, 2), and (-2, 2). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

10 / Constructed response

A 10 m by 11 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 7 m. Sketch the remaining polygon and find its area.

Day 111 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Build and Interpret Nets

Standards: 6.G.A.4

Learning target: I can draw a net and account for every face of a solid.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

A net unfolds all faces into a plane without overlap. Rectangular prisms have three matching pairs of rectangles. Pyramids have a polygonal base and triangular sides. Add face areas, not edge lengths.

Worked example 1

Draw a net for a closed 7 × 6 × 2 cm rectangular prism. Label the six faces and calculate total area.

1. A net unfolds all faces into a plane without overlap.

2. Rectangular prisms have three matching pairs of rectangles.

3. Add 2(42) + 2(14) + 2(12) = 136.

4. State the result in the form requested: 2 faces each of 7×6, 7×2, 6×2; 136 cm².

Answer: 2 faces each of 7×6, 7×2, 6×2; 136 cm²

Worked example 2

A closed rectangular prism is 7 cm by 2 cm by 2 cm. Select all true descriptions of its net.
A. Two faces are 7 by 2.
B. Its net contains exactly four triangles.
C. Two faces are 7 by 2 and two are 2 by 2.
D. Adding just the three distinct face areas counts every face.

1. A net unfolds all faces into a plane without overlap.

2. A closed rectangular prism has 6 rectangular faces in 3 matching pairs.

3. Count both copies of each face.

Answer: A, C - Two faces are 7 by 2.; Two faces are 7 by 2 and two are 2 by 2.

Day 111 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Draw a net for a closed 4 × 4 × 4 cm rectangular prism. Label the six faces and calculate total area.

02 / Constructed response

Draw a net for a closed 8 × 3 × 2 cm rectangular prism. Label the six faces and calculate total area.

03 / Constructed response

Draw a net for a closed 7 × 5 × 2 cm rectangular prism. Label the six faces and calculate total area.

04 / Constructed response

Draw a net for a closed 3 × 7 × 5 cm rectangular prism. Label the six faces and calculate total area.

05 / Constructed response

Draw a net for a closed 8 × 7 × 4 cm rectangular prism. Label the six faces and calculate total area.

06 / Constructed response

Draw a net for a closed 2 × 4 × 3 cm rectangular prism. Label the six faces and calculate total area.

07 / Constructed response

Draw a net for a triangular prism. Its two congruent right-triangle bases have legs 3 cm and 4 cm and hypotenuse 5 cm; prism length is 4 cm. Label all faces and find surface area.

08 / Multiple select

A closed rectangular prism is 6 cm by 4 cm by 4 cm. Select all true descriptions of its net.
A. Two faces are 6 by 4.
B. Its net contains exactly four triangles.
C. Two faces are 6 by 4 and two are 4 by 4.
D. Adding just the three distinct face areas counts every face.

09 / Multiple select

A closed rectangular prism is 6 cm by 5 cm by 3 cm. Select all true descriptions of its net.
A. Its net contains exactly four triangles.
B. Adding just the three distinct face areas counts every face.
C. Two faces are 6 by 5.
D. Two faces are 6 by 3 and two are 5 by 3.

10 / Constructed response

Draw a net for a triangular prism. Its two congruent right-triangle bases have legs 3 cm and 4 cm and hypotenuse 5 cm; prism length is 6 cm. Label all faces and find surface area.

Day 112 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Find Prism Surface Area

Standards: 6.G.A.4

Learning target: I can use a rectangular-prism net to find surface area.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it.

Worked example 1

A 4 cm by 2 cm base and 4 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

1. A closed rectangular prism has two faces of area lw, two of lh, and two of wh.

2. Add the six face areas: SA = 2lw + 2lh + 2wh.

3. Closed surface area = 2(8+16+8) = 64.

4. State the result in the form requested: 64 cm².

Answer: 64 cm²

Worked example 2

A closed box is 3 cm by 6 cm by 4 cm. Part A: enter its surface area. Part B: the 3-by-6 top is removed; enter the remaining surface area and explain.

1. A closed rectangular prism has two faces of area lw, two of lh, and two of wh.

2. Closed area = 2(18+12+24) = 108.

3. Subtract exactly one top face, 18, for the open box.

Answer: 108 cm²; open box 90 cm²

Day 112 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A 4 cm by 7 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

02 / Constructed response

A 8 cm by 3 cm base and 6 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

03 / Constructed response

A 6 cm by 5 cm base and 2 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A 3 cm by 4 cm base and 2 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

05 / Constructed response

A 5 cm by 6 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

06 / Constructed response

A 8 cm by 4 cm base and 5 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

07 / Constructed response

A 6 cm by 6 cm base and 2 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Two-part numeric entry

A closed box is 4 cm by 2 cm by 5 cm. Part A: enter its surface area. Part B: the 4-by-2 top is removed; enter the remaining surface area and explain.

09 / Two-part numeric entry

A closed box is 2 cm by 5 cm by 4 cm. Part A: enter its surface area. Part B: the 2-by-5 top is removed; enter the remaining surface area and explain.

10 / Constructed response

A 7 cm by 3 cm base and 2 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

Day 113 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Find Pyramid Surface Area

Standards: 6.G.A.4

Learning target: I can find pyramid surface area from its triangular faces and base.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

For a square pyramid, draw one square and four triangles. Each triangle has area base edge × face height / 2. The face height is measured on the triangular face, not vertically inside the solid.

Worked example 1

A square pyramid has base edge 6 cm. Each triangular face has perpendicular face height 6 cm. Draw a net and calculate total surface area.

1. For a square pyramid, draw one square and four triangles.

2. Each triangle has area base edge × face height / 2.

3. Base = 36

4. each triangle = 6 × 6 ÷ 2 = 18

5. add four triangles to get 108.

6. State the result in the form requested: 108 cm².

Answer: 108 cm²

Worked example 2

A square pyramid net has base side 8 cm and four triangular faces, each with perpendicular face height 7 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

1. For a square pyramid, draw one square and four triangles.

2. One face = (8 × 7)/2 = 28.

3. Four faces total 112.

4. Add square base area 64.

Answer: Face 28 cm²; total 176 cm²

Day 113 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A square pyramid has base edge 12 cm. Each triangular face has perpendicular face height 8 cm. Draw a net and calculate total surface area.

02 / Constructed response

A square pyramid has base edge 10 cm. Each triangular face has perpendicular face height 12 cm. Draw a net and calculate total surface area.

03 / Constructed response

A square pyramid has base edge 12 cm. Each triangular face has perpendicular face height 13 cm. Draw a net and calculate total surface area.

04 / Constructed response

A square pyramid has base edge 4 cm. Each triangular face has perpendicular face height 6 cm. Draw a net and calculate total surface area.

05 / Constructed response

A square pyramid has base edge 8 cm. Each triangular face has perpendicular face height 8 cm. Draw a net and calculate total surface area.

06 / Constructed response

A square pyramid has base edge 8 cm. Each triangular face has perpendicular face height 7 cm. Draw a net and calculate total surface area.

07 / Constructed response

A square pyramid has base edge 6 cm. Each triangular face has perpendicular face height 7 cm. Draw a net and calculate total surface area. Explain why the vertical pyramid height is not needed.

08 / Net reasoning and numeric entry

A square pyramid net has base side 12 cm and four triangular faces, each with perpendicular face height 8 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

09 / Net reasoning and numeric entry

A square pyramid net has base side 12 cm and four triangular faces, each with perpendicular face height 10 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

10 / Constructed response

A square pyramid has base edge 12 cm. Each triangular face has perpendicular face height 11 cm. Draw a net and calculate total surface area. Explain why the vertical pyramid height is not needed.

Day 114 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Find Volume with Fractional Edge Lengths

Standards: 6.G.A.2

Learning target: I can calculate prism volume with fractional edge lengths.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

A rectangular prism has edges 6, 1, and 2 cm. Sketch it and find its volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Fractional edge lengths still multiply.

3. B = 6 × 1 = 6 cm²

4. V = B × 2 = 12 cm³.

5. State the result in the form requested: 12 cm³.

Answer: 12 cm³

Worked example 2

A prism measures 13/2 m by 3 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 13/2 × 3 × 3/2 = 117/4.

3. The edge counts are 13, 6, 3, giving 234 cubes.

4. Each cube is 1/8 m³

5. 234/8 = 117/4.

Answer: 117/4 m³; 234 cubes

Day 114 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A rectangular prism has edges 2, 2, and 5 cm. Sketch it and find its volume.

02 / Constructed response

A rectangular prism has edges 4, 1, and 5 cm. Sketch it and find its volume.

03 / Constructed response

A rectangular prism has edges 2, 3, and 3 cm. Sketch it and find its volume.

04 / Constructed response

A rectangular prism has edges 2 1/2, 2 1/2, and 2 cm. Sketch it and find its volume.

05 / Constructed response

A rectangular prism has edges 1 1/2, 2 1/2, and 3 cm. Sketch it and find its volume.

06 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 5 cm. Sketch it and find its volume.

07 / Constructed response

A rectangular prism has edges 3, 2 1/2, and 1 cm. Sketch it and find its volume. Show both lwh and Bh.

08 / Numeric entry and model reasoning

A prism measures 7/2 m by 3/2 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Numeric entry and model reasoning

A prism measures 15/2 m by 3/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

10 / Constructed response

A rectangular prism has edges 1, 2 1/2, and 3 cm. Sketch it and find its volume. Show both lwh and Bh.

Day 115 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Connect Surface Area and Volume Models

Standards: 6.G.A.4, 6.G.A.2

Learning target: I can connect Surface Area and Volume Models, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

A 2 cm by 5 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

1. A closed rectangular prism has two faces of area lw, two of lh, and two of wh.

2. Add the six face areas: SA = 2lw + 2lh + 2wh.

3. Closed surface area = 2(10+6+15) = 62.

4. State the result in the form requested: 62 cm².

Answer: 62 cm²

Worked example 2

A prism measures 11/2 m by 2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 11/2 × 2 × 3/2 = 33/2.

3. The edge counts are 11, 4, 3, giving 132 cubes.

4. Each cube is 1/8 m³

5. 132/8 = 33/2.

Answer: 33/2 m³; 132 cubes

Day 115 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A 8 cm by 6 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

02 / Constructed response

A rectangular prism has edges 4, 3 1/2, and 2 cm. Sketch it and find its volume.

03 / Constructed response

A 7 cm by 6 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A rectangular prism has edges 3, 2, and 5 cm. Sketch it and find its volume.

05 / Constructed response

A 4 cm by 4 cm base and 6 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

06 / Constructed response

A rectangular prism has edges 3 1/2, 1 1/2, and 4 cm. Sketch it and find its volume.

07 / Constructed response

A 5 cm by 3 cm base and 5 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Numeric entry and model reasoning

A prism measures 7/2 m by 2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Two-part numeric entry

A closed box is 2 cm by 2 cm by 5 cm. Part A: enter its surface area. Part B: the 2-by-2 top is removed; enter the remaining surface area and explain.

10 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 1 1/2 cm. Sketch it and find its volume. Show both lwh and Bh.

Day 116 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Connect Fractional Unit Cubes and Volume

Standards: 6.G.A.2

Learning target: I can connect fractional unit cubes with a prism volume formula.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit. Divide each prism edge by 1/n to count cubes along it. Multiply the counts, then multiply by the volume of one small cube.

Worked example 1

Fill a prism 3/2 × 7/2 × 3 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

1. For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit.

2. Divide each prism edge by 1/n to count cubes along it.

3. Counts along edges: 3, 7, 6.

4. Product 126

5. 126 × 1/8 = 63/4, also lwh.

6. State the result in the form requested: 126 cubes; each 1/8 cubic unit; prism 63/4 cubic units.

Answer: 126 cubes; each 1/8 cubic unit; prism 63/4 cubic units

Worked example 2

A prism measures 11/2 m by 1 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. For cubes with edge 1/n, each cube has volume 1/n^3 cubic unit.

2. Multiply edges: 11/2 × 1 × 1 = 11/2.

3. The edge counts are 11, 2, 2, giving 44 cubes.

4. Each cube is 1/8 m^3

5. 44/8 = 11/2.

Answer: 11/2 m^3; 44 cubes

Day 116 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Fill a prism 3 × 5/2 × 3/2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

02 / Constructed response

Fill a prism 3/2 × 3/2 × 5/2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

03 / Constructed response

Fill a prism 5/2 × 7/2 × 2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

04 / Constructed response

Fill a prism 5/2 × 5/2 × 5/2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

05 / Constructed response

Fill a prism 7/2 × 2 × 3 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

06 / Constructed response

Fill a prism 4 × 3/2 × 2 units with cubes of edge 1/2 unit. How many cubes fit? Use their volumes to verify lwh.

07 / Constructed response

Fill a prism 5/3 × 4/3 × 5/3 units with cubes of edge 1/3 unit. How many cubes fit? Use their volumes to verify lwh.

08 / Numeric entry and model reasoning

A prism measures 13/2 m by 3/2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Numeric entry and model reasoning

A prism measures 15/2 m by 2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

10 / Constructed response

Fill a prism 5/3 × 1 × 2 units with cubes of edge 1/3 unit. How many cubes fit? Use their volumes to verify lwh.

Day 117 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Use Base Area to Find Prism Volume

Standards: 6.G.A.2

Learning target: I can calculate prism volume with fractional edge lengths.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

A rectangular prism has edges 4, 2 1/2, and 4 cm. Sketch it and find its volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Fractional edge lengths still multiply.

3. B = 4 × 5/2 = 10 cm²

4. V = B × 4 = 40 cm³.

5. State the result in the form requested: 40 cm³.

Answer: 40 cm³

Worked example 2

A prism measures 9/2 m by 3/2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 9/2 × 3/2 × 3/2 = 81/8.

3. The edge counts are 9, 3, 3, giving 81 cubes.

4. Each cube is 1/8 m³

5. 81/8 = 81/8.

Answer: 81/8 m³; 81 cubes

Day 117 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A rectangular prism has edges 7, 3 1/2, and 3 cm. Sketch it and find its volume.

02 / Constructed response

A rectangular prism has edges 7, 2, and 4 cm. Sketch it and find its volume.

03 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 3 cm. Sketch it and find its volume.

04 / Constructed response

A rectangular prism has edges 2, 1 1/2, and 6 cm. Sketch it and find its volume.

05 / Constructed response

A rectangular prism has edges 2 1/2, 2 1/2, and 6 cm. Sketch it and find its volume.

06 / Constructed response

A rectangular prism has edges 3, 1, and 6 cm. Sketch it and find its volume.

07 / Constructed response

A rectangular prism has edges 4, 1, and 2 cm. Sketch it and find its volume. Show both lwh and Bh.

08 / Numeric entry and model reasoning

A prism measures 5/2 m by 3/2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Numeric entry and model reasoning

A prism measures 7/2 m by 3/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

10 / Constructed response

A rectangular prism has edges 2 1/2, 2, and 2 cm. Sketch it and find its volume. Show both lwh and Bh.

Day 118 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Distinguish Area, Surface Area, and Volume

Standards: 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can distinguish Area, Surface Area, and Volume, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2. A slanted side is not necessarily the height. Label area in square units. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

Sketch a triangle with base 8 cm and perpendicular height 5 cm. Find its area.

1. Two matching triangles form a parallelogram, so A = (base × perpendicular height)/2.

2. A slanted side is not necessarily the height.

3. A = bh/2 = 8 × 5 ÷ 2 = 20.

4. Two such triangles make a parallelogram.

5. State the result in the form requested: 20 cm².

Answer: 20 cm²

Worked example 2

A prism measures 11/2 m by 5/2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 11/2 × 5/2 × 3/2 = 165/8.

3. The edge counts are 11, 5, 3, giving 165 cubes.

4. Each cube is 1/8 m³

5. 165/8 = 165/8.

Answer: 165/8 m³; 165 cubes

Day 118 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Sketch a triangle with base 8 cm and perpendicular height 7 cm. Find its area.

02 / Constructed response

A 7 cm by 4 cm base and 4 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

03 / Constructed response

A rectangular prism has edges 6, 3, and 4 cm. Sketch it and find its volume.

04 / Constructed response

Sketch a triangle with base 6 cm and perpendicular height 7 cm. Find its area.

05 / Constructed response

A 6 cm by 2 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

06 / Constructed response

A rectangular prism has edges 2, 1, and 3 cm. Sketch it and find its volume.

07 / Constructed response

Sketch a triangle with base 10 cm and perpendicular height 7/2 cm. Find its area. Explain why the slanted side is not needed.

08 / Two-part numeric entry

A closed box is 5 cm by 3 cm by 4 cm. Part A: enter its surface area. Part B: the 5-by-3 top is removed; enter the remaining surface area and explain.

09 / Numeric entry and model reasoning

A prism measures 5/2 m by 2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

10 / Constructed response

Sketch a triangle with base 4 cm and perpendicular height 3 cm. Find its area. Explain why the slanted side is not needed.

Day 119 | Unit 08 | Lesson

Solve Area, Surface Area, and Volume Problems

Model a Geometry Design

Standards: 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can model a Geometry Design, and explain and check my reasoning.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

01 / Learn

Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. Sketch the figure and label every given length. Split the region into nonoverlapping rectangles and triangles, or subtract a cutout from a larger rectangle. Add only the areas that belong to the figure. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

Plot and connect (0,0), (7,0), (7,8), (0,8). Find the side lengths and area.

1. Plot vertices in order and connect them, including the last to the first.

2. Equal x-values make a vertical side; equal y-values make a horizontal side.

3. Horizontal length 7 - (0) = 7

4. vertical length 8 - (0) = 8

5. area = 7 × 8.

6. State the result in the form requested: Sides 7 and 8; area 56 square units.

Answer: Sides 7 and 8; area 56 square units

Worked example 2

A prism measures 13/2 m by 5/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 13/2 × 5/2 × 1 = 65/4.

3. The edge counts are 13, 5, 2, giving 130 cubes.

4. Each cube is 1/8 m^3

5. 130/8 = 65/4.

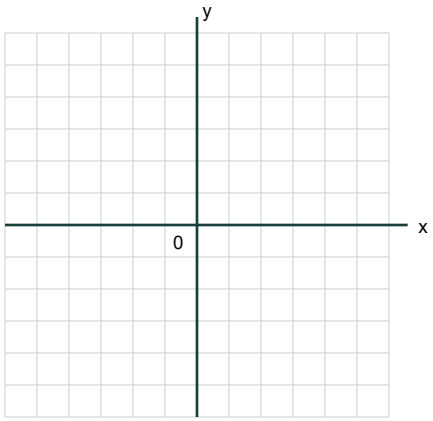
Answer: 65/4 m^3; 130 cubes

Day 119 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Plot and connect (0,0), (11,0), (11,8), (0,8). Find the side lengths and area.



Label the scale and units.

02 / Constructed response

An L-shaped garden is a 9 m by 10 m rectangle with an 6 m by 7 m corner rectangle removed. Sketch and find the garden area.

03 / Constructed response

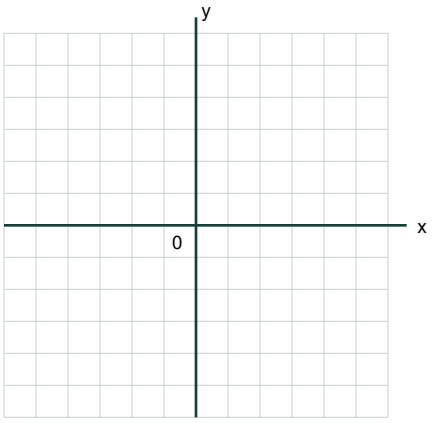
A 8 cm by 5 cm base and 3 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A rectangular prism has edges 2, 1, and 2 cm. Sketch it and find its volume.

05 / Constructed response

Plot and connect (-5,-5), (2,-5), (2,2), (-5,2). Find the side lengths and area.



Label the scale and units.

06 / Constructed response

An L-shaped garden is a 9 m by 9 m rectangle with an 3 m by 3 m corner rectangle removed. Sketch and find the garden area.

07 / Constructed response

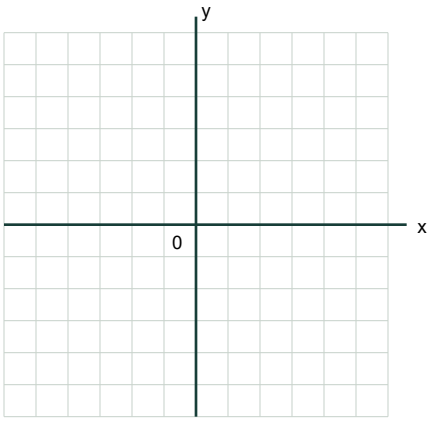
A 2 cm by 3 cm base and 2 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Numeric entry and model reasoning

A prism measures 7/2 m by 3/2 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Coordinate plotting and two-part response

A rectangle has vertices (-7, -4), (4, -4), (4, 4), and (-7, 4). Plot and connect them. Part A: find its side lengths. Part B: enter its area in square units.



Label the scale and units.

10 / Constructed response

A 11 m by 9 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 5 m. Sketch the remaining polygon and find its area.

Day 120 | Unit 08 | Unit Test

Solve Area, Surface Area, and Volume Problems

Unit 8 Test

Standards: 6.RP.A.3c, 6.G.A.1, 6.G.A.3, 6.G.A.4, 6.G.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 7: Solve Area, Surface Area, and Volume Problems | Spectrum companion Chapter 6

Directions

The first 4 questions (20%) review Unit 7. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Write 25% as a simplified fraction and decimal.

02 / Previous unit review | Constructed response

Write 10% as a simplified fraction and decimal.

03 / Previous unit review | Constructed response

Write 290% as a simplified fraction and a decimal. Explain why its decimal is greater than 1.

04 / Previous unit review | Constructed response

Find 50% of 100.

05 / Constructed response

Sketch a triangle with base 12 cm and perpendicular height 6 cm. Find its area.

06 / Constructed response

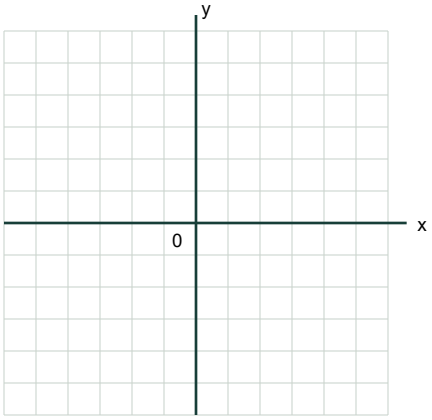
A parallelogram has base 5 m and perpendicular height 2 m. Sketch how it can be rearranged into a rectangle and find its area.

07 / Constructed response

An L-shaped garden is a 4 m by 7 m rectangle with an 2 m by 5 m corner rectangle removed. Sketch and find the garden area.

08 / Constructed response

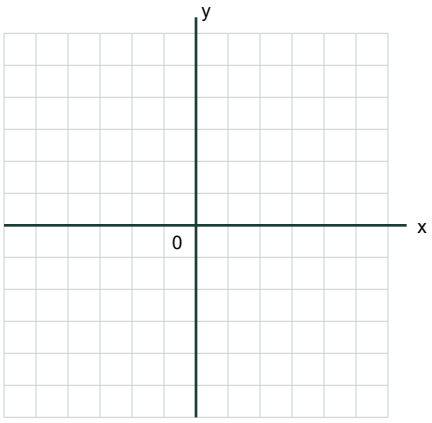
Plot and connect (-5,-5), (2,-5), (2,2), (-5,2). Find the side lengths and area.



Label the scale and units.

09 / Constructed response

Plot and connect (-6,-3), (5,-3), (5,5), (-6,5). Find the side lengths and area.



Label the scale and units.

10 / Constructed response

Draw a net for a closed 4 × 7 × 5 cm rectangular prism. Label the six faces and calculate total area.

11 / Constructed response

A 5 cm by 6 cm base and 5 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

12 / Net reasoning and numeric entry

A square pyramid net has base side 12 cm and four triangular faces, each with perpendicular face height 10 cm. Part A: enter one triangular face area. Part B: enter the entire surface area.

13 / Numeric entry and model reasoning

A prism measures 9/2 m by 1 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

14 / Two-part numeric entry

A closed box is 6 cm by 4 cm by 5 cm. Part A: enter its surface area. Part B: the 6-by-4 top is removed; enter the remaining surface area and explain.

15 / Numeric entry and model reasoning

A prism measures 11/2 m by 3/2 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

16 / Numeric entry and model reasoning

A prism measures 7/2 m by 1 m by 3/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

17 / Two-part numeric entry

A closed box is 5 cm by 6 cm by 3 cm. Part A: enter its surface area. Part B: the 5-by-6 top is removed; enter the remaining surface area and explain.

18 / Two-part numeric entry

A 10 m by 11 m rectangle has a right-triangle corner removed with perpendicular legs 5 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

19 / Multiple select

A triangle has base 8 cm and perpendicular height 3 cm. Select all expressions that give its area in cm².
A. 4 × 3
B. (8 + 3) ÷ 2
C. (8 × 3) ÷ 2
D. 8 × 3

20 / Numeric entry and error analysis

A trapezoid has parallel bases 6 m and 14 m, and perpendicular height 3 m. Part A: enter its area. Part B: a student multiplies the longer base by the height; explain why that overcounts.

Day 121 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Recognize Statistical Questions

Standards: 6.SP.A.1

Learning target: I can recognize a statistical question and the expected variability.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

A statistical question anticipates different answers in a collection of data. Name the group, the measured attribute, and the expected variation. A question about one fixed fact is not statistical.

Worked example 1

Is “What are the wingspans, in centimeters, of the birds observed at a feeder?” statistical? Explain the group, attribute, units, and variation.

1. A statistical question anticipates different answers in a collection of data.

2. Name the group, the measured attribute, and the expected variation.

3. The question anticipates a collection of measurements.

4. Each bird can have a different wingspan.

5. Use the same unit for every observation.

6. State the result in the form requested: Yes; observed birds; wingspan; centimeters; wingspans can vary..

Answer: Yes; observed birds; wingspan; centimeters; wingspans can vary.

Worked example 2

A class studies a collection of seedlings. Select all statistical questions, and identify the expected variability.
A. How does height vary among the seedlings?
B. What is 6 + 2?
C. What is the height in centimeters of each of the seedlings?
D. How many centimeters are in 6 meters?

1. A statistical question anticipates different answers in a collection of data.

2. Different seedlings can have different height.

3. The questions anticipate variation across a collection.

4. The conversion and arithmetic questions have fixed answers.

Answer: A, C - How does height vary among the seedlings?; What is the height in centimeters of each of the seedlings?

Day 121 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Is “How many pages did each student read last night?” statistical? Explain.

02 / Constructed response

Is “How many centimeters are in one meter?” statistical? Explain.

03 / Constructed response

Is “What are the heights of the plants in our garden?” statistical? Explain.

04 / Constructed response

Identify the group, attribute, and units in: How many minutes do students in our class sleep each night?

05 / Constructed response

Rewrite “How long is my pencil?” as a statistical question.

06 / Constructed response

A student asks only one person’s age once. Explain why “How old is this person today?” is not statistical.

07 / Constructed response

Write a statistical question about travel to school. Name the group, attribute, units, and why answers may vary.

08 / Multiple select

A class studies a collection of students. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 4 meters?
B. What is the reading time in minutes of each of the students?
C. What is 4 + 5?
D. How does reading time vary among the students?

09 / Multiple select

A class studies a collection of backpacks. Select all statistical questions, and identify the expected variability.
A. How does mass vary among the backpacks?
B. How many centimeters are in 3 meters?
C. What is the mass in kilograms of each of the backpacks?
D. What is 3 + 4?

10 / Constructed response

Critique: “A question is statistical whenever it contains numbers.” Give a counterexample and a corrected rule.

Day 122 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Summarize Data with Mean and Median

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c

Learning target: I can summarize Data with Mean and Median, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

The mean is a fair share: add all values and divide by the number of observations. Report the count and measurement units. A large unusual value can pull the mean toward it. Order the data first. For an odd count, choose the middle value. For an even count, average the two middle values. The median often resists an unusually high or low value better than the mean.

Worked example 1

Reading times in minutes are 5, 6, 6, 7, 7, 8. Report n, units, and mean.

1. The mean is a fair share: add all values and divide by the number of observations.

2. Report the count and measurement units.

3. Sum = 39

4. mean = 39 ÷ 6 = 13/2.

5. State the result in the form requested: n = 6; minutes; mean 13/2 min.

Answer: n = 6; minutes; mean 13/2 min

Worked example 2

Six students recorded reading times (minutes): 4, 6, 6, 8, 8, 10. Select all true statements.
A. The median is always the largest observation.
B. The median is 7 minutes.
C. The median is 8 minutes.
D. The median is the average of the 3rd and 4th ordered values.

1. Order the data first.

2. Six ordered observations have middle positions 3 and 4, so median = (6 + 8)/2 = 7.

Answer: B, D - The median is 7 minutes.; The median is the average of the 3rd and 4th ordered values.

Day 122 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Report n, units, and mean.

02 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Find the median. Show the middle positions.

03 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Report n, units, and mean.

04 / Constructed response

Reading times in minutes are 8, 12, 12, 16, 16, 20. Find the median. Show the middle positions.

05 / Constructed response

Reading times in minutes are 5, 8, 8, 11, 11, 14. Report n, units, and mean.

06 / Constructed response

Reading times in minutes are 3, 6, 6, 9, 9, 12. Find the median. Show the middle positions.

07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Report n, units, and mean. If one more student reads 6 minutes, find the new mean.

08 / Multiple select

Six students recorded reading times (minutes): 3, 8, 8, 13, 13, 18. Select all true statements.
A. The median is the average of the 3rd and 4th ordered values.
B. The median is 23/2 minutes.
C. The median is always the largest observation.
D. The median is 21/2 minutes.

09 / Two-part numeric entry

Six students recorded reading times (minutes): 3, 9, 9, 15, 15, 21. Part A: enter the mean. Part B: a seventh student reads 27 minutes. Enter the new mean and explain the denominator.

10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Find the median and mean. Explain the difference in this set.

Day 123 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Display and Interpret Dot Plots

Standards: 6.SP.B.4

Learning target: I can construct and interpret a dot plot on a number line.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count.

Worked example 1

Reading times in minutes are 7, 8, 8, 9, 9, 10. Draw a labeled dot plot. Check the number of dots.

1. Draw an equally spaced number line and stack one dot per observation above its value.

2. Repeated values stack vertically.

3. Each observation contributes one dot

4. stack repeated values at their number-line position.

5. State the result in the form requested: 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6.

Answer: 7: 1 dot; 8: 2 dots; 9: 2 dots; 10: 1 dot; n = 6

Worked example 2

A frequency table records books read: 0 books: 1 students; 1: 5; 2: 4; 3: 1. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.

1. Draw an equally spaced number line and stack one dot per observation above its value.

2. Place one dot per student, not one per distinct value.

3. Total dots = 1 + 5 + 4 + 1 = 11.

4. Label the number line books read.

Answer: 11 students; stacks at 0, 1, 2, 3 have 1, 5, 4, 1 dots.

Day 123 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Draw a labeled dot plot. Check the number of dots.

02 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Draw a labeled dot plot. Check the number of dots.

03 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Draw a labeled dot plot. Check the number of dots.

04 / Constructed response

Reading times in minutes are 4, 6, 6, 8, 8, 10. Draw a labeled dot plot. Check the number of dots.

05 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Draw a labeled dot plot. Check the number of dots.

06 / Constructed response

Reading times in minutes are 5, 11, 11, 17, 17, 23. Draw a labeled dot plot. Check the number of dots.

07 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Draw a labeled dot plot. Describe a cluster or gap and report the count.

08 / Data display and numeric entry

A frequency table records books read: 0 books: 3 students; 1: 5; 2: 3; 3: 1. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

09 / Data display and numeric entry

A frequency table records books read: 0 books: 4 students; 1: 4; 2: 5; 3: 5. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

10 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 9, 18. Draw a labeled dot plot. Describe a cluster or gap and report the count.

Day 124 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Display Data in Box Plots

Standards: 6.SP.B.4, 6.SP.B.5c

Learning target: I can create a box plot from a five-number summary.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Find minimum, Q1, median, Q3, and maximum. Draw a box from Q1 to Q3 with a median line and whiskers to the endpoints. Use an equally spaced number line with units.

Worked example 1

Reading times in minutes are 4, 5, 5, 6, 6, 7. Find the five-number summary and draw a labeled box plot.

1. Find minimum, Q1, median, Q3, and maximum.

2. Draw a box from Q1 to Q3 with a median line and whiskers to the endpoints.

3. Box runs from 5 to 6, median line at 11/2, whiskers to 4 and 7.

4. State the result in the form requested: (4, 5, 11/2, 6, 7).

Answer: (4, 5, 11/2, 6, 7)

Worked example 2

Six students recorded reading times (minutes): 5, 10, 10, 15, 15, 20. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.

1. Find minimum, Q1, median, Q3, and maximum.

2. Minimum and maximum anchor whiskers.

3. Box ends are 10, 15

4. median line is 25/2.

5. IQR = 15 - 10.

Answer: (5, 10, 25/2, 15, 20); IQR 5 min

Day 124 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Find the five-number summary and draw a labeled box plot.

02 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Find the five-number summary and draw a labeled box plot.

03 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Find the five-number summary and draw a labeled box plot.

04 / Constructed response

Reading times in minutes are 2, 6, 6, 10, 10, 14. Find the five-number summary and draw a labeled box plot.

05 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Find the five-number summary and draw a labeled box plot.

06 / Constructed response

Reading times in minutes are 2, 5, 5, 8, 8, 11. Find the five-number summary and draw a labeled box plot.

07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Find the five-number summary and draw a labeled box plot. State the IQR.

08 / Box plot and numeric entry

Six students recorded reading times (minutes): 7, 11, 11, 15, 15, 19. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

09 / Box plot and numeric entry

Six students recorded reading times (minutes): 2, 7, 7, 12, 12, 17. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Find the five-number summary and draw a labeled box plot. State the IQR.

Day 125 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Connect Data Values and Data Displays

Standards: 6.SP.B.4, 6.SP.B.5c

Learning target: I can connect Data Values and Data Displays, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count. Find minimum, Q1, median, Q3, and maximum. Draw a box from Q1 to Q3 with a median line and whiskers to the endpoints. Use an equally spaced number line with units.

Worked example 1

Reading times in minutes are 6, 7, 7, 8, 8, 9. Draw a labeled dot plot. Check the number of dots.

1. Draw an equally spaced number line and stack one dot per observation above its value.

2. Repeated values stack vertically.

3. Each observation contributes one dot

4. stack repeated values at their number-line position.

5. State the result in the form requested: 6: 1 dot; 7: 2 dots; 8: 2 dots; 9: 1 dot; n = 6.

Answer: 6: 1 dot; 7: 2 dots; 8: 2 dots; 9: 1 dot; n = 6

Worked example 2

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.

1. Find minimum, Q1, median, Q3, and maximum.

2. Minimum and maximum anchor whiskers.

3. Box ends are 5, 8

4. median line is 13/2.

5. IQR = 8 - 5.

Answer: (2, 5, 13/2, 8, 11); IQR 3 min

Day 125 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Draw a labeled dot plot. Check the number of dots.

02 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Find the five-number summary and draw a labeled box plot.

03 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Draw a labeled dot plot. Check the number of dots.

04 / Constructed response

Reading times in minutes are 3, 7, 7, 11, 11, 15. Find the five-number summary and draw a labeled box plot.

05 / Constructed response

Reading times in minutes are 8, 10, 10, 12, 12, 14. Draw a labeled dot plot. Check the number of dots.

06 / Constructed response

Reading times in minutes are 8, 10, 10, 12, 12, 14. Find the five-number summary and draw a labeled box plot.

07 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Draw a labeled dot plot. Describe a cluster or gap and report the count.

08 / Box plot and numeric entry

Six students recorded reading times (minutes): 3, 5, 5, 7, 7, 9. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

09 / Data display and numeric entry

A frequency table records books read: 0 books: 4 students; 1: 5; 2: 2; 3: 2. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Find the five-number summary and draw a labeled box plot. State the IQR.

Day 126 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Use Frequency Tables and Histograms

Standards: 6.SP.B.4

Learning target: I can construct and interpret a histogram with equal-width bins.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Group numerical values into adjacent, nonoverlapping intervals. Count each observation in exactly one interval, draw touching bars, and label the numerical axis and the frequency axis.

Worked example 1

Daily reading times in minutes: 13, 16, 14, 15, 0, 14. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

1. Group numerical values into adjacent, nonoverlapping intervals.

2. Count each observation in exactly one interval, draw touching bars, and label the numerical axis and the frequency axis.

3. Count every value once.

4. Bars touch and have equal width

5. horizontal axis is minutes, vertical axis is frequency.

6. State the result in the form requested: Frequencies 1, 0, 3, 2.

Answer: Frequencies 1, 0, 3, 2

Worked example 2

Reading-time frequencies (minutes) are 0–4: 3, 5–9: 4, 10–14: 2, 15–19: 2. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

1. Group numerical values into adjacent, nonoverlapping intervals.

2. Use four equal-width touching bars with heights 3, 4, 2, 2.

3. Frequencies sum to 11.

4. Include every interval with frequency 4.

Answer: 11 observations; highest intervals 5–9

Day 126 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Daily reading times in minutes: 5, 18, 18, 13, 9, 17. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

02 / Constructed response

Daily reading times in minutes: 18, 5, 14, 0, 19, 18. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

03 / Constructed response

Daily reading times in minutes: 18, 17, 14, 6, 13, 1. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

04 / Constructed response

Daily reading times in minutes: 0, 2, 5, 7, 2, 4, 15, 16, 9, 5. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

05 / Constructed response

Daily reading times in minutes: 14, 5, 9, 17, 7, 0, 15, 18, 8, 16. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

06 / Constructed response

Daily reading times in minutes: 14, 6, 10, 0, 1, 18, 6, 2, 11, 1. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

07 / Constructed response

Daily reading times in minutes: 7, 9, 13, 2, 15, 12, 16, 0, 17, 8. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes. Which interval has the greatest frequency? Include ties.

08 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 2, 5–9: 3, 10–14: 5, 15–19: 5. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

09 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 4, 5–9: 2, 10–14: 1, 15–19: 2. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

10 / Constructed response

Daily reading times in minutes: 16, 13, 0, 7, 10, 1, 2, 1, 4, 18. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes. Which interval has the greatest frequency? Include ties.

Day 127 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Find Quartiles and Interquartile Range

Standards: 6.SP.A.3, 6.SP.B.5c

Learning target: I can calculate quartiles and interquartile range.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Sort the data. Find the median, then the median of each half for Q1 and Q3. For an odd count, exclude the overall median from both halves in this course. IQR = Q3 - Q1; range = maximum - minimum.

Worked example 1

Reading times in minutes are 2, 3, 3, 4, 4, 5. Find Q1, Q3, IQR, and range.

1. Sort the data.

2. Find the median, then the median of each half for Q1 and Q3.

3. Lower half [2, 3, 3] has median 3

4. upper half [4, 4, 5] has median 4.

5. Subtract Q1 from Q3.

6. State the result in the form requested: Q1 3, Q3 4; IQR 1 min; range 3 min.

Answer: Q1 3, Q3 4; IQR 1 min; range 3 min

Worked example 2

Six students recorded reading times (minutes): 5, 11, 11, 17, 17, 23. Part A: enter Q1 and Q3. Part B: enter IQR and explain which portion of the data it describes.

1. Sort the data.

2. Lower-half median is 11

3. upper-half median is 17.

4. IQR = 17 - 11 = 6, the span of the middle half of the ordered data.

Answer: Q1 = 11; Q3 = 17; IQR = 6 min

Day 127 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Find Q1, Q3, IQR, and range.

02 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Find Q1, Q3, IQR, and range.

03 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Find Q1, Q3, IQR, and range.

04 / Constructed response

Reading times in minutes are 5, 11, 11, 17, 17, 23. Find Q1, Q3, IQR, and range.

05 / Constructed response

Reading times in minutes are 2, 5, 5, 8, 8, 11. Find Q1, Q3, IQR, and range.

06 / Constructed response

Reading times in minutes are 2, 7, 7, 12, 12, 17. Find Q1, Q3, IQR, and range.

07 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Find Q1, Q3, IQR, and range. Explain what IQR describes.

08 / Multi-field numeric entry and explanation

Six students recorded reading times (minutes): 7, 11, 11, 15, 15, 19. Part A: enter Q1 and Q3. Part B: enter IQR and explain which portion of the data it describes.

09 / Multi-field numeric entry and explanation

Six students recorded reading times (minutes): 6, 11, 11, 16, 16, 21. Part A: enter Q1 and Q3. Part B: enter IQR and explain which portion of the data it describes.

10 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 6, 15. Find Q1, Q3, IQR, and range. Explain what IQR describes.

Day 128 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Find Mean Absolute Deviation

Standards: 6.SP.B.5c

Learning target: I can calculate and interpret mean absolute deviation.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Find the mean. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations. MAD describes a typical distance from the mean in the original units.

Worked example 1

Reading times in minutes are 6, 7, 7, 8, 8, 9. Calculate the mean and MAD. Show all absolute deviations.

1. Find the mean.

2. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations.

3. Distances: 3/2, 1/2, 1/2, 1/2, 1/2, 3/2

4. sum 5

5. divide by 6 = 5/6.

6. State the result in the form requested: Mean 15/2 min; MAD 5/6 min.

Answer: Mean 15/2 min; MAD 5/6 min

Worked example 2

Six students recorded reading times (minutes): 4, 7, 7, 10, 10, 13. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

1. Find the mean.

2. Mean = 51/6 = 17/2.

3. Absolute deviations: 9/2, 3/2, 3/2, 3/2, 3/2, 9/2.

4. Their sum divided by 6 is 5/2.

5. This is the average distance from the mean.

Answer: Mean 17/2 min; MAD 5/2 min

Day 128 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Calculate the mean and MAD. Show all absolute deviations.

02 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Calculate the mean and MAD. Show all absolute deviations.

03 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Calculate the mean and MAD. Show all absolute deviations.

04 / Constructed response

Reading times in minutes are 3, 6, 6, 9, 9, 12. Calculate the mean and MAD. Show all absolute deviations.

05 / Constructed response

Reading times in minutes are 5, 9, 9, 13, 13, 17. Calculate the mean and MAD. Show all absolute deviations.

06 / Constructed response

Reading times in minutes are 8, 14, 14, 20, 20, 26. Calculate the mean and MAD. Show all absolute deviations.

07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Calculate the mean and MAD. Show all absolute deviations. Interpret MAD in a sentence.

08 / Numeric entry and interpretation

Six students recorded reading times (minutes): 2, 8, 8, 14, 14, 20. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

09 / Numeric entry and interpretation

Six students recorded reading times (minutes): 4, 6, 6, 8, 8, 10. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

10 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Calculate the mean and MAD. Show all absolute deviations. Interpret MAD in a sentence.

Day 129 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Choose Appropriate Statistical Measures

Standards: 6.SP.B.5d

Learning target: I can choose a measure of center and explain why.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Use the shape and context. Mean with mean absolute deviation can describe fairly symmetric data. Median with interquartile range often describes skewed data or data with an unusual value more fairly.

Worked example 1

Reading times in minutes are 5, 6, 6, 7, 7, 8. Choose a center and matching variability measure. Calculate both and justify your choice.

1. Use the shape and context.

2. Mean with mean absolute deviation can describe fairly symmetric data.

3. Mean = 13/2, median = 13/2, IQR = 7 - 6 = 1, MAD = 5/6.

4. Accept another justified choice tied to shape and purpose.

5. State the result in the form requested: Mean 13/2 min and MAD 5/6 min; roughly symmetric data.

Answer: Mean 13/2 min and MAD 5/6 min; roughly symmetric data

Worked example 2

Six students recorded reading times (minutes): 3, 9, 9, 15, 15, 21. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. Use the shape and context.

2. Order the observations.

3. Middle positions give median 12 and half medians give IQR 6.

4. Mean is 72/6

5. averaging distances gives MAD 5.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 12, IQR 6; mean 12, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Day 129 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Choose a center and matching variability measure. Calculate both and justify your choice.

02 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Choose a center and matching variability measure. Calculate both and justify your choice.

03 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Choose a center and matching variability measure. Calculate both and justify your choice.

04 / Constructed response

Reading times in minutes are 2, 8, 8, 14, 14, 20. Choose a center and matching variability measure. Calculate both and justify your choice.

05 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Choose a center and matching variability measure. Calculate both and justify your choice.

06 / Constructed response

Reading times in minutes are 5, 9, 9, 13, 13, 17. Choose a center and matching variability measure. Calculate both and justify your choice.

07 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 6, 15. Choose a center and matching variability measure. Calculate both and justify your choice.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 4, 8, 8, 12, 12, 16. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 19. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 10, 19. Choose a center and matching variability measure. Calculate both and justify your choice.

Day 130 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Critique Statistical Summaries

Standards: 6.SP.B.5d, 6.SP.A.2, 6.SP.A.3

Learning target: I can critique Statistical Summaries, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Use the shape and context. Mean with mean absolute deviation can describe fairly symmetric data. Median with interquartile range often describes skewed data or data with an unusual value more fairly. A distribution shows which values occur and how often. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

Worked example 1

Reading times in minutes are 3, 4, 4, 5, 5, 6. Choose a center and matching variability measure. Calculate both and justify your choice.

1. Use the shape and context.

2. Mean with mean absolute deviation can describe fairly symmetric data.

3. Mean = 9/2, median = 9/2, IQR = 5 - 4 = 1, MAD = 5/6.

4. Accept another justified choice tied to shape and purpose.

5. State the result in the form requested: Mean 9/2 min and MAD 5/6 min; roughly symmetric data.

Answer: Mean 9/2 min and MAD 5/6 min; roughly symmetric data

Worked example 2

Six students recorded reading times (minutes): 6, 12, 12, 18, 18, 24. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A distribution shows which values occur and how often.

2. Order the observations.

3. Middle positions give median 15 and half medians give IQR 6.

4. Mean is 90/6

5. averaging distances gives MAD 5.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 15, IQR 6; mean 15, MAD 5. Roughly balanced; mean/MAD or median/IQR with justification.

Day 130 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Choose a center and matching variability measure. Calculate both and justify your choice.

02 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Describe a typical value, range, shape, and any striking value.

03 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Choose a center and matching variability measure. Calculate both and justify your choice.

04 / Constructed response

Reading times in minutes are 2, 6, 6, 10, 10, 14. Describe a typical value, range, shape, and any striking value.

05 / Constructed response

Reading times in minutes are 7, 11, 11, 15, 15, 19. Choose a center and matching variability measure. Calculate both and justify your choice.

06 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Describe a typical value, range, shape, and any striking value.

07 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 9, 18. Choose a center and matching variability measure. Calculate both and justify your choice.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 3, 5, 5, 7, 7, 9. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 4, 8, 8, 12, 12, 24. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 11, 20. Describe a typical value, range, shape, and any striking value.

Day 131 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Describe Center, Spread, and Shape

Standards: 6.SP.A.2, 6.SP.A.3

Learning target: I can describe a distribution by center, spread, and shape.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

A distribution shows which values occur and how often. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

Worked example 1

Reading times in minutes are 3, 4, 4, 5, 5, 6. Describe a typical value, range, shape, and any striking value.

1. A distribution shows which values occur and how often.

2. Describe a typical value, how spread out the values are, and whether the shape is roughly symmetric, skewed, clustered, or has unusual values.

3. The values balance around the middle.

4. Range = 6 - 3.

5. State the result in the form requested: Center near 9/2 min; range 3 min; roughly symmetric; no isolated extreme.

Answer: Center near 9/2 min; range 3 min; roughly symmetric; no isolated extreme

Worked example 2

Six students recorded reading times (minutes): 5, 7, 7, 9, 9, 11. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A distribution shows which values occur and how often.

2. Order the observations.

3. Middle positions give median 8 and half medians give IQR 2.

4. Mean is 48/6

5. averaging distances gives MAD 5/3.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 8, IQR 2; mean 8, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Day 131 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Describe a typical value, range, shape, and any striking value.

02 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Describe a typical value, range, shape, and any striking value.

03 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Describe a typical value, range, shape, and any striking value.

04 / Constructed response

Reading times in minutes are 8, 10, 10, 12, 12, 14. Describe a typical value, range, shape, and any striking value.

05 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Describe a typical value, range, shape, and any striking value.

06 / Constructed response

Reading times in minutes are 8, 11, 11, 14, 14, 17. Describe a typical value, range, shape, and any striking value.

07 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 9, 18. Describe a typical value, range, shape, and any striking value.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 6, 8, 8, 10, 10, 12. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 10, 10, 15, 15, 28. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 8, 17. Describe a typical value, range, shape, and any striking value.

Day 132 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Summarize Data in Context

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can write a statistical summary tied to its context.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value.

Worked example 1

Reading times in minutes are 8, 9, 9, 10, 10, 11. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Explain why the chosen center and spread fit the data and mention any striking value.

3. Calculate the center and spread before interpreting them in context.

4. Other accurate, justified summaries are acceptable.

5. State the result in the form requested: 6 observations of reading time in minutes. Mean 19/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable..

Answer: 6 observations of reading time in minutes. Mean 19/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Worked example 2

Six students recorded reading times (minutes): 2, 4, 4, 6, 6, 8. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Order the observations.

3. Middle positions give median 5 and half medians give IQR 2.

4. Mean is 30/6

5. averaging distances gives MAD 5/3.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 5, IQR 2; mean 5, MAD 5/3. Roughly balanced; mean/MAD or median/IQR with justification.

Day 132 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

02 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 6, 7. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

03 / Constructed response

Reading times in minutes are 5, 6, 6, 7, 7, 8. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

04 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

05 / Constructed response

Reading times in minutes are 2, 8, 8, 14, 14, 20. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

06 / Constructed response

Reading times in minutes are 3, 9, 9, 15, 15, 21. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

07 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 3, 6, 6, 9, 9, 12. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 9, 9, 13, 13, 25. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 11, 20. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

Day 133 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Compare Data Distributions

Standards: 6.SP.A.2, 6.SP.B.5d

Learning target: I can compare centers and variability in two data sets.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

Compute comparable measures for both groups. Compare typical values and consistency using the same units. State what the data support without claiming that every member of one group exceeds every member of the other.

Worked example 1

Reading times (min): Group A [6, 7, 7, 8, 8, 9], Group B [8, 9, 9, 10, 10, 11]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

1. Compute comparable measures for both groups.

2. Compare typical values and consistency using the same units.

3. Adding 2 shifts all values and the median by 2, but does not change Q3 - Q1.

4. State the result in the form requested: A median 15/2, B median 19/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability..

Answer: A median 15/2, B median 19/2 min; both IQR 1 min. B is typically 2 min higher; equal middle-half variability.

Worked example 2

Six students recorded reading times (minutes): 2, 4, 4, 6, 6, 8. A second group has every corresponding time 5 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

1. Compute comparable measures for both groups.

2. Adding 5 shifts every position and both quartiles by 5.

3. The median increases by 5

4. Q3 - Q1 remains 2.

Answer: First median 5; second 10 min; both IQR 2 min.

Day 133 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times (min): Group A [8, 9, 9, 10, 10, 11], Group B [10, 11, 11, 12, 12, 13]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

02 / Constructed response

Reading times (min): Group A [4, 5, 5, 6, 6, 7], Group B [6, 7, 7, 8, 8, 9]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

03 / Constructed response

Reading times (min): Group A [7, 8, 8, 9, 9, 10], Group B [9, 10, 10, 11, 11, 12]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

04 / Constructed response

Reading times (min): Group A [8, 12, 12, 16, 16, 20], Group B [10, 14, 14, 18, 18, 22]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

05 / Constructed response

Reading times (min): Group A [6, 9, 9, 12, 12, 15], Group B [8, 11, 11, 14, 14, 17]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

06 / Constructed response

Reading times (min): Group A [6, 8, 8, 10, 10, 12], Group B [8, 10, 10, 12, 12, 14]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

07 / Constructed response

Reading times (min): Group A [4, 5, 5, 6, 7, 16], Group B [6, 7, 7, 8, 9, 18]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

08 / Two-part comparison and reasoning

Six students recorded reading times (minutes): 5, 10, 10, 15, 15, 20. A second group has every corresponding time 4 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

09 / Two-part comparison and reasoning

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. A second group has every corresponding time 2 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

10 / Constructed response

Reading times (min): Group A [6, 7, 7, 8, 9, 18], Group B [8, 9, 9, 10, 11, 20]. Compare median and IQR. Which group reads longer typically? Which is more consistent?

Day 134 | Unit 09 | Lesson

Display, Describe, and Summarize Data

Plan and Explain a Statistical Investigation

Standards: 6.SP.A.1, 6.SP.B.4, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d

Learning target: I can plan and Explain a Statistical Investigation, and explain and check my reasoning.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

01 / Learn

A statistical question anticipates different answers in a collection of data. Name the group, the measured attribute, and the expected variation. A question about one fixed fact is not statistical. Draw an equally spaced number line and stack one dot per observation above its value. Repeated values stack vertically. Add an axis label with units and check that the dot count equals the data count. The mean is a fair share: add all values and divide by the number of observations. Report the count and measurement units. A large unusual value can pull the mean toward it. A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value.

Worked example 1

Is “What are the wingspans, in centimeters, of the birds observed at a feeder?” statistical? Explain the group, attribute, units, and variation.

1. A statistical question anticipates different answers in a collection of data.

2. Name the group, the measured attribute, and the expected variation.

3. The question anticipates a collection of measurements.

4. Each bird can have a different wingspan.

5. Use the same unit for every observation.

6. State the result in the form requested: Yes; observed birds; wingspan; centimeters; wingspans can vary..

Answer: Yes; observed birds; wingspan; centimeters; wingspans can vary.

Worked example 2

Six students recorded reading times (minutes): 3, 6, 6, 9, 9, 12. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Order the observations.

3. Middle positions give median 15/2 and half medians give IQR 3.

4. Mean is 45/6

5. averaging distances gives MAD 5/2.

6. Tie the chosen summary to the displayed shape and context.

Answer: 6 observations; minutes; median 15/2, IQR 3; mean 15/2, MAD 5/2. Roughly balanced; mean/MAD or median/IQR with justification.

Day 134 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Is “How many pages did each student read last night?” statistical? Explain.

02 / Constructed response

Reading times in minutes are 7, 8, 8, 9, 9, 10. Draw a labeled dot plot. Check the number of dots.

03 / Constructed response

Reading times in minutes are 6, 7, 7, 8, 8, 9. Report n, units, and mean.

04 / Constructed response

Reading times in minutes are 2, 7, 7, 12, 12, 17. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

05 / Constructed response

Rewrite “How long is my pencil?” as a statistical question.

06 / Constructed response

Reading times in minutes are 3, 8, 8, 13, 13, 18. Draw a labeled dot plot. Check the number of dots.

07 / Constructed response

Reading times in minutes are 4, 5, 5, 6, 7, 16. Report n, units, and mean. If one more student reads 8 minutes, find the new mean.

08 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

09 / Multiple select

A class studies a collection of backpacks. Select all statistical questions, and identify the expected variability.
A. How does mass vary among the backpacks?
B. How many centimeters are in 5 meters?
C. What is the mass in kilograms of each of the backpacks?
D. What is 5 + 3?

10 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 5, 14. Draw a labeled dot plot. Describe a cluster or gap and report the count.

Day 135 | Unit 09 | Unit Test

Display, Describe, and Summarize Data

Unit 9 Test

Standards: 6.G.A.1, 6.G.A.3, 6.SP.A.1, 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.4, 6.SP.A.3, 6.SP.B.5d, 6.SP.A.2

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 8: Display, Describe, and Summarize Data | Spectrum companion Chapter 7

Directions

The first 4 questions (20%) review Unit 8. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Sketch a triangle with base 8 cm and perpendicular height 4 cm. Find its area.

02 / Previous unit review | Constructed response

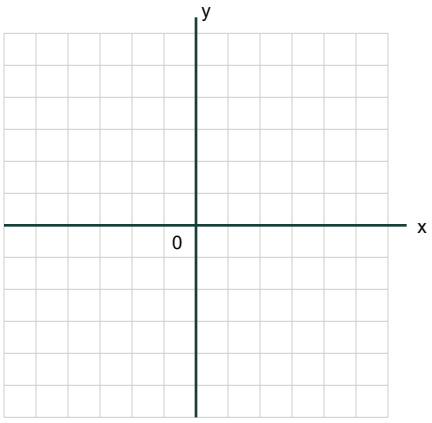
A parallelogram has base 2 m and perpendicular height 3 m. Sketch how it can be rearranged into a rectangle and find its area.

03 / Previous unit review | Constructed response

An L-shaped garden is a 6 m by 6 m rectangle with an 2 m by 2 m corner rectangle removed. Sketch and find the garden area.

04 / Previous unit review | Constructed response

Plot and connect (0,0), (9,0), (9,9), (0,9). Find the side lengths and area.



Label the scale and units.

05 / Constructed response

Is “What are the heights of the plants in our garden?” statistical? Explain.

06 / Constructed response

Reading times in minutes are 3, 4, 4, 5, 5, 6. Report n, units, and mean.

07 / Constructed response

Reading times in minutes are 8, 9, 9, 10, 10, 11. Draw a labeled dot plot. Check the number of dots.

08 / Constructed response

Reading times in minutes are 7, 9, 9, 11, 11, 13. Find the five-number summary and draw a labeled box plot.

09 / Constructed response

Reading times in minutes are 6, 10, 10, 14, 14, 18. Draw a labeled dot plot. Check the number of dots.

10 / Constructed response

Daily reading times in minutes: 12, 15, 14, 19, 12, 12, 6, 15, 8, 5. Draw a histogram using bins 0–4, 5–9, 10–14, and 15–19. Label axes.

11 / Constructed response

Reading times in minutes are 5, 11, 11, 17, 17, 23. Find Q1, Q3, IQR, and range.

12 / Numeric entry and interpretation

Six students recorded reading times (minutes): 2, 5, 5, 8, 8, 11. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

13 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 7, 7, 9, 9, 11. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

14 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 7, 13, 13, 19, 19, 25. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

15 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 11, 11, 17, 17, 23. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

16 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 6, 11, 11, 16, 16, 21. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

17 / Two-part comparison and reasoning

Six students recorded reading times (minutes): 6, 8, 8, 10, 10, 12. A second group has every corresponding time 4 minutes larger. Part A: compare medians. Part B: compare IQRs and justify without listing all new data.

18 / Data display and numeric entry

A frequency table records books read: 0 books: 1 students; 1: 3; 2: 4; 3: 5. Part A: draw a dot plot. Part B: enter the observation count and explain how to verify it.



Choose and label an equal-interval scale.

19 / Multiple select

A class studies a collection of students. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 2 meters?
B. What is the reading time in minutes of each of the students?
C. What is 2 + 5?
D. How does reading time vary among the students?

20 / Multiple select

Six students recorded reading times (minutes): 6, 8, 8, 10, 10, 12. Select all true statements.

A. The median is the average of the 3rd and 4th ordered values.

B. The median is 10 minutes.

C. The median is always the largest observation.

D. The median is 9 minutes.

Day 136 | Unit 10 | Lesson

AASA Applications and Cumulative Review

Connect Rational Numbers, Expressions, and Ratios

Standards: 6.NS.A.1, 6.NS.C.7a, 6.NS.C.7b, 6.EE.A.2c, 6.RP.A.3a

Learning target: I can connect Rational Numbers, Expressions, and Ratios, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

01 / Learn

If (a/b) ÷ (c/d) = q, then (c/d) × q = a/b. Multiplying by d/c undoes multiplying by c/d, so divide by a nonzero fraction by multiplying by its reciprocal. Simplify the product. A number farther right is greater. Among negative numbers, the value closer to zero is greater. Use a common denominator or familiar benchmarks to compare negative fractions. Replace every variable with its assigned value, using parentheses. Apply the order of operations. In formulas, label the answer with the appropriate measurement units. Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order.

Worked example 1

Calculate 1/2 ÷ 1/2. Simplify.

1. Rename the numbers as fractions: 1/2 = 1/2 and 1/2 = 1/2.

2. Keep the first fraction and multiply by the reciprocal of the second: 1/2 × 2.

3. Multiply numerators and denominators: (1 × 2) / (2 × 1) = 2/2.

4. Simplify by dividing top and bottom by 2: 1.

5. Check the result: 1/2 × 1 = 1/2, so the quotient is correct.

Answer: 1

Worked example 2

A recipe uses 5 cups oats for 3 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (26, 15)
B. (15, 9)
C. (10, 7)
D. (10, 6)

1. Multiply both values in a ratio by the same scale factor.

2. Equivalent ratios multiply both quantities by the same factor.

3. A uses 2

4. B uses 3.

5. C and D change only one scaled quantity.

Answer: B, D - (15, 9); (10, 6)

Day 136 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Calculate 1/4 ÷ 1/2. Simplify.

02 / Constructed response

Compare -2 ___ -7 using <, >, or =. Sketch a number line to justify.

03 / Constructed response

Evaluate 4x + 3 for x = 4.

04 / Constructed response

A mix uses 3 cups of oats for 7 cups of seeds. Complete the table (oats, seeds): (3, 7), (6, ___), (12, ___).

05 / Constructed response

Calculate 1/2 ÷ 1. Simplify.

06 / Constructed response

Compare -3/2 ___ -7/5 using <, >, or =. Sketch a number line to justify.

07 / Constructed response

A rectangle has length x + 8 and width 2. Use A = lw to write its area expression and find A when x = 5.

08 / Multiple select

A recipe uses 3 cups oats for 4 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (9, 12)
B. (6, 9)
C. (6, 8)
D. (16, 20)

09 / Multiple choice

Calculate 2/3 ÷ 1/5. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 10/3
B. 7/3
C. 3/10
D. 13/3

10 / Constructed response

Temperatures are -7/3°C and -5/2°C. Write an inequality comparing them and explain which is warmer.

Day 137 | Unit 10 | Lesson

AASA Applications and Cumulative Review

Connect Equations, Rates, and Percent

Standards: 6.EE.B.6, 6.EE.B.7, 6.RP.A.3b, 6.RP.A.3c

Learning target: I can connect Equations, Rates, and Percent, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

01 / Learn

Define the unknown quantity. Translate the relationship into an equation before calculating. Use the inverse operation and check whether the answer makes sense in the situation. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating. Write part = decimal rate × whole. Divide the part by the decimal rate to find the whole. Check by taking the stated percent of your answer. Divide the part by the whole, then multiply by 100. Keep the same units in part and whole. The percent can be above 100 if the part is larger than the reference whole.

Worked example 1

A collection grows by 5 cards to 10 cards. Let x be the starting count. Write and solve an equation.

1. Define the unknown quantity.

2. Translate the relationship into an equation before calculating.

3. Subtract 5 from both sides

4. 5 + 5 = 10.

5. State the result in the form requested: x + 5 = 10; x = 5 cards.

Answer: x + 5 = 10; x = 5 cards

Worked example 2

A quantity changes from a reference whole of 120 to a compared amount of 30. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

1. Divide the part by the whole, then multiply by 100.

2. Percent = (30 ÷ 120) × 100 = 25%.

3. The denominator is the reference whole, not automatically the larger number.

Answer: 25%; denominator 120

Day 137 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

A collection grows by 5 cards to 35 cards. Let x be the starting count. Write and solve an equation.

02 / Constructed response

A bus travels at 40 km/h. How long does 200 km take?

03 / Constructed response

50% of a number is 40. Find the number and check.

04 / Constructed response

144 is what percent of 240?

05 / Constructed response

7 identical lengths of rope total 28 m. Let x be the length of one piece. Write and solve an equation, then check.

06 / Constructed response

A bus travels at 40 km/h. How long does 120 km take?

07 / Constructed response

125% of a number is 225. Find the number and check.

08 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 75. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

09 / Equation entry and two-part response

2 identical ropes have total length 3 meters. Part A: write and solve an equation for one rope length x. Part B: show a substitution check.

10 / Constructed response

A bus travels at 35 km/h. How long does 175/2 km take?

Day 138 | Unit 10 | Lesson

AASA Applications and Cumulative Review

Connect Coordinates and Geometry Models

Standards: 6.NS.C.8, 6.G.A.3, 6.G.A.1, 6.G.A.4, 6.G.A.2

Learning target: I can connect Coordinates and Geometry Models, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

01 / Learn

Plot both points. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units. On opposite sides of zero, add distances to zero. On the same side, subtract magnitudes. Plot vertices in order and connect them, including the last to the first. Equal x-values make a vertical side; equal y-values make a horizontal side. Count or subtract coordinates to get the side lengths. A closed rectangular prism has two faces of area lw, two of lh, and two of wh. Add the six face areas: SA = 2lw + 2lh + 2wh. If a face is missing, do not include it. A rectangular prism has volume V = lwh = Bh, where B is base area. Fractional edge lengths still multiply. Use cubic units; surface area and volume measure different attributes.

Worked example 1

Plot (-3, 5) and (-8, 5). Find their distance.

1. Plot both points.

2. If x-coordinates match, count the vertical units; if y-coordinates match, count horizontal units.

3. Only one coordinate changes.

4. Same side of 0: subtract magnitudes to get 5.

5. State the result in the form requested: 5 units.

Answer: 5 units

Worked example 2

A prism measures 13/2 m by 2 m by 2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

1. A rectangular prism has volume V = lwh = Bh, where B is base area.

2. Multiply edges: 13/2 × 2 × 2 = 26.

3. The edge counts are 13, 4, 4, giving 208 cubes.

4. Each cube is 1/8 m^3

5. 208/8 = 26.

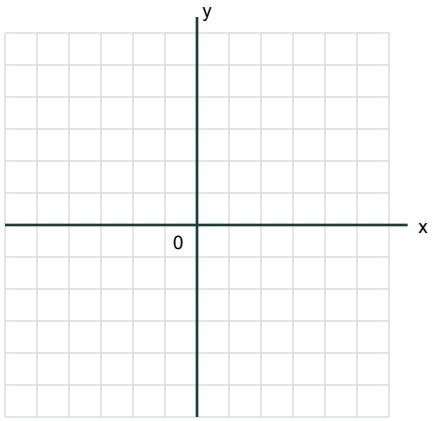
Answer: 26 m^3; 208 cubes

Day 138 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

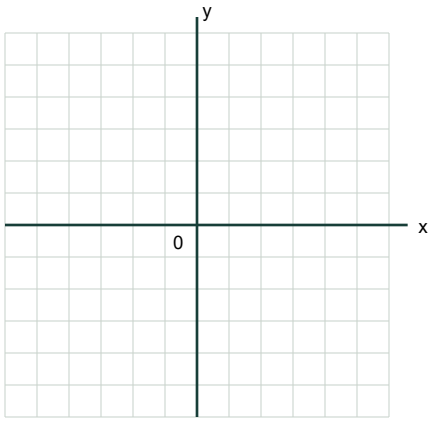
Plot (4, -6) and (4, -10). Find their distance.



Label the scale and units.

02 / Constructed response

Plot and connect (0,0), (12,0), (12,8), (0,8). Find the side lengths and area.



Label the scale and units.

03 / Constructed response

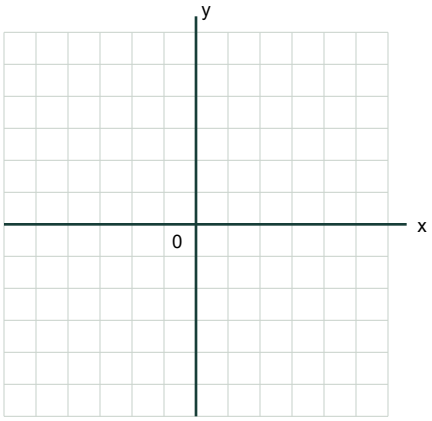
A 5 cm by 5 cm base and 6 cm height form a closed box. Draw its net and find the material area. Ignore tabs and overlap.

04 / Constructed response

A rectangular prism has edges 3, 3, and 3 cm. Sketch it and find its volume.

05 / Constructed response

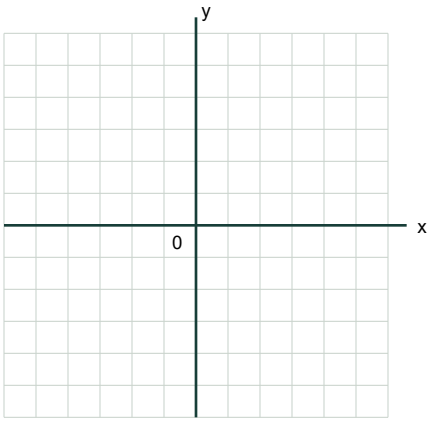
Plot (6, -4) and (6, 5). Find their distance.



Label the scale and units.

06 / Constructed response

Plot and connect (-6,-4), (4,-4), (4,4), (-6,4). Find the side lengths and area.



Label the scale and units.

07 / Constructed response

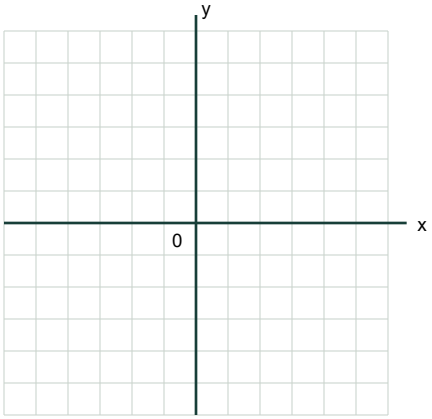
A 4 cm by 2 cm base and 5 cm height form an open-top box. Draw its net and find the material area. Ignore tabs and overlap.

08 / Numeric entry and model reasoning

A prism measures 5/2 m by 1 m by 5/2 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

09 / Coordinate plotting and numeric entry

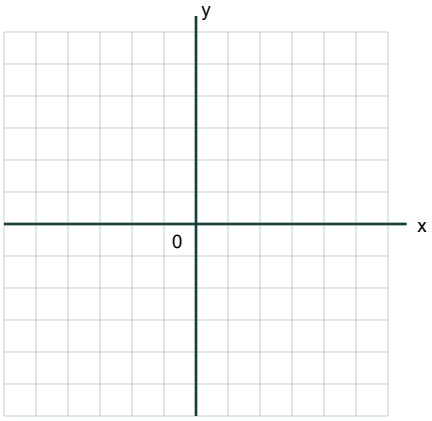
Plot A(-7, -6), B(5, -6), and C(-7, 6). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

10 / Constructed response

Plot and connect (-3,-6), (3,-6), (3,3), (-3,3). Find the side lengths and area. Find the perimeter too.



Label the scale and units.

Day 139 | Unit 10 | Lesson

AASA Applications and Cumulative Review

Explain Data and Critique Mathematical Claims

Standards: 6.SP.B.5a, 6.SP.B.5b, 6.SP.B.5c, 6.SP.B.5d, 6.EE.A.4, 6.EE.B.5, 6.EE.B.8

Learning target: I can explain Data and Critique Mathematical Claims, and explain and check my reasoning.

Cumulative application across enVision Topics 1–8 | Spectrum cross-chapter review

01 / Learn

A useful summary names the group and attribute, units, observation count, center, variability, and shape. Explain why the chosen center and spread fit the data and mention any striking value. Use operation properties to rewrite an expression. Matching values at one input do not prove equivalence for all inputs. A single counterexample is enough to show two expressions are not equivalent. Substitute each value and decide whether the comparison is true. A continuous number-line ray represents infinitely many real-number solutions. A real situation may restrict useful solutions to whole numbers. Find the mean. Calculate each value’s distance from the mean, add those distances, then divide by the number of observations. MAD describes a typical distance from the mean in the original units.

Worked example 1

Reading times in minutes are 6, 7, 7, 8, 8, 9. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

1. A useful summary names the group and attribute, units, observation count, center, variability, and shape.

2. Explain why the chosen center and spread fit the data and mention any striking value.

3. Calculate the center and spread before interpreting them in context.

4. Other accurate, justified summaries are acceptable.

5. State the result in the form requested: 6 observations of reading time in minutes. Mean 15/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable..

Answer: 6 observations of reading time in minutes. Mean 15/2 min, MAD 5/6 min; roughly symmetric, so mean/MAD are reasonable.

Worked example 2

Six students recorded reading times (minutes): 3, 8, 8, 13, 13, 18. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

1. Find the mean.

2. Mean = 63/6 = 21/2.

3. Absolute deviations: 15/2, 5/2, 5/2, 5/2, 5/2, 15/2.

4. Their sum divided by 6 is 25/6.

5. This is the average distance from the mean.

Answer: Mean 21/2 min; MAD 25/6 min

Day 139 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

Reading times in minutes are 2, 3, 3, 4, 4, 5. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

02 / Constructed response

Are 5(x + 3) and 5x + 15 equivalent for every x? Use a property or a counterexample.

03 / Constructed response

Which values in {9, 10, 11} satisfy x < 10? Test each by substitution.

04 / Constructed response

Reading times in minutes are 7, 10, 10, 13, 13, 16. Calculate the mean and MAD. Show all absolute deviations.

05 / Constructed response

Reading times in minutes are 5, 7, 7, 9, 9, 11. Write a summary with count, attribute, units, center, variability, shape, and a justified choice of measures.

06 / Constructed response

Are 6(x + 2) and 6x + 12 equivalent for every x? Use a property or a counterexample.

07 / Constructed response

Which values in {10, 11, 12} satisfy x < 11? Does the real-number inequality have only these solutions? Explain.

08 / Numeric entry and interpretation

Six students recorded reading times (minutes): 3, 7, 7, 11, 11, 15. Part A: enter the mean. Part B: enter the MAD and interpret it in minutes.

09 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 5, 11, 11, 17, 17, 31. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.

10 / Constructed response

Are 3(x + 4) and 3x + 12 equivalent for every x? Use a property or a counterexample.

Course Final Test

Grade 6 Mathematics

Name: _____ Date: _____

Directions

This test has 48 questions in two test units of 24 questions. Work independently without a calculator. Read each question carefully. For multiple choice, select one answer; for multiple select, select every correct answer and no incorrect answer. Write exact numbers or expressions when requested, and label graph scales and units. Use the spaces provided for work.

This is original Math Coach practice modeled on public AASA item formats. It is not an official Arizona assessment.

Test Unit 1

Questions 1–24 | No calculator

01 / Constructed response

Use the standard algorithm: 39.5 + 8.2. Show your regrouping or division steps.

02 / Constructed response

Write 4^3 as repeated multiplication and evaluate.

03 / Constructed response

A tray has 5 red tiles and 5 blue tiles. Write the red-to-blue ratio in simplest form.

04 / Constructed response

A marker is 20 m below sea level. Write its signed elevation and explain what elevation 0 means.

05 / Constructed response

For 2x + 7y + 4, list terms, coefficients, and the constant.

06 / Multiple select

A class studies a collection of dogs. Select all statistical questions, and identify the expected variability.
A. How many centimeters are in 4 meters?
B. How does mass vary among the dogs?
C. What is 4 + 5?
D. What is the mass in kilograms of each of the dogs?

07 / Constructed response

Use long division: 300 ÷ 5. Show the algorithm and check.

08 / Constructed response

Solve x + 2 = 6. Check in the original equation.

09 / Multiple choice

Use the standard algorithm: 2.6 - 1.63. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 97/10
B. 97/100
C. 197/100
D. -3/100

10 / Multiple select

A recipe uses 3 cups oats for 2 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (6, 5)
B. (7, 4)
C. (6, 4)
D. (9, 6)

11 / Numeric entry and error analysis

Evaluate $3^2 + 4(5 - 3)$. Part A: enter the value. Part B: a student begins by adding 3^2 and 4; explain why that changes the expression.

12 / Multiple choice

Use the standard algorithm: 1.92 × 1.4. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 461/125
B. 211/125
C. 336/125
D. 672/25

13 / Multiple select

A triangle has base 14 cm and perpendicular height 5 cm. Select all expressions that give its area in cm².
A. 14 × 5
B. 7 × 5
C. (14 × 5) ÷ 2
D. (14 + 5) ÷ 2

14 / Expression entry and reasoning

A student buys x markers at $5 each and y folders at $2 each, then uses one $2 coupon. Part A: enter an expression for the total cost. Part B: explain why the coupon is subtracted only once.

15 / Multiple choice

Use the standard algorithm: 13.75 ÷ 2.5. Show your regrouping or division steps. Select one answer, then justify it with a model or calculation.
A. 55
B. 9/2
C. 13/2
D. 11/2

16 / Double number line and explanation

Draw a double number line for 3 liters of paint covering 3 m². Part A: align 0, 1, 2, and 7 batches. Part B: enter the area covered by 21 liters and explain why the scales can be different.

17 / Numeric entry and reasoning

Let x = 3/2 and y = 3. Evaluate 7x + y². Part A: enter the value. Part B: show substitution and the order of operations.

18 / Multiple choice

Draw a bar partitioned into 4 equal parts. Shade 2 parts. Circle groups of 1 parts. Use the model to find 1/2 ÷ 1/4. Select one answer, then justify it with a model or calculation.
A. 3
B. 2
C. 1
D. 20

19 / Multiple select

5 notebooks cost $25. Select all statements at the same price per notebook.
A. 3 notebooks cost $16.
B. Each notebook costs $125.
C. 3 notebooks cost $15.
D. Each notebook costs $5.

20 / Multiple select

Select all expressions equivalent to 2(x + 4).
A. 2x + 4
B. 2x + 8
C. 8 + 2x
D. 6x

21 / Multiple choice

Calculate 7/4 ÷ 3/5. Simplify. Select one answer, then justify it with a model or calculation.
A. 12/35
B. 23/12
C. 47/12
D. 35/12

22 / Two-part numeric entry

A 10 m by 10 m rectangle has a right-triangle corner removed with perpendicular legs 4 m and 6 m. Part A: enter the removed area. Part B: enter the remaining area and justify subtraction.

23 / Multiple select

Select all expressions equivalent to 5x + 6x + 5.
A. 16x
B. 11x + 6
C. (5 + 6)x + 5
D. 11x + 5

24 / Multiple choice

Calculate 5 ÷ 2 1/2. Simplify. Select one answer, then justify it with a model or calculation.
A. 2
B. 3
C. 1
D. 1/2

Test Unit 2

Questions 25–48 | No calculator

25 / Two-part numeric entry and reasoning

Store A sells 6 notebooks for $24. Store B sells 7 notebooks for $35. Part A: enter the unit price at each store. Part B: choose the better buy and explain why package totals alone are insufficient.

26 / Multiple select

Select all expressions equivalent to 6(x + 4).
A. 10x
B. 6x + 24
C. 6x + 4
D. 24 + 6x

27 / Multiple select

Select all true statements about 12 and 18. Explain why the greatest factor fits the kit situation.
A. 6 is the largest number of identical no-leftover kits from 12 pencils and 18 erasers.
B. 30 divides both counts.
C. 7 is their greatest common factor.
D. 6 divides both 12 and 18.

28 / Box plot and numeric entry

Six students recorded reading times (minutes): 5, 8, 8, 11, 11, 14. Part A: draw a box plot and label its five-number summary. Part B: enter the IQR.



Choose and label an equal-interval scale.

29 / Multiple select

Solve x - 5/2 = 2. Select every representation of the solution.
A. 18/4
B. 9/2
C. 11/2
D. 13/2

30 / Multiple select

A bicycle travels 40 km in 2 hours at constant speed. Select all true statements.
A. The speed is 42 km per hour.
B. In 2 hours it travels 40 km.
C. In 1 hour it travels 21 km.
D. The speed is 20 km per hour.

31 / Multiple select

Temperatures are -9 °C and -7 °C. Select all true comparisons.
A. -9 is warmer than -7.
B. -9 < -7
C. -7 lies to the right of -9 on a number line.
D. -9 > 0

32 / Multiple select

Solve 2 × x = 7. Select every representation of the solution.
A. 14/4
B. 9/2
C. 11/2
D. 7/2

33 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 7/2 feet.
A. 42 inches
B. 43 inches
C. 7/2 × 12 inches
D. 7/24 inches

34 / Multiple select

Consider -9 and 7. Select all true statements.
A. -9 is farther from zero than 7.
B. |-9| = -9
C. -9 > 7
D. |-9| = 9

35 / Multiple select

Solve x ÷ 3 = 13/6. Select every representation of the solution.
A. 26/4
B. 17/2
C. 13/2
D. 15/2

36 / Numeric entry and equation

A school club has 80 members; 15% take part in a project. Part A: enter the number of participants. Part B: write an equation and explain whether the answer is greater or less than the whole.

37 / Two-part numeric entry

A closed box is 6 cm by 2 cm by 3 cm. Part A: enter its surface area. Part B: the 6-by-2 top is removed; enter the remaining surface area and explain.

38 / Data display and explanation

Reading-time frequencies (minutes) are 0–4: 3, 5–9: 4, 10–14: 5, 15–19: 4. Part A: draw a histogram. Part B: enter the observation count and name every highest-frequency interval.

39 / Model and constructed response

A jug contains 2 L. Each full bottle needs 4/3 L. How many full bottles can be filled, and how much liquid remains?

40 / Multiple select

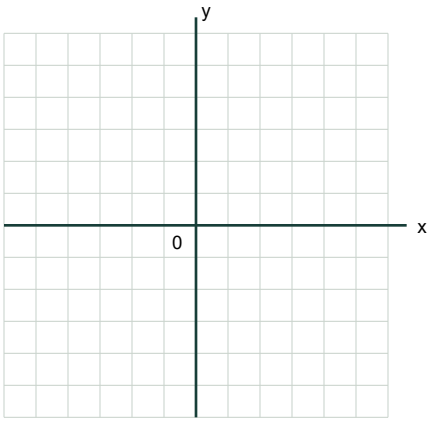
Select all values of x from the list that make x + 2 > 5 true. Explain why each rejected value fails.
A. 5
B. 3
C. 2
D. 4

41 / Multiple select

75 is 125% of an unknown quantity w. Select all equations that find w.
A. (125/100)w = 75
B. w = 75 + 125
C. w = 75 ÷ (125/100)
D. w = 75 × (125/100)

42 / Coordinate plotting and explanation

Plot P(-7, 4), Q(7, -4), and R(0, -4). Part A: name the quadrant of P and Q. Part B: explain why R is not in a quadrant.



Label the scale and units.

43 / Multiple select

A club needs at least 4 members. Select all true statements for its membership count n.

A. n ≥ 4

B. n > 4

C. 4 is an allowed membership count.

D. 3 is an allowed membership count.

44 / Numeric entry and model reasoning

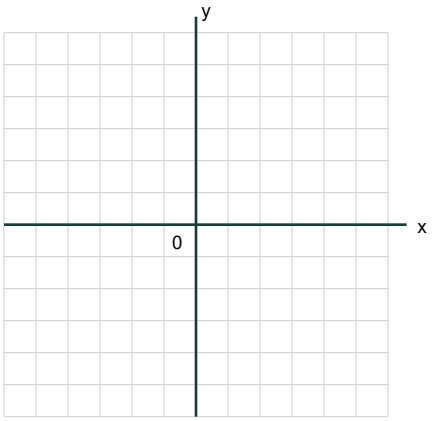
A prism measures 13/2 m by 5/2 m by 1 m. Part A: enter its volume. Part B: how many cubes with edge 1/2 m fill it? Explain why the cube count differs from the cubic-meter volume.

45 / Numeric entry and explanation

A quantity changes from a reference whole of 100 to a compared amount of 20. Part A: the compared amount is what percent of the reference whole? Part B: explain which number belongs in the denominator.

46 / Coordinate plotting and numeric entry

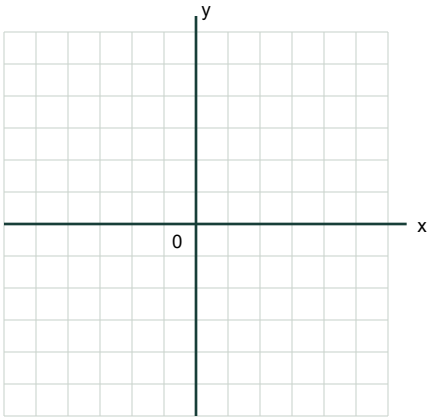
Plot A(-2, -3), B(3, -3), and C(-2, 3). Part A: enter AB. Part B: enter AC and explain how zero helps find each distance.



Label the scale and units.

47 / Table completion, equation, and graph

A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

48 / Statistical reasoning and explanation

Six students recorded reading times (minutes): 2, 8, 8, 14, 14, 28. Part A: describe center and spread using appropriate measures. Part B: describe shape, count, units, and any striking value; justify the chosen measures.